

GEODESY PART 2:

THE GEODESIC ON THE ELLIPSOID
GEODETIC MAIN TASKS
THE PROJECTION OF THE ELLIPSOID

by professor Jon Holsen

Geodesic
Geodetic main tasks
Gauss-Krüger conformal projection

FOR USE IN THE COURSE:
TBA4245 GEODESY

Geomatics
Institutt for bygg- og miljøteknikk (IBM)
Department of Civil and Environmental Engineering
NTNU

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About the compendium, contents

The contents of the compendium:

- It is the mapping of the *ellipsoid* which is dealt with.
- A page containing actual formulas for use at the exam is also included. See the next page of this preface.

See also chapter 8 in the textbook "GNSS" (Hofmann-Wellenhof et al. (2008) for the Gauss-Krüger conformal projection ("Transverse Mercator projection")):

- See especially about the latitude *footpoint* used in the computation from x and y coordinates in the projection plane to latitude and longitude on the spheroid, as the formulas for calculations of the footpoint is not covered in this compendium.
- The formulas of the projection are the same as in this compendium, but the formulas in the GNSS book have generally 1 extra term.

NB: In Hofmann-Wellenhof is the coordinate axis y towards north, in this compendium x is towards north.

2009: In chapter 2 the contents are new, as another method is described compared to the original text. The new general method to compute the geodetic main tasks in chapter 2 also replaces the method described in chapter 3. Therefore chapter 3 is left out in this compendium.

2015: Chapters 1, 2 and 4 are written as word files. Some more explanations, figures and text are added compared to the original text.

NTNU, 13.3.2015
Terje Skogseth
Geomatics NTNU

February/March 2018: A few misprints are corrected by TS. Latest version is from 19.03.2018.

Formulas (see also chapter 2.1) and definitions

DATUM

Datum	a	b	f	Ellipsoid
NGO1948	6377492.0176	6356173.5083	1/299.15281285	Norw. Bessel
ED50 / ED87	6378388.0000	6356911.9461	1/297.000	International
WGS84 / EUREF89	6378137.0000	6356752.3142	1/298.257223562	WGS84

CORRECTIONS FROM THE ELLIPSOID TO THE GAUSS-KRÜGER PROJECTION ON THE PLANE (formulas for the sphere, see chapter 4 for extended formulas):

Directions from s to t: $\delta = -\rho \cdot (x_s - x_t) \cdot (2 \cdot y_s + y_t) / (6 \cdot R^2)$

Distance from s to t: $\Delta S = d = +S_e \cdot (y_s^2 + y_s \cdot y_t + y_t^2) / (6 \cdot R^2)$

R is the radius of curvature and S_e is the ellipsoidal distance.

RADIUS OF CURVATURE, PRIME VERTICAL: $N = R_N = a / (1 - e^2 \cdot \sin^2 B)^{0.5}$

RADIUS OF CURVATURE, MERIDIAN: $M = R_M = a \cdot (1 - e^2) / (1 - e^2 \cdot \sin^2 B)^{1.5}$

RADIUS OF CURVATURE in direction with azimuth α : $R = \frac{M \cdot N}{N \cdot \cos^2 \alpha + M \cdot \sin^2 \alpha}$

(The bearing can often be used instead of the azimuth, α).

THE GAUSSIAN SPHERE: $R = \sqrt{(N \cdot M)}$

FLATTENING: $f = (a - b) / a$

ECCENTRICITY (e is the 1st numerical eccentricity): $e^2 = (a^2 - b^2) / a^2 = f \cdot (2 - f)$

LENGTH OF THE MERIDIAN FROM EQUATOR (integrated from Equator to the latitude B):

$$BL = \int M \cdot dB = a \cdot (1 - e^2) \cdot \int (1 - e^2 \cdot \sin^2 B)^{-1.5} \cdot dB$$

THE GAUSSIAN MERIDIAN CONVERGENCE, where $\eta^2 = e^2 \cdot \cos^2 B / (1 - e^2)$:

$$c = l \cdot \sin B + (l^3/3) \cdot \sin B \cdot \cos^2 B \cdot (1 + 3 \cdot \eta^2 + 2 \cdot \eta^4) + (l^5/15) \cdot \sin B \cdot \cos^4 B \cdot (2 - \tan^2 B)$$

(l^5 term can be left out if $|l| < 2,5^\circ$ (ca!?). Can be left out at exam!!)

ORIGIN OF THE NORTH AXES IN THE DATUM NGO1948

Axis	Latitude (°)	Longitude (°)	Axis	Latitude (°)	Longitude (°)
1	58	6.05625000	5	58	16.88958334
2	58	8.38958334	6	58	20.88958334
3	58	10.72291667	7	58	24.88958334
4	58	13.22291667	8	58	29.05625000

UTM: The zones are 6° broad. The zone 31 is between 0° and 6° east of Greenwich. (0° east is the Greenwich meridian). Detailed maps of Norway are covered by the zones 32, 33 and 35. Zone 34 is left out as 33 and 35 are extended.

GEODESY PART 2
LECTURES IN GEODESY

by

Jon Holsen
Professor in geodesy

NORCONSULT INTERNATIONAL AS

In cooperation with

THE UNIVERSITY OF TRONDHEIM

Norwegian Institute of Technology

Division of Geodesy and Photogrammetry

Trondheim, Norway, 1985

“Geodesy part 1” and “Geodesy part 2” are for the main parts a translation from my lectures in geodesy: “Geodesi del 1” and “Geodesi del 2” written in Norwegian.

Jon Holsen

Misprints (errata) found in first Word version of 25.2.2015

Now: Continuous page numbering, for instance page 54 instead of page 4.1 etc. in the original compendium.

Page	New text	Correction
0.1, 2.1, 4.1	Ellipsoid of revolution instead of using Spheroid	Spheroid instead of ellipsoid
1.2, heading	Normal section on the ellipsoid	Ellipsoid instead of spheroid

This compendium is only in digital form (as a pdf file). Therefore new students in the course TBA4245 Geodesy are using the latest version of this compendium. This is the reason for not making a list of the changes made after 2015 to the previous version from 2015.

Geodesy part 2

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CHAPTER 0

0.1 Introduction to chapter 1, 2 and 4

The focus in “Geodesy part 2” is calculations on- and mapping of the ellipsoid of revolution.

Ellipsoid of revolution instead of using spheroid

Scientific literature (particularly geodesy) often uses “*ellipsoid*” in place of “*ellipsoid of revolution*” and only applies the adjective “tri-axial” when treating the general case. Older literature uses “*spheroid*” in place of ‘*ellipsoid of revolution*’.

In this new version of the textbook “Geodesy part 2” we will use *ellipsoid* and mean ellipsoid of revolution. We will not use spheroid, spheroid was used in the old/original version of the textbook.

Chapter 1:

The shortest distance between two points on a sphere is along the great circle arc. Equivalent, the shortest path on an ellipsoid is the geodesic (“geodetisk linje”). The flattening of the ellipsoid makes the curvature dependent on the latitude. The radius of curvature on the ellipsoid is defined in two main directions. 1) The radius of curvature in the prime vertical (“perpendikulærkrumningsradius») R_N also denoted N . And 2). Radius of curvature in the meridian (“meridiankrumningsradius») R_M also denoted M . R_N can be interpreted geometrically as the distance along the ellipsoid normal from the rotational axis to the ellipsoid surface, while R_M has no geometrical interpretation.

At the end of chapter 1, relations between some differentials are derived. These relations will be used to derive equations/formulas from series expansions, for calculations on the ellipsoid, e.g. to find azimuth for and distance along the geodesic, with the Bessel-Helmert method in chapter 2.

Chapter 2:

Holsen explains the Bessel-Helmerts method to solve the two geodetic main tasks (direct or indirect geodetic problem). The method make use of a simplification, projecting curves or angles from the ellipsoid surface up to a sphere with radius equal to the major ellipsoid semi axis a .

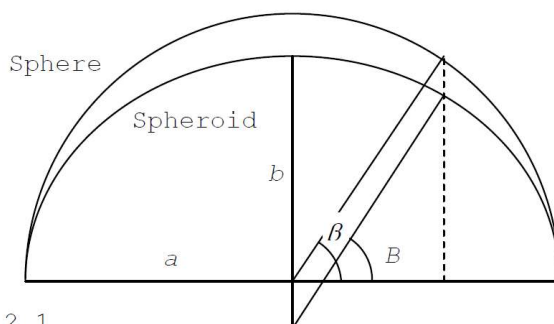


Figure 2.1

Recent years new methods and most important, the number of terms from series expansion to find equations to calculate between a sphere, to/from a ellipsoid surface have been improved. (Karney 2013) ensures/claim precession less than one millimeter using his Matlab toolbox:

<http://www.mathworks.com/matlabcentral/fileexchange/39108-geodesics-on-an-ellipsoid-of-revolution>

Chapter 3: Not covered

Old and approximate methods for calculations on the ellipsoid. Not covered.

Chapter 4: The mapping, corrections, number of terms

Mapping

Chapter 4 explains how to convert from/to ellipsoidal latitude and longitude to/from north and east coordinates in the Gauss-Krüger map projection.

The equations are derived, and Holsen derives the equations based on the auxiliary quantities «hjelpstørrelsene», footpoint «fotpunkts-» and isometric latitude «isometrisk bredde».

Recent years, faster converging methods have been developed also here. (Karney 2011) claims to achieve a precession better than five nanometers (10^{-9} m) in a distance within 3900 km from the central meridian: <http://www.mathworks.se/matlabcentral/fileexchange/15285-geodetic-toolbox> (2017).

Corrections

By using the equations of the map projection we can derive formulas for the meridian convergence, the corrections on the directions and distances and the scale factor. These formulas are for instance used when observed directions and distances on the ellipsoid are planned to be used on the plane for calculations of UTM coordinates.

The number of terms in the equations

The equations are power series with terms of various order. Higher order terms might not contribute to the final results, as they might be equal 0. We need more terms of higher order the further away the point we want to map is from the central meridian, as the size of the terms are very dependent on the distance from the central meridian (the north axis). The maximum difference or distance in longitude is ca. $3-3,5^\circ$ for the UTM projection.

Various textbooks have various ways of presenting the equations regarding the number of terms. In this new version of chapter 4 we have extended the number of terms compared to the old version.

Thus, the mapping equations should have the relevant terms for sub cm accuracy on rectangular coordinates and an accuracy of $0,0001''$ (arc second) = $0,000\ 00003^\circ$ on geographical coordinates.

The same accuracies are obtained by the given equations for the meridian convergence, the corrections on the directions and distances and the scale factor.

Today, the series can be implemented on computer programs, and terms of higher order can easily be included in the equations.

Equations from the various textbooks and articles are compared and used, see the bibliography at the back of this chapter.

This chapter 0 is written by Trond Arve Haakonsen. A few sentences with additional information are added by Terje Skogseth

CHAPTER 1

THE GEODESIC ON THE ELLIPSOID

The formulas in chapter 2.1 are also used in chapter 1.

Version 13.03.2015, TS

1.0 Introduction to chapter 1

The focus in “Geodesy part 2” is calculations on and mapping of the ellipsoid of revolution.

Ellipsoid of revolution instead of using spheroid

Scientific literature (particularly geodesy) often uses “*ellipsoid*” in place of “*ellipsoid of revolution*” and only applies the adjective “tri-axial” when treating the general case.

Older literature uses “*spheroid*” in place of ‘*ellipsoid of revolution*’.

In this new version of the textbook “Geodesy part 2” we will use *ellipsoid* and mean ellipsoid of revolution. We will not use spheroid, spheroid was used in the old/original version of the textbook.

This chapter is extended with more explanations, figures and examples compared to the original content written by Jon Holsen.

Terje Skogseth

Chapter 1:

The shortest distance between two points on a sphere is along the great circle arc. Equivalent, the shortest path on an ellipsoid is the geodesic (“geodetisk linje”). The flattening of the ellipsoid makes the curvature dependent on the latitude.

The radius of curvature on the ellipsoid is defined in two main directions:

1) The radius of curvature in the prime vertical (“perpendikulærkrumningsradius») R_N also denoted N .

2). Radius of curvature in the meridian (“meridiankrumningsradius») R_M also denoted M .

R_N can be interpreted geometrically as the distance along the ellipsoid normal from the rotational axis to the ellipsoid surface, while R_M has no geometrical interpretation.

At the end of chapter 1, relations between some differentials are derived. These relations will be used to derive equations/formulas from series expansions, for calculations at the ellipsoid, e.g. to find azimuth for and distance along the geodesic, with the Bessel-Helmert method in chapter 2.

1.1 Normal sections on the ellipsoid

In this chapter the *geodesic* is introduced as the solution of the problem with defining an unambiguous line or path between two points on the ellipsoid. The geodesic is an unambiguous path between two points on surface of the ellipsoid, and the geodesic is also the shortest path between two points on a curved surface like an ellipsoid or a sphere. This line is also known as a *geodetic line*, “*geodetisk linje*” in Norwegian.

1.1.1 Definitions

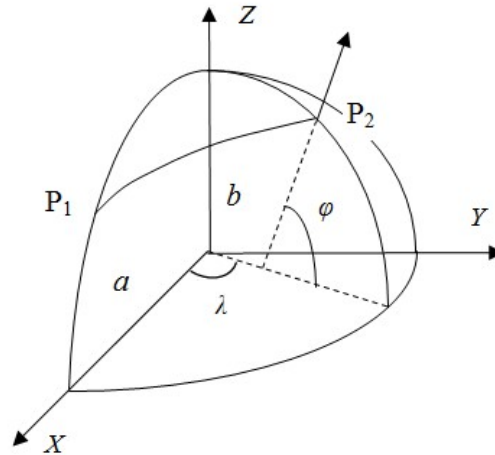


Figure 1.1

From figure 1.1 we can define some lines, angles and coordinates on the ellipsoid. Most of the definitions are also valid on the sphere, see figure 1.2a:

- *Meridian*: A curve between the North Pole and the South Pole. The longitude is constant along the meridian. The meridian through Greenwich is called the Prime Meridian.
- *Parallel circles*, (“*Breddesirkler*”): Circles of latitude (the latitude is constant). The parallel circles are parallel to each other and to Equator, which is also a parallel circle.
- *Semi major and semi minor axes of the ellipsoid*: a and b
- *Normal, surface normal, vertical*: A line forming a 90° angle with the (ellipsoid) surface, is a vector perpendicular to the surface. See the line through point P_2 on figure 1.1.
- *Cartesian ellipsoidal coordinates*: X, Y, Z ., where Z is the rotation axis, X is through the Greenwich meridian. The origin is at the center of the ellipsoid.
- *Latitude* (“*breddegrad*”) in point P : The angle from Equator to the normal through point P , the angle φ . B is also used as a symbol in this book.
- *Longitude* (“*lengdegrad*”) in point P : The angle from the Prime Meridian to the meridian through point P , the angle λ . L is also used as a symbol in this book.
- *Normal plane, vertical plane*: A plane containing the surface normal / vertical of a point on the surface (ellipsoid or sphere).
- *Normal section, vertical section*: The curve produced by the *intersection* of the surface (ellipsoid or sphere) and the normal plane.
- *Normal section arc, vertical section arc* (“*normalsnittbue, vertikalsnittbue*”): A curve from a point to another point on the ellipsoid or the sphere defined by the *normal section*.

- *Radius of the curvature*: From the principal radii R_M (along the meridian) and R_N (along the parallel circle) we can compute other radii of curvature in a point (along a curve or the mean value), see chapter 1 in “Geodesy part 1”.
- *Geodesic, geodetic line* (“geodetisk linje”): The shortest line and an unambiguous line between two points P_1 and P_2 on the ellipsoid or the sphere.

1.1.2 The sphere. Surface normal, normal section, geodesic

See figure 1.2a. On a sphere all surface normals intersect in one point, the centre of the sphere, C . There are infinite vertical planes containing the surface normal at point P_1 . The vertical plane in P_1 in the direction of P_2 will contain P_2 and the surface normal in P_2 .

On a sphere the three points C , P_1 and P_2 define a plane that contains the two surface normals in point P_1 and P_2 . As both the normals in P_1 and P_2 are in the same plane, the normal section arc P_1 - P_2 is the same as the normal section arc P_2 - P_1 on the sphere. Thus the normal section arc will be an unambiguous path between P_1 and P_2 on the sphere.

On the sphere the geodesic will be identical with the normal section arc.

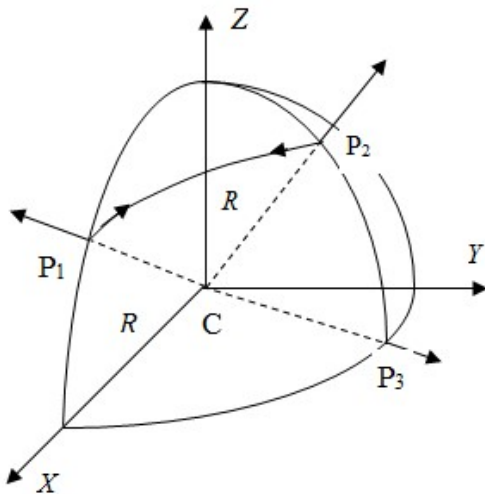


Figure 1.2a The sphere

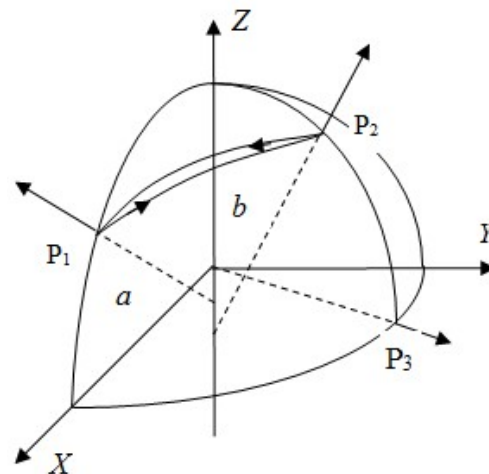


Figure 1.2b The ellipsoid

1.1.3 The ellipsoid. Surface normal, normal section arc, geodesic

See figure 1.2b. On the ellipsoid, however, the surface normals at two different points generally are wry or twisted relative to each other. Exceptions are normals at points of the same latitude and thus also normals at points on the equator. These normals in points of the same latitude will intersect at the same place (in a point on the Z axis, the rotation axis). On figure 1.2b the normal in point P_3 and all other points on the equator intersect the rotation axis Z in the centre of the ellipsoid.

The flattening of the *ellipsoid of rotation* makes the surface normals to intersect the rotation axis Z on different places, depending on the latitude of the points. On the northern hemisphere the intersection will be below the equator plane, see figure 1.2b.

There are infinite vertical planes containing the surface normal at point P_1 . The vertical plane in P_1 in the direction of P_2 will contain P_2 but will not contain the surface normal in P_2 , given that the two points have different values of latitude. Visa versa, the vertical plane in P_2 in the direction of P_1 will contain P_1 but will not contain the surface normal in P_1 . See figure 1.2b.

We will therefore have two alternative normal section arcs between P_1 and P_2 depending on in which point we start.

In the example shown on figure 1.2b the difference between the two paths between P_1 and P_2 is very exaggerated, see chapter 1.3 for a numerical example.

When measuring horizontal angles with a total station, we in principle measure angles between vertical planes. Consequently the measured direction from P_1 to P_2 and the direction from P_2 to P_1 do not refer to the same curve on the ellipsoid. The radius of the curvature at P_1 is less than the radius of the curvature at P_2 , if P_2 is at higher latitude than P_1 .

$$\text{Then } R_{N2} > R_{N1} \quad \text{and} \quad R_{M2} > R_{M1}.$$

R_N and R_M are the principal radii (R_N along the parallel circle and R_M along the meridian), and they are respectively the maximum and the minimum of the radius of curvature of the normal sections at a point.

Therefore we have the largest radius of curvature in point P_2 for the curve P_1 - P_2 . Indicating the azimuth of the normal sections with A and the latitudes with B , we have:

$$R_{A_{21}} > R_{A_{12}} \text{ if } B_2 > B_1$$

This is illustrated on figure 1.3. The plane P_1 - P_2 - K_1 contains point P_2 and the surface normal in P_1 , and the plane P_1 - P_2 - K_2 contains point P_1 and the surface normal in P_2 . The radius of the curvature in P_1 , P_1 - K_1 , is smaller than the radius in P_2 , P_2 - K_2 .

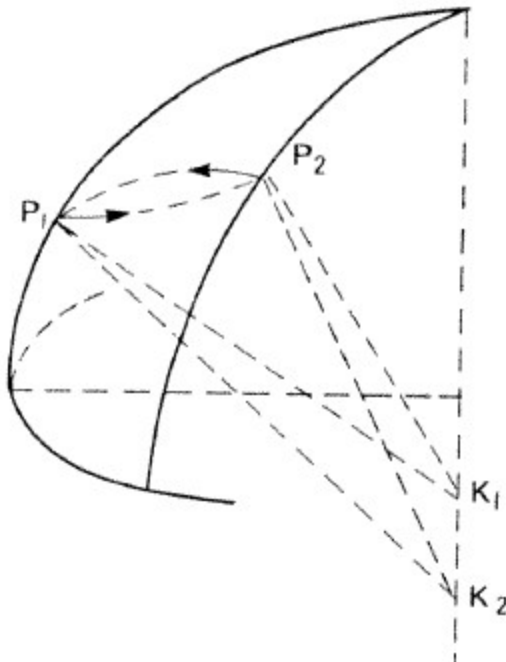


Figure 1.3 The ellipsoid and the normal section arcs

The geodesic (“geodetisk linje”)

The ambiguity of having two different normal section arcs between P_1 and P_2 is eliminated by introducing the geodesic and correcting the measured directions in P_1 and P_2 to the geodesic between P_1 and P_2 .

The geodesic lies in general inside the two normal sections.

1.2 Some geometric properties of the geodesic

In chapter 1.1 we have defined the geodesic as a curve on a spatial surface. The geometric properties of a curve are characterized by the curvature of the curve, and we can define the geodesic by its curvature.

First we shall look into some differential geometry of spatial twisted curves in general, and thereafter specify the characteristics of the geodesic concerning the curvature.

1.2.1 Curvature and torsion of space curves

Any curve in the three-dimensional space is called a space curve.

A curve in two dimensions, a plane curve, lies in a single plane and is a variant of a space curve.

A space curve is characterised by its curvature and its torsion.

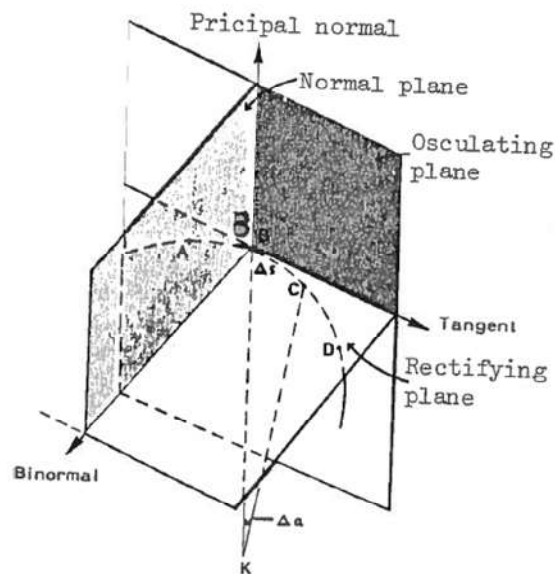


Figure 1.4 Space curve. Planes and lines

With reference to figure 1.4 we define some planes and lines:

1. *A plane defined by three points:* The points A, B and C are three neighbouring points on a space curve. These three points determine or define a plane.
2. *The osculating plane ("smygplan"):* Let points A and C at the same time move up to and coincide with point B. Then we have a limiting case of the plane A-B-C, which is called the osculating plane of the curve at point B.
3. *The tangent of the curve:* The tangent of the curve in point B lies in the osculating plane.
4. *Principal normal ("hovednormal") of the curve:* The normal to the curve in point B lies in the osculating plane, and this principal normal is perpendicular to the tangent in point B.
5. *The normal plane:* A plane normal to the osculating plane and containing the principal normal at B is called the normal plane of the curve at B.
6. *The rectifying plane ("rektifiserende plan"):* A plane containing the tangent at B and which is normal to the osculating and the normal planes is called the rectifying plane of the curve at point B.
7. *The binormal of the curve:* The intersection between the normal plane and the rectifying plane is called the binormal.

We have now at each point of the space curve three mutually orthogonal lines:

- The *tangent*, the *principal normal* and the *binormal*

This configuration of these three lines (or vectors) forms a system of axes, and it is called the *moving trihedral*. In general this trihedral will be twisted when we move along a space curve.

Curvature and torsion

A twisted (3D) curve will normally curve in two planes. A space curve is characterised by its curvature and its torsion. The curvature is used for defining the geodesic, see chapter 1.2.2.

Curvature

Plane curves:

- The curvature of a straight line is defined to be zero, and the radius of the straight line is infinite.
- The curvature of a circle is dependent on the size of the radius, R . The curvature is large if R is small, and the curvature is small if R is large.
- A general plane curve C , see figure 1.5. The osculating plane of the plane curve C lies in the plane. At point P the curvature can be described by a circle in the osculating plane. This circle is an approximation of the curve in and near P , and is called the osculating circle at P .

These examples indicate that the curvature, k , of a curve is defined to be the reciprocal of the radius of the osculating circle, R :

$$k = \frac{1}{R}$$

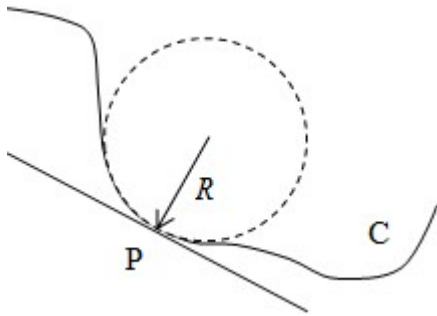


Figure 1.5 Plane curve

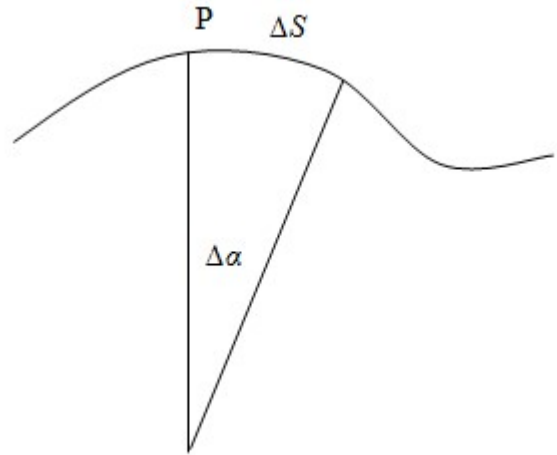


Figure 1.6 Space curve

Space curve:

Generally the curvature at point P is defined by this limiting case, see also figure 1.6:

$$k_1 = \frac{1}{R_1} = \lim_{\Delta S \rightarrow 0} \frac{\Delta \alpha}{\Delta S} = \frac{d\alpha}{dS}$$

This is similar to the definition of the osculating plane on figure 1.4. The radius of curvature is in the osculating plane, we have the osculating circle.

Torsion

Space curve

The curvature is defined in each point of the curve. When we move from point to point along a space curve generally the osculating plane will be twisted. This is illustrated in figure 1.8 where the axes of the two trihedrals are pointing in different directions.

The axes or vectors of a trihedral are the tangent, the principal normal and the binormal. The tangent and the principal normal are in the osculating plane where the curvature is defined. The binormal is orthogonal to the osculating plane, and the binormal will rotate with the osculating plane along the space curve.

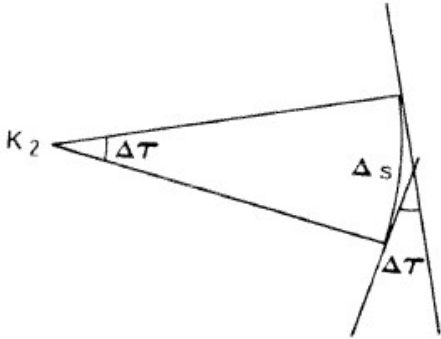


Figure 1.7 Torsion

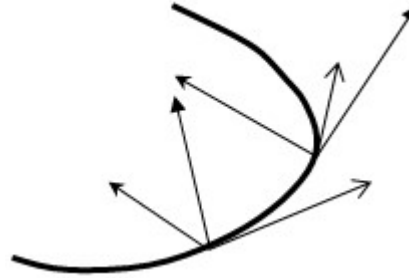


Figure 1.8 Space curve. Tangent ($\rightarrow$), principle normal ($\rightarrow$), binormal ($\rightarrow$)

Animation (2017): <http://en.wikipedia.org/wiki/Curvature>

The rotation of the binormal vector is measured by the torsion. The torsion k_2 of a curve is defined as follows, see figure 1.7 where $\Delta\tau$ is the differential angle between two osculating planes, and the two corresponding binormals form the same angle:

$$k_2 = \frac{1}{R_2} = \lim_{\Delta s \rightarrow 0} \frac{\Delta\tau}{\Delta s} = \frac{d\tau}{ds}$$

Plane curve

We have defined the curvature of a plane curve. The osculating plane is the plane of the curve, we have the same osculating plane for all the points on the plane curve. Along a plane curve all the binormals will be parallel.

Therefore the torsion of a plane curve is 0, there is no torsion.

1.2.2 Geodesic curvature and definition of the geodesic

We will use figure 1.9 to define the geodesic. On figure 1.9:

- Points A, B and C are neighbouring points of a space curve on a surface
- The tangent plane of the surface in point B
- The osculating plane of the curve in point B
- The points a, b and c are the normal projections of A, B and C on the tangent plane

Geodesic curvature

The geodesic is defined to be a curve on a surface. In geodesy we normally use ellipsoid, cone, sphere or a plane as surfaces.

The curve is projected into the tangent plane of the surface, points a, b and c.

We use the points a, b and c to define the geodesic curvature, k_g of the curve.

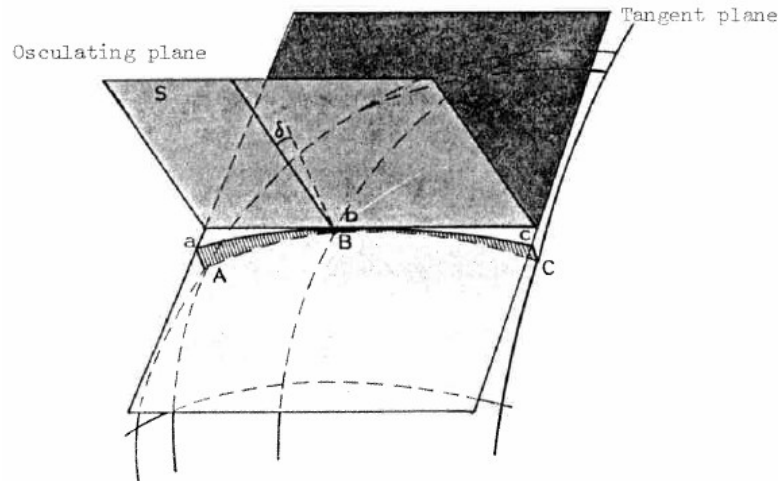


Figure 1.9

We draw a circle through a, b and c on the tangent plane. Let points A and C on the surface at the same time move up to and coincide with point B.

The curvature of this circle through a, b and c on the tangent plane in this limiting case is called the geodesic curvature of the curve on the surface. The radius of the geodesic curvature, R_g , is the reciprocal of the curvature:

$$k_g = \frac{1}{R_g}$$

Let then B wander from P_1 to P_2 , in such way that $k_g = 0$ along the whole curve, and we obtain the geodesic.

The geodesic (“geodetisk linje”):

A curve on a surface whose geodesic curvature in every point is equal to zero, is called a geodesic.

Conditions to obtain a geodesic:

A geodesic curvature of zero means that the geodesic radius is infinite. Thus the points a, b and c in the tangent plane must form a straight line in the limiting case when a and c moves towards b..

Point b coincides with B on figure 1.9. To obtain a straight line, the normal projections of A and C into the tangent plane in the differential mode, must be in the intersection between the tangent plane and the osculating plane. This is possible when the osculating plane is normal to the tangent plane and thus contains the principle normal of the surface.

NB: In figure 1.9 the differential curve A-B-C is not a geodesic: The normal projected points a and c are not in the osculating plane. In the limiting case a, b and c must form a straight line.

A geodesic has the property that its osculating plane at every point of the curve must contain the principle normal of the surface.

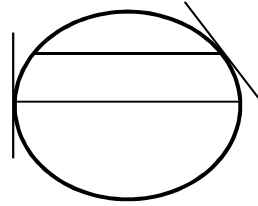
The shortest distance between two points on a surface is a geodesic.

The geodesic is an unique/unambiguous path between two points on a surface.

Examples:

Condition for being a geodesic: The osculating plane of the curve must be normal to the tangent plane of the surface.

- Plane: A straight line is a geodesic
- Plane: A circle is a geodesic
- Sphere: A great circle is a geodesic, as the osculating plane is normal to the tangent plane. The osculating plane is the circle plane.
- Sphere and ellipsoid: Equator is a geodesic, as the osculating plane is normal to the tangent plane. The osculating plane is the Equator plane. See the figure below.
- Ellipsoid: A meridian is a geodesic, as the osculating plane is normal to the tangent plane. The osculating plane is the meridian plane.
- Ellipsoid and sphere: A circle of latitude (a parallel circle) is not a geodesic. The osculating plane is the circle of latitude, and the osculating plane is not normal to the tangent plane. This is illustrated on the figure to the right.

**1.2.3 An example of computing geodesic curvature**

In the examples above we used the geometry to decide if a curve is a geodesic or not.

In this example we calculate the geodesic radius, R_g , of a parallel circle on the ellipsoid. The value of R_g will show if the parallel curve is a geodesic or not.

The example will also repeat some definitions from “Geodesy part 1”, see figure 1.10b:

The definition of geographic latitude, B , and the radius of curvature in the prime vertical, R_N or N on the figure

Figure 1.10a shows three neighbouring points on a parallel circle (PC) of an ellipsoid: P, P' and P''. P' and P'' are placed symmetrical around P and very close to P.

The ellipsoid of rotation is used. Therefore a curve with constant latitude must be a circle. The osculating plane is the parallel circle, also called the circle of latitude.

The radius of the curvature at P of the parallel circle is r , see figure 1.10b which is a normal section of the ellipsoid, normal to the osculating plane of the parallel circle at P.

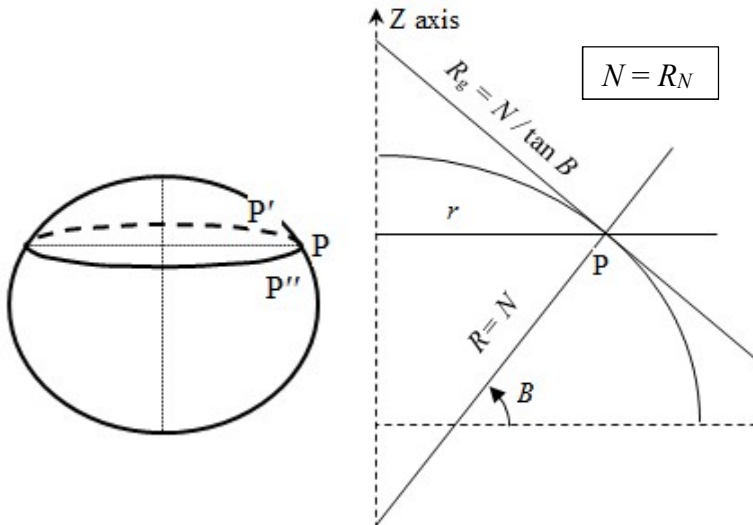


Figure 1.10a Points

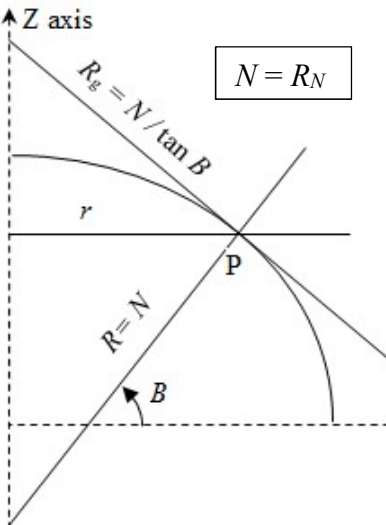


Figure 1.10b Normal section through P

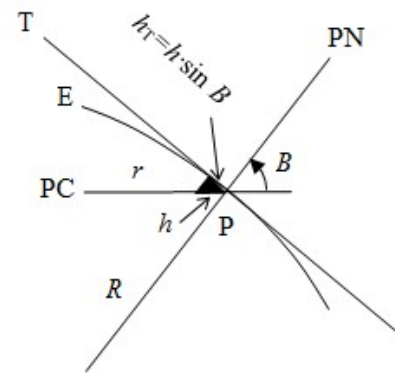


Figure 1.10c Details

With reference to the figures 1.10a,b,c, here are explanations of the symbols:

- PN: Principal normal through P
- T: Tangent plane and the tangent through P
- PC: Parallel plane through P
- B : Geographic latitude of P, and the angle between the osculating plane of the parallel circle and the surface normal in P
- r : Radius of the parallel circle = radius of the curvature of the parallel circle
- R_g : Geodesic radius of curvature in point P
- $R = N$: N is R_N . Radius of curvature in the prime vertical
- h, h_T : Bow height (“pilhøyde”), see also figure 1.11 a and b
- s, s_T : Chord (“kårde”), see figure 1.11 a and b

On figure 1.10b the tangent plane, the normal plane and the parallel plane (= the osculating plane) are normal to the page.

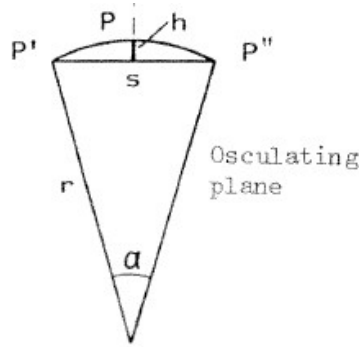


Figure 1.11a On the parallel plane (= the osculating plane)

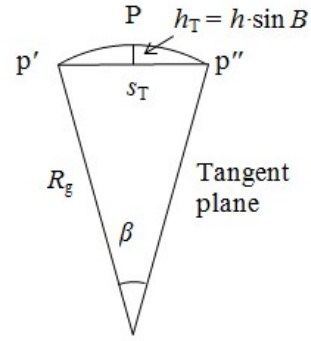


Figure 1.11b On the tangent plane

The points P' and P'' are projected into the tangent plane, the points p' and p'' on figure 1.11b. Using differential geometry the points P, P' and P'' are close to each other, and the distance s and s_T on the figures 1.11 a and b are short. Thus we can use some approximate formulas in the deviation of the geodesic radius of curvature in point P, R_g .

From figure 1.10b, the radius of the parallel circle and the radius of curvature, N :

$$r = R \cdot \cos B = R_N \cdot \cos B \quad (1.1)$$

The bow height h expressed by s and α , figure 1.11 a:

$$h = r - r \cdot \cos \frac{\alpha}{2} = r \cdot \left(1 - \cos \frac{\alpha}{2}\right) \quad (1.2a)$$

The angle α expressed by s and r , figure 1.11 a, approximate formulas:

$$\alpha \approx \frac{s}{r} \quad \cos \frac{\alpha}{2} \approx 1 - \frac{\alpha^2}{8} \approx 1 - \frac{s^2}{8 \cdot r^2} \quad (1.2b)$$

Combining (1.2a) and (1.2b), we have the radius of the curvature of the circle of constant latitude in point P, r :

$$\begin{aligned}
h &= r \cdot \left(1 - \cos \frac{\alpha}{2}\right) \approx \frac{s^2}{8 \cdot r} \\
\Rightarrow r &\approx \frac{s^2}{8 \cdot h}
\end{aligned} \tag{1.3}$$

We defined the geodesic curvature, k_g , and the geodesic radius of curvature, R_g , to be in the tangent plane, where the projected points p' and p'' are. Point P is also in the tangent plane. See figure 1.11b.

A circle through the three points P , p' and p'' in the tangent plane, will define R_g .

The projection of the bow height h into the tangent plane T is, see figure 1.10c:

$$h_T = h \cdot \sin B$$

Replacing h in equation (1.3) with the bow height h_T in the tangent plane and replacing s with s_T , see figures 1.11a and b, and we obtain the geodesic radius of curvature of the parallel circle of constant latitude B , R_g :

$$R_g \approx \frac{s_T^2}{8 \cdot h_T} \approx \frac{s^2}{8 \cdot h \cdot \sin B} \approx \frac{r}{\sin B} \tag{1.4}$$

By combining (1.1) and (1.4) we have the radius of geodesic curvature R_g of the circle of constant latitude on the ellipsoid, expressed by the radius of curvature in the direction of prime vertical, R_N , and the geographical latitude B :

$$R_g = \frac{R_N}{\tan B} \tag{1.5}$$

Geometric interpretation of R_g :

See figure 1.10b. R_g is the distance from point P along the tangent plane to the rotation axis Z .

From equation (1.5) we can calculate the geodesic radius of curvature, R_g , and the corresponding reciprocal value, the geodesic curvature, k_g , when the geographical latitude varies:

- If the latitude $B > 0$, then $R_g > 0$ and $k_g = \frac{1}{R_g} > 0$
- If $B = 0$, then $R_g \rightarrow \infty$ and $k_g = \frac{1}{R_g} = 0$

Conclusion, using the values above, obtained by equation (1.5) and remembering the definition of a geodesic (a curve on a surface whose geodesic curvature in every point is equal to zero, is called a geodesic):

- The parallel circles (circles of constant latitude) are not geodesics, as the geodesic curvature is not zero along the parallel circles.
- Equator is a geodesic, as the geodesic curvature along equator is zero.

These conclusions are the same as we got in the examples at the end of chapter 1.2.2, where we had a geometrical approach and used the conditions for being a geodesic: The osculating plane of the curve must be normal to the tangent plane of the surface.

1.2.4 Another definition of the geodesic

Theoretically the geodesic can be set out by using a theodolite (total station).

Assumption: The surface normals of the ellipsoid are coinciding with the direction of gravity (the plumb lines).

Imagine we set out a straight line on the ellipsoid with a theodolite (total station) in the following way, see figure 1.12:

With the theodolite at point A we aim at point O. Then we turn the telescope exactly 180° and mark point B.

Mounting up the theodolite in B, set out a straight line between A and C by using the same procedure as in A, and continuing to set out until the last point (here D) is set out.

The crosswise deviation between a and b etc. is very small, see the numerical example in ch. 1.3.1.

Imagine the number of stations is approaching infinity and the distances between the stations are approaching zero. Then the normal section arcs (a and b between A and B and so on) will collapse to one curve, which is the geodesic. The dotted line on the figure 1.12 to the right is the geodesic.

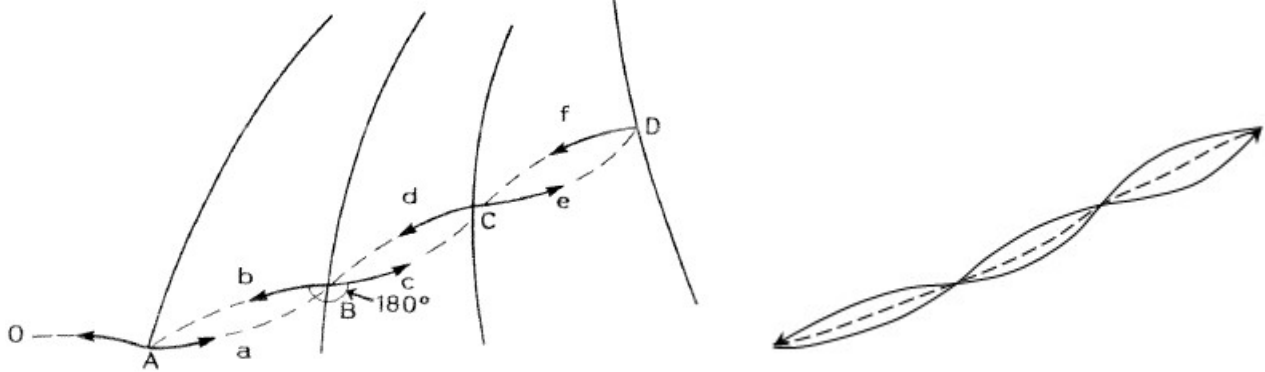


Figure 1.12 Normal section arcs. The dotted line on the figure to the right is the geodesic

1.3 The geodesic and normal sections. Differences, corrections

When we measure angles with a theodolite (total station) we measure angles between the tangents of the normal section arcs. More precise: We mount up the theodolite with its vertical axis in the plumb line, and then correct the measurements from the plum line to the surface normal in the station point.

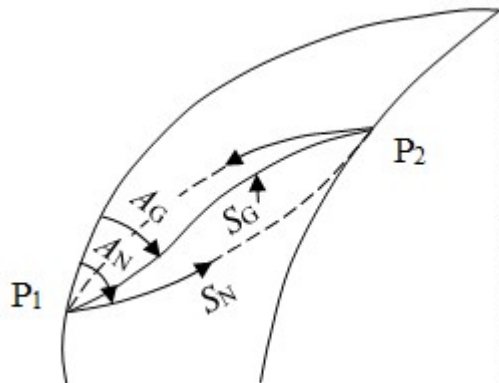


Figure 1.13 Two normal section arcs and the geodesic (in middle) between P_1 and P_2 on the ellipsoid

Measured azimuths and distances are referred to the normal section arcs.

In the computations on the ellipsoid we use the geodesics and not the normal sections, because the geodesic is a unique path on the ellipsoid between two points, while the normal section arc from P_1 to P_2 differ from the normal section arc from P_2 to P_1 . See figure 1.13.

Therefore we need formulas to correct the measurements from the normal section arc to the geodesic. The ellipsoidal azimuths and distances are referred to the geodesic. On figure 1.13 the difference between a normal section arcs and the geodesic is extremely exaggerated. The difference in azimuths on shorter distances is negligible, see a numerical example in chapter 1.3.1.

See chapters 1.4 and 1.5 for other corrections on the azimuth and the distance.

1.3.1 Difference in azimuth

On figure 1.13 the following azimuths are shown:

- A_N is the azimuth in P_1 of the normal section that contains P_2 on the ellipsoid.
- A_G is the azimuth in P_1 of the geodesic between P_1 and P_2 on the ellipsoid.

The correction of the azimuth from A_N to A_G can be computed by equation (1.6a) or (1.6b).

In (1.6a) e and a are ellipsoidal parameters (first numerical eccentricity and semi major axis), B is the geographical latitude of the station point P_1 and S is the distance on the ellipsoid between the two points.

As this correction is very small we can use either B or the reduced latitude β , either A_N or A_G and either the length of the normal section arc or the geodesic distance S in the equation without corrupting the result.

The correction on the azimuth:

$$\Delta A = A_G - A_N = -\frac{e^2 \cdot S^2}{12 \cdot a^2} \cdot \cos^2 B \cdot \sin 2A + \dots \quad (1.6a)$$

NB: In (1.6a) and (1.6b) the difference ΔA is in radians. We must multiply ΔA with a factor $(180/\pi)$ to obtain ΔA in degrees.

If η and R_N are introduced instead of a and e , the correction on the azimuth is:

$$\Delta A = A_G - A_N = -\frac{\eta^2 \cdot S^2}{12 \cdot R_N^2} \cdot \sin 2A + \dots \quad (1.6b)$$

See chapter 2.1 for formulas of η and the radius of the curvature in the prime vertical, R_N .

Numerical example:

$$S_G = 100 \text{ km}$$

$$B = 64^\circ$$

$$A_G = 45^\circ$$

The WGS84 ellipsoid, half axes: $a = 6378137,0000 \text{ m}$ $b = 6356752,3142 \text{ m}$

The correction ΔA is, by using (1.6a): $-0,000\ 00151^\circ = 0^\circ\ 0' -0,0054''$ (arc seconds)

⇒ The difference ΔA , even for a very long distance, 100 km, is far smaller than what is achievable accuracy with a high precision theodolite (total station).

This correction can be neglected for almost all practical cases.

1.3.2 Difference in length

On figure 1.13 the following distances are shown:

- S_N is the distance from P_1 to P_2 of the normal section in P_1 that contains P_2 on the ellipsoid
- S_G is the distance of the geodesic between P_1 and P_2 on the ellipsoid.

The correction of the length from the normal section arc, S_N , to the length of the geodesic, S_G can be computed by equation (1.7a) or (1.7b).

In (1.7a) e and a are ellipsoidal parameters (first numerical eccentricity and semi major axis), B is the geographical latitude of the station point P_1 , A is the azimuth in P_1 to P_2 , and S is the distance on the ellipsoid between the two points.

As this correction is very small we can use either B or the reduced latitude β , either the azimuth A_N or A_G and either the length of the normal section arc or the geodesic distance S in the equation without corrupting the result.

The correction on the distance:

$$\Delta S = S_G - S_N = -\frac{e^4 \cdot S^5}{360 \cdot a^4} \cdot \cos^4 B \cdot \sin^2 2A + \dots \quad (1.7a)$$

If η and R_N are introduced instead of a and e , the correction on the distance is:

$$\Delta S = S_G - S_N = -\frac{\eta^4 \cdot S^5}{360 \cdot R_N^4} \cdot \sin^2 2A + \dots \quad (1.7b)$$

See chapter 2.1 for formulas of η and the radius of the curvature in the prime vertical, R_N .

Numerical example:

$S_G = 3000$ km $B = 64^\circ$ $A_G = 45^\circ$
 The WGS84 ellipsoid, half axes: $a = 6378137,0000$ m $b = 6356752,3142$ m

The correction ΔS is, by using (1.7a): $-0,0007$ m = $-0,7$ mm

⇒ The difference ΔS , even for a very long distance, 3000 km, is far smaller than what is achievable accuracy with a high precision instrument.

This correction can be neglected for almost all practical cases.

1.4 Correction of measured azimuth caused by height above ellipsoid

The skew-normal correction.

The azimuth for use in the calculations is the ellipsoidal azimuth.

If a point, H on figure 1.14, is situated at a height h above the ellipsoid, we have to check if the skew normal in H and P_2 will give significant corrections to the measured azimuth in point P_1 towards H.

On figure 1.14 P_1 , P_2 and P_2'' are points on the ellipsoid, and point H is the height h above the ellipsoid.

The plane P_1 -H- K_1 is the normal section plane in P_1 containing H.

The plane H- P_1 - K_2 is the normal section plane in H containing P_1 .

This correction is called the *skew-normal correction* as the two surface normal P_1 - K_1 and H- K_2 are not in the same plane.

The normal section from P_1 to P_2 has the azimuth A_{N1} (β_1 on the figure).

Point P_2 on the ellipsoid is on the surface normal of point H . A normal plane in P_1 containing point H intersects the ellipsoid along the curve P_1 - P_2'' , and this normal section arc has the azimuth A_H (γ_1 on the figure).

The azimuth A_h (γ_1 on the figure) is, if it is corrected for the deviation of the vertical (plumb line), equal to the measured azimuth at P_1 and should be corrected to A_N (β_1 on the figure).

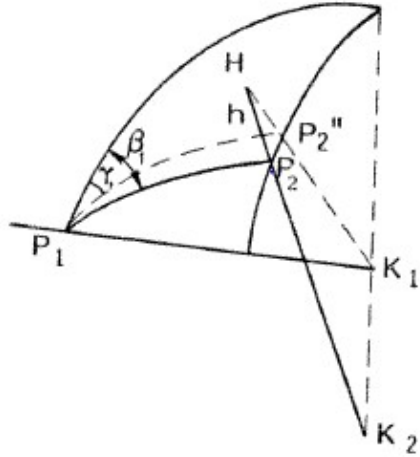


Figure 1.14 Skew-normal correction of the azimuth because of the height above the ellipsoid

The correction from the measured azimuth, A_h , to the ellipsoidal azimuth A_N , can be computed from the following equation (1.8):

The correction on the azimuth, the skew-normal correction:

$$\Delta A_h = (\beta - \gamma) A_N - A_h = \frac{e^2 \cdot h}{2 \cdot b} \cdot \cos^2 B \cdot \sin 2A + \dots \quad (1.8)$$

In (1.8) e and b are ellipsoidal parameters (first numerical eccentricity and semi minor axis), B is the geographical latitude of the station point P_1 and h is the height above the ellipsoid of the point that is aimed at.

As this correction is very small we can use either B or the reduced latitude β and either A_h or A_N in the equation without corrupting the result.

NB: In (1.8) the difference ΔA is in radians. We must multiply ΔA with a factor $(180/\pi)$ to obtain ΔA in degrees.

Numerical example:

$$h = 2000 \text{ m}$$

$$B = 64^\circ$$

$$A_G = 45^\circ$$

$$\text{The WGS84 ellipsoid, half axes: } a = 6378137,0000 \text{ m} \quad b = 6356752,3142 \text{ m}$$

$$\text{The correction } \Delta A_h \text{ is, by using (1.8): } 0,000\,0116^\circ = 0^\circ\,0'\,0,042'' \text{ (arc seconds)}$$

⇒ The difference ΔA_h , even for a high value of the height h , 2000 m, is smaller than what is achievable accuracy with a high precision theodolite (total station).

In most of the cases this correction is negligible.

1.5 Other reductions or corrections

Measured quantities on the physical surface of the earth must be corrected to quantities which would have been obtained if the measurements were carried out on the ellipsoid itself. As we already have seen in chapters 1.3 and 1.4, some of these reductions or corrections are in most cases negligible. In addition to the corrections in chapters 1.3 and 1.4, we shall now look into a few more.

1.5.1 Corrections to the measured azimuth

See chapter 1.3.1 for the correction from the normal section arc to the geodesic.

Geodetic instruments like total stations (theodolites) are generally having a level bubble mounted on it. Generally, when we use the level bubble to mount up an instrument, the measurements are referenced to the physical plumb line (the local vertical) instead of being oriented along the ellipsoidal normal. The corrections in **a** and **b** below are related to rotations between the astronomic to the ellipsoidal coordinate system, while correction in **c** is caused by the refraction of the line of sight.

a: Caused by the deflection of the vertical (“loddavvik”)

The deflection θ (theta) is the angle between the normal of the ellipsoid and the vertical (plumb line). We can express it by, see also figure 1.15:

(1) The magnitude θ and the azimuth A_V from the north axis to the plumb line

(2) More used, the two components of the deflection θ :

ξ (xi, ksi): The component in the north-south direction (+ value on figure 1.15)

η (eta): The component in the east-west direction (+ value on figure 1.15)

For the horizontal direction: It is the component of the deviation of the vertical *orthogonal* to the line of sight that is of interest.

This correction caused by θ can be computed by the formula which expresses the error of a horizontal direction caused by an obliqueness of the vertical axis of the theodolite (total station). For more, see a textbook in basic land surveying.

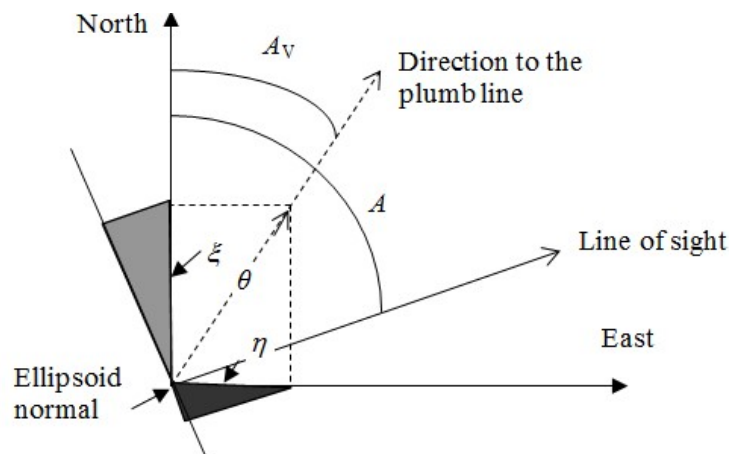


Figure 1.15 The deflection of the vertical. Correction on the horizontal direction

In equation (1.9) the correction is *from* the plumb line *to* the ellipsoid normal. The equation is derived in chapter 14 of “Basic GPS” which is used in the course TBA4245 Geodesy (2018).

The correction on the measured horizontal direction is:

$$\Delta A_V = A_E - A_A = (\eta \cdot \cos A - \xi \cdot \sin A) \cdot \frac{1}{\tan z} \quad (1.9)$$

- z : Ellipsoidal zenith angle
- A_E : Ellipsoidal azimuth
- A_A : Astronomical azimuth (referred to astronomic latitude and longitude)

In mountainous areas with steep lines of sights the value of this correction, dependent of the zenith angle, can be so large that it must be considered.

Numerical example, deflection of the vertical

The example on figure 1.14:

- ξ : +0,0038°
- η : +0,0013°
- A : 60°
- z : 95,146001 gon (grads)

For this relatively steep line of sight the correction on *the measured horizontal direction* is:

$$\Delta A_V = A_E - A_A = (0,0013 \cdot \cos 60 - 0,0039 \cdot \sin 60) \cdot \frac{1}{\tan 95,146001} = +0,000200^\circ$$

For an almost horizontal line of sight with $z = 99,750001$ gon, the correction is:

$$\Delta A_V = A_E - A_A = (0,0013 \cdot \cos 60 - 0,0039 \cdot \sin 60) \cdot \frac{1}{\tan 99,750001} = +0,000010^\circ$$

b: Caused by different orientation of coordinate systems. Laplace correction

The Laplace correction, the Laplace equation.

Geodetic instruments like total stations (theodolites) are generally having a level bubble mounted on it. Generally, when we use the level bubble to mount up an instrument, the measurements are referenced to the physical plumb line (the local vertical) instead of being oriented to the ellipsoidal normal. We need to correct the astronomic azimuth to the geodetic azimuth (= ellipsoidal azimuth) by using the Laplace correction or the Laplace equation.

The astronomic azimuth is converted to the geodetic azimuth by equation (1.10).

The relationship among astronomic and geodetic azimuth is, *the Laplace correction*, where we correct an astronomic azimuth to an ellipsoidal (geodetic) azimuth:

$$\Delta A_L = A_E - A_A = -\eta \cdot \tan B \quad (1.10)$$

The equation is derived in chapter 14 of “Basic GPS” which is used in the course TBA4245 Geodesy (2018).

The Laplace correction in (1.10) is independent of the observed azimuth and corresponds to a twist in the observed directions in a point.

A horizontal angle is a difference of two observed horizontal directions. The horizontal angles measured by theodolites are not affected by this change of orientation, as the Laplace correction is eliminated in the difference between two horizontal directions.

Baselines between two receivers obtained from GPS (GNSS) observations are originally referenced to the ellipsoid normal and not to the plumb line.

The Laplace equation is originally used in defining a classical datum, for instance the old Norwegian datum NGO1948, which is now replaced with EUREF89, and ED50.

In NGO1948 the corresponding ellipsoid (Bessel modified) was placed and linked to the geoid by setting some conditions to the rotations in a control point in Oslo, the datum origin

(“fundamentalkpunkt”):

- $\Delta A_L = 0$ $\Rightarrow A_E = A_A$, where A_A to another control point was observed
- $\xi = 0$ and $\eta = 0$ $\Rightarrow$ The ellipsoid normal coincide with the plumb line

Numerical example, deflection of the vertical

$$B = 64^\circ$$

$$\eta = +0,0013^\circ$$

The correction ΔA_L is, by using (1.10): $-0,002665^\circ$

c: Caused by lateral refraction

When measuring by using a theodolite (total station) the refraction of the light beam must be considered for correcting the measured horizontal direction and the zenith angle.

This refraction is caused mainly by the component of the temperature gradient:

- For correcting the measured zenith angle, the *vertical component* of the temperature gradient must be considered. This correction is treated elsewhere (in a basic text book in surveying).
- Lateral refraction in the horizontal plane is caused mainly by the *horizontal component* of the temperature gradient.
In general the order of lateral refraction is about $\pm 0,2'' = 0,000278^\circ$, in extreme cases $\pm 2''$ (bow seconds).

We very seldom have necessary data for computing a correction caused by lateral refraction.

1.5.2 Corrections on the measured distance

See chapter 1.3.2 for the correction from the normal section arc to the geodesic.

Corrections on the measured distance from the measured distance on the physical surface of the earth down to a distance on the ellipsoid are treated elsewhere. For the course TBA4245 Geodesy, see for instance the text book “Basic GPS”.

This is a geometric problem only, if we assume that the measured distance is corrected for the effects caused by the atmosphere, as the speed of light is dependent on the temperature, pressure and the water vapour.

1.6 The differential equations of the geodesic. Clairaut's equation.

In chapter 2 we will derive the equations of the two geodetic main tasks. *The first geodetic main task* (also named *the direct problem*) is transferring of geographical coordinates from one point to another point, and with *the second geodetic main task* (also named the *inverse problem*), the inverse task of no. 1, we can compute the geodesic between two known points and its azimuths at these two points.

In this chapter we derive Clairaut's equation which also is used for computations on the surface of the ellipsoid. With Clairaut's equation we can calculate the azimuths in points along a geodesic if we know the latitude of the points and an azimuth of one of the points.

1.6.1 The differential equations of the geodesic on the ellipsoid

As often used in deriving useful equations in geodesy, we start with differential equations of the geodesic. These equations can be derived by help of simple geometrical figures. From these differential equations we obtain the practical equations.

We want to find how a differential change in the distance along the geodesic (dS) is affecting differential changes in the longitude (dL), latitude (dB) and azimuth (dA).

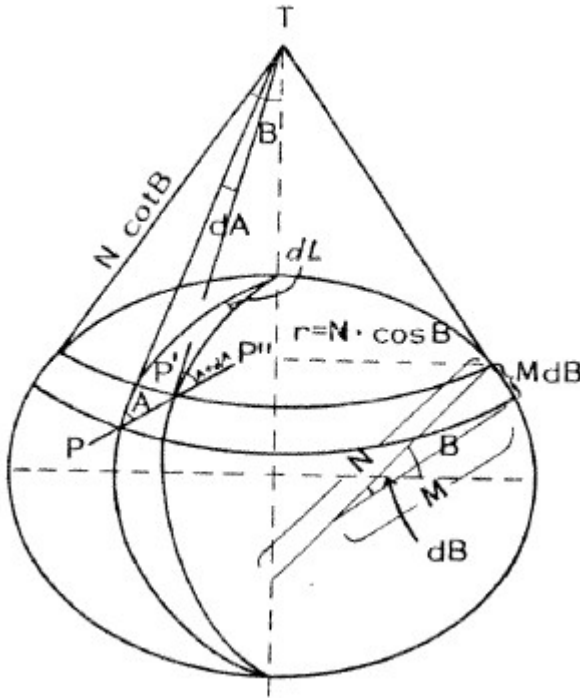


Figure 1.16a

$$\begin{aligned} N &= R_N \\ M &= R_M \end{aligned}$$

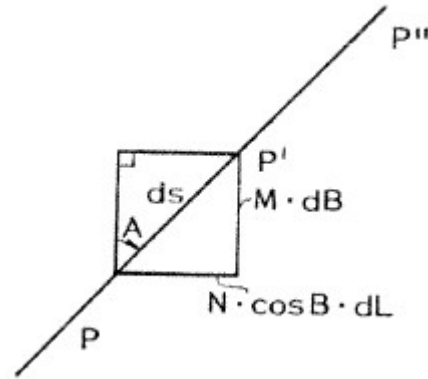


Figure 1.16b

On the figures 1.16 $N = R_N$, $M = R_M$ (the two principal radii of curvature)

On figure 1.16a the cone with apex T is touching the ellipsoid along a parallel of latitude B, where P' is placed. The points P, P' and P'' are the normal projections of three neighbouring points on a geodesic through P' on the tangent plane at P'. The geodesic is a curve on the ellipsoid and the normal projections are in the tangent plane of point P'.

If we choose the three points on the geodesic sufficiently close enough to each other, then P, P' and P'' are lying on a straight line (see figure 1.16b to the right), because the geodetic curvature of the geodesic is zero, and the geodetic curvature is defined in the tangent plane.

a: The differential quotients dB/dS , dL/dS and dA/dS

From figure 1.16b we see directly:

A differential length along the meridian, where A is the azimuth, R_M is the radius of curvature along the meridian and B is the latitude:

$dS \cdot \cos A = R_M \cdot dB$	$\Rightarrow$	$\frac{dB}{dS} = \frac{\cos A}{R_M}$	(1.10)
----------------------------------	---------------	--------------------------------------	----------

A differential length along the circle of constant latitude (parallel circle), where r is the radius of the parallel circle (see equation (1.1)) and N is the radius of curvature of the prime vertical:

$dS \cdot \sin A = r \cdot dL = R_N \cdot \cos B \cdot dL$	$\Rightarrow$	$\frac{dL}{dS} = \frac{\sin A}{R_N \cdot \cos B}$	(1.11)
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The cone in figure 1.16a is slit along a generatrix and opened out flat. A generatrix is a line from apex (the top T) on the cone surface and here to the tangent circle.

The infinitesimal triangle P-P'-T then becomes a plane triangle, see figure 1.17.

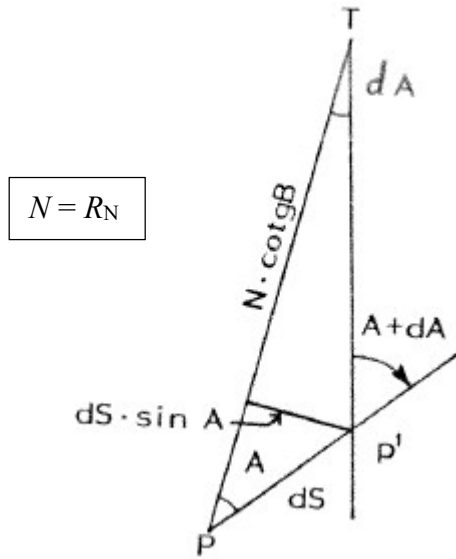


Figure 1.17

The “bow length” from P' to the line P-T on figure 1.17, and $\cotg B = \cot B = 1 / \tan B$:

$dS \cdot \sin A = R_N \cdot \cotg B \cdot dA = (R_N / \tan B) \cdot dA$	$\Rightarrow$	$\frac{dA}{dS} = \frac{1}{R_N} \cdot \sin A \cdot \tan B$	(1.12)
--	---------------	---	----------

The equations (1.10), (1.11) and (1.12) are the differential equations of the geodesic on an ellipsoid.

b: The Bessel's equation dA/dL

We use equation (1.10) to develop the Clairaut's equation for the geodesic on the ellipsoid, see chapter 1.6.2.

Dividing equation (1.12) by (1.11) we get the Bessel's equation of the geodesic on differential form:

$$\frac{dA}{dL} = \sin B \quad (1.13)$$

This equation shows how the azimuth is varying if the longitude changes with dL .

Equation (1.13) can also be derived directly from figure 1.18, compare with figure 1.16.

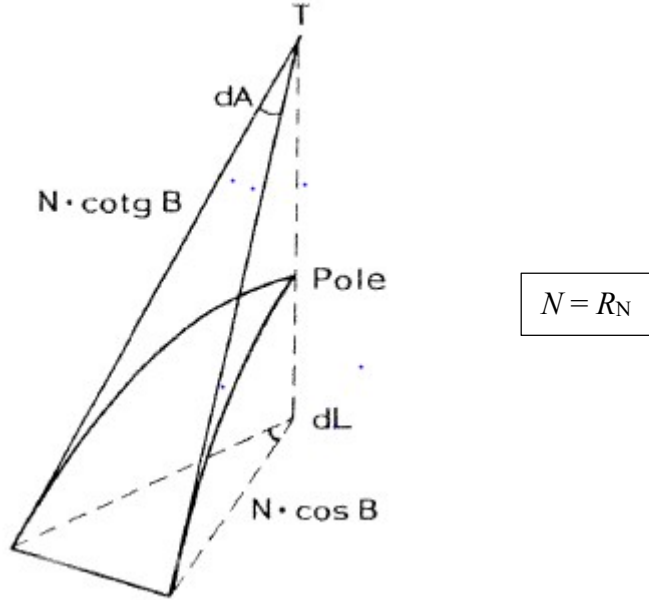


Figure 1.18

From figure 1.18 we find two formulas for the “arc length” along the parallel circle. NB: Two different arcs, but in the differential mode they are considered to be of the same length:

$$dA \cdot R_N \cdot \cotg B = (R_N / \tan B) \cdot dA = R_N \cdot \cos B \cdot dL, \quad \cotg B = \frac{1}{\tan B}$$

Reforming and we get (1.13): $\frac{dA}{dL} = \sin B$

1.6.2 Clairaut's equation for the geodesic on the ellipsoid

Reforming equation (1.12) by multiplying with $R_M \cdot dB$ in both the numerator and denominator and regrouping:

$$\frac{dA}{dS} = \frac{1}{R_N} \cdot \sin A \cdot \frac{\sin B}{\cos B} = \frac{R_M \cdot dB \cdot \sin B}{R_N \cdot \cos B} \cdot \frac{\sin A}{R_M \cdot dB} \quad (1.14a)$$

From figure 1.10b we find the radius of the parallel circle (the circle with constant latitude), r :

$$r = R_N \cdot \cos B$$

The radius r inserted into (1.14a):

$$\frac{dA}{dS} = \frac{R_M \cdot dB \cdot \sin B}{r} \cdot \frac{\sin A}{R_M \cdot dB} \quad (1.14b)$$

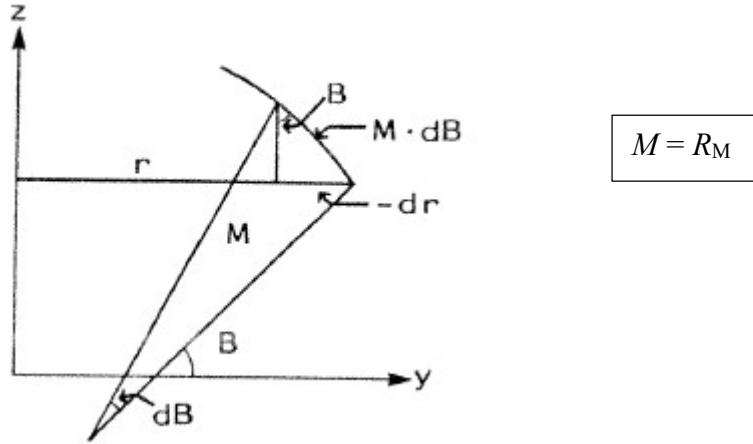


Figure 1.19 The radius of the parallel circle and a differential length along the meridian

From figure 1.19 we obtain two formulas of the differential length along the meridian.

A positive dB means we go northwards on the meridian, and then the length of r is decreasing with dr . This is the reason of the negative sign of dr .

$$R_M \cdot dB = \frac{-dr}{\sin B}$$

From (1.10):

$$dS \cdot \cos A = R_M \cdot dB$$

Combining the last two equations:

$$R_M \cdot dB = \frac{-dr}{\sin B} = \cos A \cdot dS$$

Substituting $R_M \cdot dB$ in equation (1.14b):

$$\frac{dA}{dS} = \frac{R_M \cdot dB \cdot \sin B}{r} \cdot \frac{\sin A}{R_M \cdot dB} = -\frac{dr}{r} \cdot \frac{\sin A}{\cos A \cdot dS} \quad (1.14c)$$

Reforming (1.14c), and (1.14d) is the differential form of the Clairaut's equation between dr and dA :

$$r \cdot \cos A \cdot dA = -\sin A \cdot dr$$

$$r \cdot \cos A \cdot dA + \sin A \cdot dr = 0$$

$$\frac{dA}{\tan A} = -\frac{dr}{r} \quad (1.14d)$$

a: The 1st form of Clairaut's equation for the geodesic on the ellipsoid

Integration of (1.14d) gives the 1st form of Clairaut's equation of the geodesic on the ellipsoid:

$$\ln \sin A = -\ln r + \ln C \quad C \text{ is a constant}$$

Rewritten: $\sin A = \frac{C}{r}$ where $r = R_N \cdot \cos B$, see above

Finally the 1st form of Clairaut's equation of the geodesic on the ellipsoid:

$$\boxed{r_i \cdot \sin A_i = R_{Ni} \cdot \cos B_i \cdot \sin A_i = C_1 = \text{constant}} \quad (1.15)$$

Clairaut's equation (1.15) expresses that the product ($r \cdot \sin A$) is constant for a geodesic. A is the azimuth of the geodesic at a point P_i and r is the radius of the parallel circle through the point. R_N is the radius of curvature in the prime vertical in P_i and B is the latitude of point P_i .

See below for the 2nd form of Clairaut's equation, where R_N is substituted and thus needs not to be calculated.

See further below for discussion of the *northernmost point of the geodesic* and for a *numerical example*.

b: The 2nd form of Clairaut's equation for the geodesic on the ellipsoid

For two points P_1 and P_2 on a geodesic, the first form of Clairaut's equation for the geodesic on the ellipsoid is:

$$R_{N1} \cdot \cos B_1 \cdot \sin A_1 = R_{N2} \cdot \cos B_2 \cdot \sin A_2 = C_1 = \text{constant} \quad (1.16a)$$

We want to substitute R_N from (1.15) for more simplified (?) calculations. To achieve that, the reduced latitude, β , must be used instead of the geographical latitude B .

The mathematical relation between β and B is given in equation (2.13b) in chapter 2.1, see "Geodesy part 1" for the derivation:

$$\cos B = \cos \beta \cdot W, \quad \text{where: } W = (1 - e^2 \cdot \sin^2 B)^{0.5}, \text{ see (2.6) in chapter 2.1}$$

also substituting R_N by equation (2.9) in chapter 2.1:

$$R_N = \frac{a}{W}$$

New version of (1.16a):

$$R_N \cdot \cos B \cdot \sin A = \frac{a}{W} \cdot \cos \beta \cdot W \cdot \sin A = a \cdot \cos \beta \cdot \sin A = C_1 \quad (1.16b)$$

In the formulas above a and e are the half semi major half axis and the first numerical eccentricity of the ellipsoid.

Rearranging (1.15b) and setting the constant $C_2 = C_1 / a$, and we obtain the *second form of Clairaut's equation* of the geodesic on the ellipsoid:

$$\boxed{\cos \beta_i \cdot \sin A_i = C_2 = \text{constant}} \quad (1.17)$$

Used in this form, we do not need to compute the radius of curvature of the prime vertical, R_N .

c: The northernmost point of the geodesic

See also page 37 for more about the northernmost point on the geodesic. On the northern hemisphere we can use equation (1.15) to find the latitude of the northernmost point of the geodesic under

consideration. At the northernmost point (nm) the geodesic is crossing a meridian at a right angle. In this case the azimuth is $A = 90^\circ$ and $\sin A = 1$, and the constants are:

$$C_1 = R_{N\text{ nm}} \cdot \cos B_{\text{nm}} \quad (1.18a)$$

$$C_2 = \cos \beta_{\text{nm}} \quad (1.18b)$$

Numerical example no. 1

The curve P_0 - P_1 - P_2 on figure 1.19 is a geodesic on the ellipsoid. Notice the direction of the geodesic, the curve starts at P_0 .

Known:

Geographical latitude: $B_1 = 64,00000000^\circ$ and $B_2 = 60,00000000^\circ$

Azimuth of the geodesic in P_1 towards P_2 : $A_1 = 105,0000000^\circ$

The WGS84 ellipsoid, half axes: $a = 6378137,0000$ m $b = 6356752,3142$ m

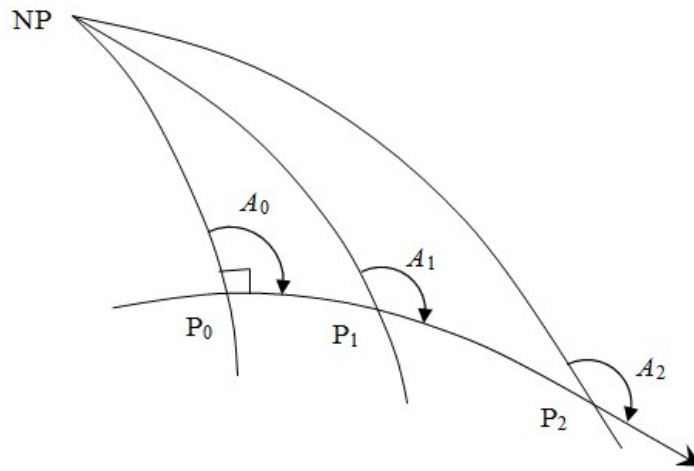


Figure 1.19

The reduced latitude β calculated from B by using equation (2.12) in chapter 2.1:

$$\tan \beta = \frac{b}{a} \cdot \tan B \quad \Rightarrow \quad \beta_1 = 63,92410530^\circ \quad \text{and} \quad \beta_2 = 59,91660780^\circ$$

The radius of the curvature in the prime vertical in point P_1 and P_2 , by using equations (2.9) and (2.6):

$$R_{N1} = \frac{a}{W} = 6395453,513 \text{ m and} \quad R_{N2} = \frac{a}{W} = 6394209,174 \text{ m}$$

We know the latitude B or β and the azimuth A_1 in point P_1 . Inserting these values into the Clairaut's equations to calculate the constants C_1 and C_2 of the two Clairaut's equations (1.15) and (1.17):

$$C_1 = R_{N1} \cdot \cos B_1 \cdot \sin A_1 = 2708052,5407$$

$$C_2 = \cos \beta_1 \cdot \sin A_1 = 0,42458362695$$

Now we can use the Clairaut's equations (1.15) and (1.17) for calculation of latitudes and azimuths along the geodesic in this example, both are used in this example (not necessary in actual projects):

The azimuth A_2 in point P_2 ?

$$(1.15): \sin A_2 = C_1/(R_{N2} \cdot \cos B_2) = 0,84703282830$$

$$(1.17): \sin A_2 = C_2/\cos \beta_2 = 0,84703282830$$

The azimuth in point P_2 , NB: pointing southeast, see the positive direction of the geodesic on figure 1.19:

$$A_2 = \text{asin } 0,84703282830 = 57,89039520^\circ$$

If the azimuth in P_2 towards P_1 is wanted, we have to “turn A_1 around” by adding 180° :

$$A_{2-1} = 57,89039520 + 180 = 237,89039520^\circ$$

The latitude B of the northernmost point of the geodesic?

As described in the text above, $A_0 = 90^\circ$.

We cannot use (1.15) or (1.18a) because we need the latitude to calculate the radius of the curvature, R_N , in point P_0 .

Using the 2nd for of Clairaut’s equation, (1.17) or (1.18b):

$$\cos \beta_0 = C_2/\sin A_0 = C_2/\sin 90 = C_2 = 0,42458362695$$

The reduced latitude of the northernmost point on the actual geodesic, $\cos^{-1} = \text{acos}$:

$$\beta_0 = \cos^{-1} 0,42458362695 = 64,87568880^\circ$$

The geographical latitude of the northernmost point of the actual geodesic, by using equation (2.12) in chapter 2.1:

$$B_0 = 64,94957993^\circ$$

Numerical example no. 2

See example exercise no. 2 (2018) in the course TBA4245 . This example is more complete than example no. 1 above.

CHAPTER 2

SOLUTIONS OF THE GEODETIC MAIN TASKS OVER LONG DISTANCES

The Bessel-Helmert method

The formulas in chapter 2.1 are also used in chapters 1 and 4.

2.0 Introduction to chapter 2

The focus in “Geodesy part 2” is calculations on- and mapping of the ellipsoid of revolution.

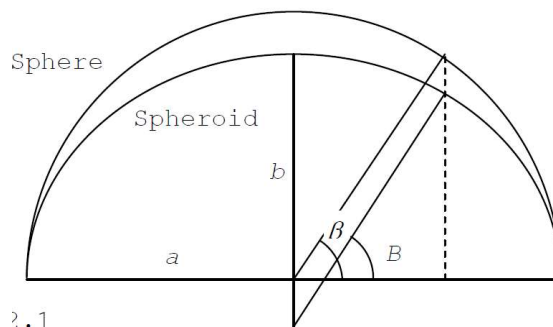
Ellipsoid of revolution instead of using Spheroid

Scientific literature (particularly geodesy) often uses “*ellipsoid*” in place of “*ellipsoid of revolution*” and only applies the adjective “tri-axial” when treating the general case. Older literature uses “*spheroid*” in place of ‘*ellipsoid of revolution*’.

In this new version of the textbook “Geodesy part 2” we will use *ellipsoid* and mean ellipsoid of revolution. We will not use spheroid, spheroid was used in the old/original version of the textbook.

Chapter 2:

Holsen explains the Bessel-Helmerts method to solve the two geodetic main tasks (direct or indirect geodetic problem). The method make use of a simplification, projecting curves or angles from the ellipsoid surface up to a sphere with radius equal to the major ellipsoid semi axis a .



Recent years new methods and most important, the number of terms from series expansion to find equations to calculate between a sphere, to/from a ellipsoid surface have been improved. (Karney 2013) ensures/claim precession less than one millimeter using his Matlab toolbox:

<http://www.mathworks.com/matlabcentral/fileexchange/39108-geodesics-on-an-ellipsoid-of-revolution>

2.1 An overview over ellipsoidal variables

The formulas in chapter 2.1 are also used in chapters 1 and 4.

Definitions:

- **Ellipsoid:** The 2 axes in the equatorial plane are generally not alike in numerical value.
 - **Ellipsoid of revolution.** The 2 axes in the equatorial plane have the same numerical value.
 - **Spheroid:** Old name of the ellipsoid of revolution
- See chapter 1.2 in “Geodesy part 1” for details.

The ellipsoid:

The semi axes of the ellipsoid of revolution, often called the ellipsoid:

$$a \text{ (major), } b \text{ (minor)} \quad (2.1)$$

$$\text{The flattening (“flattrykning”):} \quad \alpha = f = \frac{a - b}{a} \quad (2.2)$$

$$\text{The third flattening, } n: \quad n = \frac{a-b}{a+b} \quad (2.2b)$$

$$\text{The 1st numerical eccentricity, } e: \quad e^2 = \frac{a^2 - b^2}{a^2} \quad (2.3)$$

$$\text{The 2nd numerical eccentricity, } e': \quad e'^2 = \frac{a^2 - b^2}{b^2} \quad (2.4)$$

$$(1 - e^2) \cdot (1 + e'^2) = 1 \quad (2.5a)$$

$$e^2 = \frac{e'^2}{e'^2 + 1} \quad e'^2 = \frac{e^2}{1 - e^2} \quad (2.5b)$$

Auxiliary quantities “hjelpetørrelser”:

$$W = (1 - e^2 \cdot \sin^2 B)^{1/2} \quad (2.6)$$

$$V = (1 + e'^2 \cdot \cos^2 B)^{1/2} = (1 - e^2 \cdot \cos^2 B)^{-1/2} = (1 + \eta^2)^{1/2} \quad (2.7)$$

$$\eta^2 = e'^2 \cos^2 B = \frac{e^2}{1 - e^2} \cos^2 B \quad (2.8)$$

Radius of the curvature:

$$\text{In the prime vertical: } R_N = N = \frac{a}{W} = \frac{c}{V} = \frac{a^2/b}{V} \quad (2.9)$$

$$\text{In the meridian: } R_M = M = \frac{a(1 - e^2)}{W^3} = \frac{c}{V^3} = \frac{a\sqrt{1 + e'^2}}{V^3} \quad (2.10)$$

$$\text{The curvature on the north and south poles: } c = \frac{a^2}{b} \quad (2.10b)$$

The Gaussian mean radius: $R_G = \sqrt{R_M \cdot R_N} = \sqrt{\frac{c}{V^3} \cdot \frac{c}{V}} = \sqrt{\frac{1}{V^2} \cdot \frac{c^2}{V^2}} = \frac{R_N}{V}$ (2.11a)

Radius of curvature in direction with azimuth α : $R_\alpha = \frac{R_M \cdot R_N}{R_N \cdot \cos^2 \alpha + R_M \cdot \sin^2 \alpha}$ (2.11b)

(The plane bearing can often be used instead of the azimuth, α).

Latitude

Geographic latitude: B

Geographic longitude: L

Reduced latitude: β

Spherical longitude: λ

$$\tan \beta = \frac{b}{a} \cdot \tan B = \sqrt{1 - e^2} \cdot \tan B \quad (2.12)$$

By use of trigonometric formulas the following relations between geographic (B) and reduced latitude (β) are developed:

$$\sin \beta = \frac{\sqrt{1 - e^2} \sin B}{W} \quad (2.13a)$$

$$\cos \beta = \frac{\cos B}{W} \quad (2.13b)$$

$$\sin \beta = \frac{\sin B}{V} \quad (2.13c)$$

$$\cos \beta = \frac{\sqrt{1 + e'^2} \cos B}{V} \quad (2.13d)$$

$$\cos^2 B = \frac{V^2 \cos^2 \beta}{1 + e'^2} \quad (2.13e)$$

From (2.9) and (2.13c):

$$V^2 - 1 = e'^2 \cos^2 B = \frac{e'^2 V^2 \cos^2 \beta}{1 + e'^2} = e^2 V^2 \cos^2 \beta$$

$$\frac{1}{V^2} = 1 - e^2 \cos^2 \beta \quad (2.14)$$

$$\frac{dB}{d\beta} = \frac{V^2}{\sqrt{1 + e'^2}} \quad (2.15)$$

From (2.10) and (2.11): $\frac{R_M}{R_N} = \frac{1}{V^2}$ (2.16)

From (2.13d and e): $\frac{\cos \beta}{\cos B} = \frac{\sqrt{1 + e'^2}}{V}$ (2.17)

2.2 Bessel-Helmert. The differential equations

Calculations on the ellipsoid by using the geodetic main tasks.

The geodetic main tasks are:

- *The first geodetic main task* (chapter 2.5.1), *the direct problem*. Transferring of geographical coordinates from one point to another point:
Known: The coordinates of $P_1 (B_1, L_1)$, the azimuth A_1 and the distance ΔS_{1-2} along the geodesic.
Unknown: The coordinates of $P_2 (B_2, L_2)$ and the azimuth A_2
- *The second geodetic main task* (chapter 2.5.2), *the inverse problem*. The reverse task, to compute the geodesic between two known points and its azimuths at these points:
Known: The coordinates of $P_1 (B_1, L_1)$ and $P_2 (B_2, L_2)$
Unknown: The azimuths A_1 and A_2 , the distance ΔS_{1-2} along the geodesic.

The Bessel-Helmert sphere

The Bessel-Helmert solution of the Geodetic main tasks is based on the differential equations of the geodesic. These equations are developed in chapter 1.6.

In the Bessel-Helmert solution the ellipsoid is enclosed of a sphere with radius equal to the semi major axis a , see figure 2.1. Equator is the common line (tangency line) between the sphere and the ellipsoid.

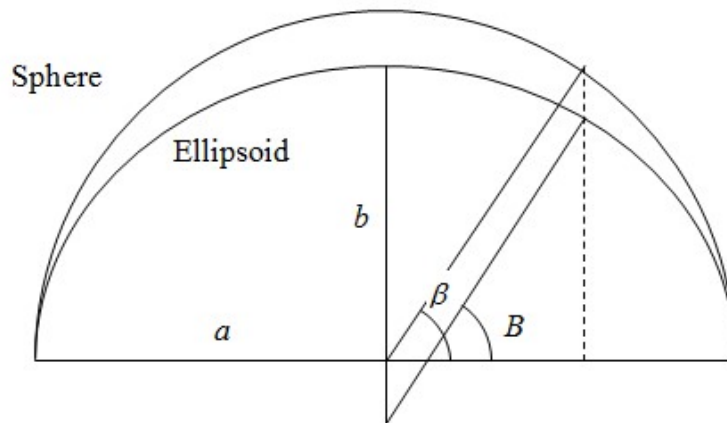


Figure 2.1

On the ellipsoid surface the polar triangle P_1-N-P_2 is formed, where the points P_1 and P_2 are the end points of the geodesic ΔS , figure 2.2.

This triangle P_1-P_2-N is *transferred to the sphere*, forming the triangle $P'_1-P'_2-N'$, see figures 2.2 and 2.3.

The sphere is a less complicated surface to do the computations on, compared to the ellipsoid. The *computations* are done on the sphere using spherical trigonometric formulas, and the results are then *transferred back to the ellipsoid*.

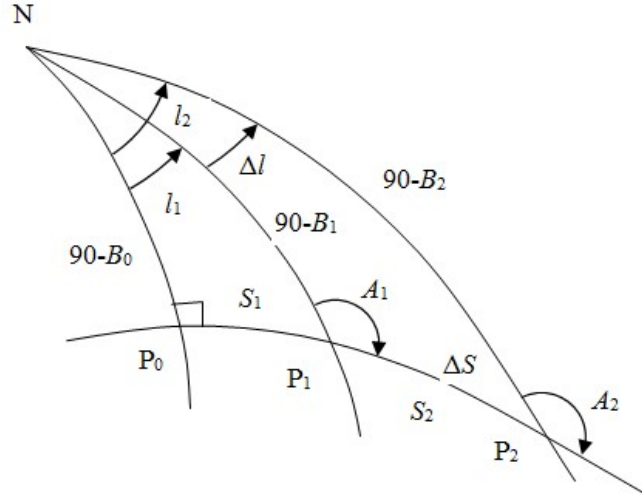


Figure 2.2 The ellipsoid. The distances (geodesics) S_1 and S_2 start at P_0 . In meter.

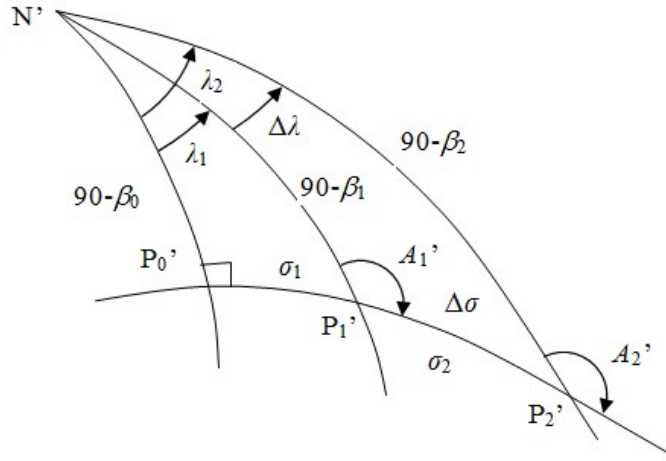


Figure 2.3 The sphere. The angles (great circles) σ_1 and σ_2 start at P_0 . In radians or grads.

The differential equations

A pointwise description of the method to transform the latitude, longitude, azimuth and distances to the sphere from the ellipsoid:

- (1) *Choosing:* Let the spherical latitudes of the points P_1 and P_2 be equal to the reduced latitudes.

$$\beta_1 (\text{sphere}) = \beta_1 (\text{ellipsoid})$$

$$\beta_2 (\text{sphere}) = \beta_2 (\text{ellipsoid})$$

- (2) *Choosing:* Let the azimuth in point P_1 have the same value both on the ellipsoid and the sphere:

$$A'_1 (\text{sphere}) = A_1 (\text{ellipsoid})$$

The choice in point (2) does not contradict the choices made in point (1). If B_1 and A_1 are given, the reduced latitude β_1 is also given. But the point P'_1 on the sphere can be moved/dislocated along the circle of latitude through P'_1 until:

$$A'_1 (\text{sphere}) = A_1 (\text{ellipsoid})$$

- (3) All the curves on the surface of the sphere on figure 2.3 are great circles. The spherical sine proportion applied to the triangle $P_1' - P_2' - N'$:

$$\frac{\sin A_1'}{\sin(90 - \beta_2)} = \frac{\sin(180 - A_2')}{\sin(90 - \beta_1)} \quad \text{is developed to:}$$

$$\cos \beta_1 \cdot \sin A_1' = \cos \beta_2 \cdot \sin A_2' \quad (2.18a)$$

- (4) The Clairaut's equation (2.17) used for the geodesic going through P_2 with azimuth A_2 :

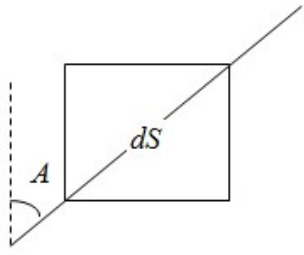
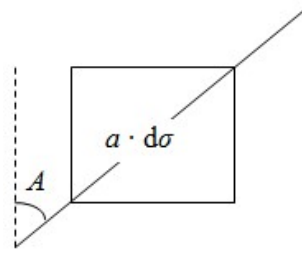
$$\cos \beta_1 \cdot \sin A_1 = \cos \beta_2 \cdot \sin A_2 \quad (2.18b)$$

- (5) Comparing (2.18a) and (2.18b), and remembering the choice in point (2):

$$A_1' (\text{sphere}) = A_1 (\text{ellipsoid})$$

$$A_2' (\text{sphere}) = A_2 (\text{ellipsoid})$$

After the choices and the derived relations in the points (1) to (5) above, we can establish some useful differential ratios between the geodesic going through P_1 and P_2 on the ellipsoid and the corresponding great circle through P_1' and P_2' on the sphere.

On the ellipsoid	On the sphere with radius a
	
$dS \cdot \cos A = R_M \cdot dB \quad (2.19a)$	$a \cdot d\sigma \cdot \cos A = a \cdot d\beta \quad (2.19c)$
$dS \cdot \sin A = R_N \cdot \cos B \cdot dl \quad (2.19b)$	$a \cdot d\sigma \cdot \sin A = a \cdot \cos \beta \cdot d\lambda \quad (2.19d)$

Ellipsoid:	ΔS :	The length of the geodesic from P_1 to P_2 . In meters.
Sphere.	$\Delta \sigma$:	The great circle from P_1' to P_2' . In radians.
Sphere:	$a \cdot \Delta \sigma$:	The length in meters of the great circle from P_1' to P_2' . In meters.
Ellipsoid:	Δl :	The difference in ellipsoidal longitude between P_1 and P_2
Sphere.	$\Delta \lambda$:	The difference in spherical longitude between P_1' and P_2'

Using the equations in the table, where R_M and R_N are the two principal radii of curvature on the ellipsoid, see chapter 2.3 in “Geodesy part 1”:

$$(2.19a) / (2.19c): \quad \frac{dS}{d\sigma} = R_M \frac{dB}{d\beta} \quad (2.20)$$

$$\begin{aligned}
 (2.19b) / (2.19d): \quad \frac{dS}{d\sigma} &= \frac{R_N \cdot \cos B}{\cos \beta} \cdot \frac{dl}{d\lambda} \\
 \frac{dl}{d\lambda} &= \frac{\cos \beta}{R_N \cdot \cos B} \cdot \frac{dS}{d\sigma}
 \end{aligned} \tag{2.21}$$

(2.20) combined with (2.21):

$$\frac{dl}{d\lambda} = \frac{\cos \beta}{\cos B} \cdot \frac{1}{R_N} \cdot \frac{R_M \cdot dB}{d\beta} \tag{2.22}$$

From chapter 2.1, also equation (2.14):

$$R_M \cdot \frac{dB}{d\beta} = (2.10) \cdot (2.15) = \frac{a}{V} = a \cdot \sqrt{1 - e^2 \cdot \cos^2 \beta}$$

The above inserted into (2.20), *the differential equation of the distance*:

$$\boxed{dS = a \cdot \sqrt{1 - e^2 \cdot \cos^2 \beta} \cdot d\sigma} \tag{2.23}$$

From chapter 2.1:

$$(2.16) \cdot (2.17) \cdot (2.15) = \frac{R_M}{R_N} \cdot \frac{\cos \beta}{\cos B} \cdot \frac{dB}{d\beta} = \frac{1}{V}$$

The above inserted into (2.22), *the differential equation of the latitude*:

$$\boxed{dl = \frac{1}{V} \cdot d\lambda = \sqrt{1 - e^2 \cdot \cos^2 \beta} \cdot d\lambda} \tag{2.24}$$

Equation (2.23) gives the relation on the differential form between the length of the geodesic on the ellipsoid and the length of the great circle on the sphere, ΔS and $\Delta \sigma$.

Quite equivalent is the equation (2.24) the relation between ellipsoidal longitude and the corresponding spherical longitude, Δl and $\Delta \lambda$.

The further work is to integrate the differential equations (2.23) and (2.24).

2.3 Integration of the differential equations of dS to $S1$ and $S2$

The differential equation of the distance, (2.23): $dS = a \cdot \sqrt{1 - e^2 \cdot \cos^2 \beta} \cdot d\sigma$

To be able to integrate equation (2.23) the reduced latitude β of point P_i must be substituted with the spherical angle (distance) σ .

Reason: The reduced latitude β is dependent on σ , and we therefore must use σ as the variable, as we are integrating (2.23) between two values of σ .

2.3.1 Definition of point, P_0 , the northernmost point of the geodesic

The geodesic P_1 - P_2 is extended until this line is crossing a meridian under a right angle (90 degrees). The intersection point P_0 (B_0, L_0) is the point on the geodesic which has the largest numerical value of the geographical latitude, B_0 , and therefore also the largest value of the reduced latitude, β_0 .

On the corresponding sphere the point P_0' is equal to P_0 , because an extension of the great circle between P_1' and P_2' is crossing the meridian under a right angle according to (2.18).

2.3.2 Manipulations on the dS equation (2.23)

Substituting the latitude β with the spherical angle (distance) σ

We consider P_0' as the starting point of the great circle going through P_1' and P_2' with azimuth 90 degrees in P_0' , and a central angle σ , see figures 2.3 and 2.5.

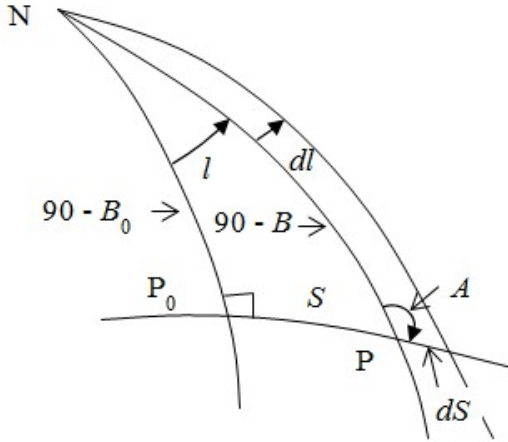


Figure 2.4 The triangle N - P_0 - P and the differentials dl and dS on the ellipsoid

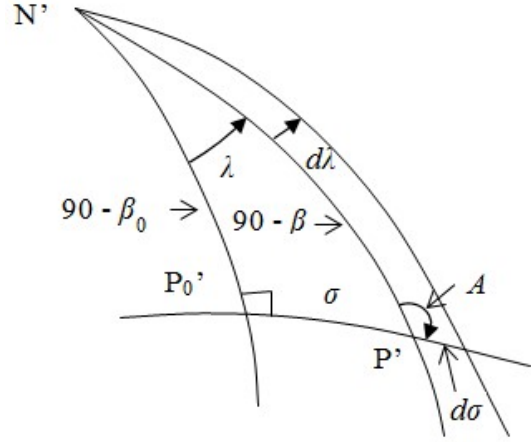


Figure 2.5 The triangle N' - P'_0 - P' and the differentials $d\sigma$ and $d\lambda$ on the sphere

From the spherical triangle N' - P'_0 - P' , the cosine sentence in a right-angled triangle, (3.15a) in “Geodesy part 1” (spherical trigonometry):

$$\sin \beta = \sin \beta_0 \cdot \cos \sigma \quad (2.25)$$

(2.25) is converted to:

$$\cos^2 \beta = 1 - \sin^2 \beta_0 \cdot (1 - \sin^2 \sigma)$$

This expression of $\cos^2 \beta$ is inserted into (2.23):

$$dS = a \cdot \sqrt{1 - e^2 \cos^2 \beta_0 - e^2 \sin^2 \beta_0 \sin^2 \sigma} \cdot d\sigma$$

Which, when $(1 - e^2 \cos^2 \beta_0)^{1/2}$ is moved outside the radical sign, get the form:

$$dS = a \cdot \sqrt{1 - e^2 \cos^2 \beta_0} \cdot \sqrt{1 - \frac{e^2 \sin^2 \beta_0}{1 - e^2 \cos^2 \beta_0} \sin^2 \sigma} \cdot d\sigma \quad (2.26)$$

Using (2.6), (2.13b), (2.13c) and (2.7) we form the following expression:

$$e^2 \cdot \cos^2 \beta_0 = \frac{e^2 \cdot \cos^2 B_0}{1 - e^2 \cdot \sin^2 B_0} \quad \text{and} \quad e^2 \cdot \sin^2 B_0 = \frac{e^2 \cdot \sin^2 \beta_0}{1 - e^2 \cdot \cos^2 \beta_0} \quad (2.27)$$

The axillary quantity or function k

Introducing the function k , see (2.27):

$$k = e \cdot \sin B_0 = \frac{e \cdot \sin \beta_0}{\sqrt{1 - e^2 \cos^2 \beta_0}} \quad (2.28)$$

k is a function of the ellipsoid (the 1st numerical eccentricity, e) and the latitude β_0 of the northernmost point of the geodesic, P_0 . For a formula of e , see chapter 2.1

k^2 is equal to the coefficient in front of $\sin^2 \sigma$ in (2.26). We want to express the factor under the first radical sign in (2.26) by k .

Developing (2.28):

$$\begin{aligned} 1 - k^2 &= 1 - e^2 \cdot \sin^2 B_0 = W_0^2 = 1 - \frac{e^2 \cdot \sin^2 \beta_0}{1 - e^2 \cdot \cos^2 \beta_0} = \frac{1 - e^2}{1 - e^2 \cdot \cos^2 \beta_0} \\ \Rightarrow \quad 1 - e^2 \cdot \cos^2 \beta_0 &= \frac{1 - e^2}{1 - k^2} \end{aligned} \quad (2.29)$$

Combining (2.28) and (2.29) with (2.26) we get an equation that is not optimal, see the next step, equation (2.31):

$$dS = a \cdot \sqrt{\frac{1 - e^2}{1 - k^2}} \cdot \sqrt{1 - k^2 \cdot \sin^2 \sigma} \cdot d\sigma = \frac{b}{\sqrt{1 - k^2}} \cdot \sqrt{1 - k^2 \cdot \sin^2 \sigma} \cdot d\sigma \quad (2.30)$$

In formula (2.30): $b^2 = a^2(1 - e^2)$.

The value of k is not dependent on σ . Therefore we can integrate (2.30).

After $\sqrt{1 - k^2 \cdot \sin^2 \sigma}$ is developed into a series, we obtain the length of the geodesic, S , expressed by the great circle curve, σ .

The axillary quantities or functions k_1 and w_0

To obtain a series that is converging faster than (2.30), we introduce a new constant, k_1 , instead of k .

Converging faster: Fewer terms in the series.

The definition of k_1 and w_0 , and see (2.28) for the definition of k :

$$\boxed{w_0 = \sqrt{1 - k^2} = \frac{1 - k_1}{1 + k_1} \quad \quad k_1 = \frac{1 - w_0}{1 + w_0}} \quad (2.31a)$$

which gives k expressed by k_1 :

$$k^2 = \frac{4k_1}{(1 + k_1)^2} \quad , \quad k = \frac{2\sqrt{k_1}}{1 + k_1} \quad (2.31b)$$

As k_1 and w_0 is derived from k , both k_1 and w_0 are functions of the ellipsoid (e) and the latitude β of the northernmost point of the geodesic, P_0 .

(2.31) inserted in (2.30), and we get an equation that is giving a series that is converging faster than a series obtained from (2.30):

$$\boxed{dS = \frac{b}{1 - k_1} \sqrt{1 + k_1^2 + 2k_1 \cdot \cos 2\sigma} \cdot d\sigma} \quad (2.32)$$

In developing (2.32) we have used:

$$\begin{aligned} \cos^2 \sigma &= 0.5 + 0.5 \cdot \cos(2\sigma) \\ \text{and} \quad \sin^2 \sigma &= 0.5 - 0.5 \cdot \cos(2\sigma) \end{aligned} \quad (2.33a)$$

For the further development of (2.32) we also have used:

$$\begin{aligned} \cos^3 2\sigma &= \frac{1}{2} \cos 2\sigma \cdot \cos 4\sigma + \frac{1}{2} \cos 2\sigma \\ \frac{1}{2} \cos 2\sigma \cdot \cos 4\sigma &= \frac{1}{4} \cos 6\sigma + \frac{1}{4} \cos 2\sigma \end{aligned}$$

$$\text{Ergo: } \cos^3 2\sigma = \frac{1}{4} \cos 6\sigma + \frac{3}{4} \cos 2\sigma \quad (2.33b)$$

Binomial series instead of radical sign

It is not possible to integrate (2.32) directly.

We have to develop the function under the radical sign into series. All terms which contains k_1 in the forth or higher power in this series, can without losing accuracy be left out. The numerical value of k_1^4 is of the order of size $2,5 \cdot 10^{-12}$.

Developing the general expression $(1 + x)^{1/2}$ in a binominal series:

$$(1 + x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} \dots$$

In our case: $x = k_1^2 + 2k_1 \cdot \cos 2\sigma$, see (2.32).

Developing (2.32) by using the binominal series and where all terms with k_1 in power of 4 or higher are left out:

$$dS = \frac{b}{1-k_1} \cdot \left\{ 1 + \frac{k_1^2}{2} + k_1 \cdot \cos 2\sigma - \frac{1}{2}k_1^3 \cdot \cos 2\sigma - \frac{1}{2}k_1^2 \cdot \cos^2 2\sigma + \frac{1}{2}k_1^3 \cdot \cos^3 2\sigma + \dots \right\} d\sigma$$

The 2nd and 3rd power of $\cos 2\sigma$ is converted according to (2.33a) and (2.33b):

$$dS = \frac{b}{1-k_1} \cdot \left\{ 1 + \frac{k_1^2}{4} + \left(k_1 - \frac{1}{8}k_1^3 \right) \cdot \cos 2\sigma - \frac{1}{4}k_1^2 \cdot \cos 4\sigma + \frac{1}{8}k_1^3 \cdot \cos 6\sigma + \dots \right\} \cdot d\sigma \quad (2.34)$$

2.3.3 Integration of the differential equation dS

Integration of (2.34a) between $\sigma = 0$ and $\sigma = \sigma$ gives the following equation. It is developed further to the final equation (2.35a) and (2.35b):

$$S = \frac{b}{1-k_1} \cdot \left\{ \left(1 + \frac{k_1^2}{4} \right) \cdot \sigma + \left(\frac{k_1}{2} - \frac{1}{16}k_1^3 \right) \cdot \sin 2\sigma - \frac{1}{16}k_1^2 \cdot \sin 4\sigma + \frac{1}{48}k_1^3 \cdot \sin 6\sigma + \dots \right\}$$

Finally, $\left(1 + \frac{k_1^2}{4} \right)$ is left outside the parenthesis. Terms within the parenthesis which will contain k_1 in power of 4 or higher are left out:

$$S = \frac{b \cdot \left(1 + \frac{k_1^2}{4} \right)}{1-k_1} \cdot \left\{ \sigma + \left(\frac{k_1}{2} - \frac{3}{16}k_1^3 \right) \cdot \sin 2\sigma - \frac{1}{16}k_1^2 \cdot \sin 4\sigma + \frac{1}{48}k_1^3 \cdot \sin 6\sigma + \dots \right\} \quad (2.35a)$$

NB: In (2.35a) and (2.35b) the great circle curve (= an angle), σ , is expressed in radians. If σ is given in degrees, we must multiply σ with the factor $(\pi/180)$ to obtain σ in radians.

The axillary quantities or functions C , D_1 , D_2 , and D_3

To simplify the writing of (2.34) C and D_i are introduced:

$$\begin{aligned} C &= \frac{b \cdot \left(1 + \frac{k_1^2}{4} \right)}{(1-k_1)} & D_1 &= \left(\frac{k_1}{2} - \frac{3}{16}k_1^3 \right) \\ D_2 &= -\frac{1}{16}k_1^2 & D_3 &= \frac{1}{48}k_1^3 \end{aligned}$$

For the definition and formula of the auxiliary quantity k_1 , see (2.31a).

Finally we have the length in meter of the geodesic, S , expressed by the great circle curve, σ :

$$S = C \cdot \left\{ \sigma + D_1 \cdot \sin 2\sigma + D_2 \cdot \sin 4\sigma + D_3 \cdot \sin 6\sigma + \dots \right\} \quad (2.35b)$$

NB: In (2.35a) and (2.35b) the great circle curve (= an angle), σ , is expressed in radians. If σ is given in degrees, we must multiply σ with the factor $(\pi/180)$ to obtain σ in radians.

2.3.4 Computations from/to S and to/from σ

As described in chapter 2.2 we transfer the coordinates, azimuths and distances on the ellipsoid to the Bessel-Helmert sphere. On this sphere we calculate new spherical coordinates, azimuths and distances. Finally we transfer the spherical values to coordinates, azimuths and distances on the ellipsoid.

Regarding the distance which is defined along the geodesic on the ellipsoid, S , and along the great circle curve on the sphere, σ , we need formulas to calculate S from σ and σ from S :

- A: The length of the great circle curve on the sphere, σ , is computed. As a final result we want the unknown length of the geodesic, S , on the ellipsoid.
- B: The length of the geodesic on the ellipsoid is known. We need the great circle curve on the sphere, σ , for the calculations on the sphere.

A: S on the ellipsoid is unknown, σ on the sphere is known:

Equations (2.35a) and (2.35b) will give S directly.

B: σ on the sphere is unknown, S on the ellipsoid is known:

Check the right hand side of (2.35b) and you will see that it is difficult to solve σ from (2.35b). Obtaining σ directly from equation (2.35b) is a problem, as the equation on the right hand side is rather complex concerning σ .

An often used method in geodesy in such cases is *iteration*. We start with an initial approximate value of σ , obtain a better value in the first iteration (round) and continue the iterations until we obtain the same value of σ as in the previous iteration.

1st approximation, the starting value or first trial:

The terms containing D_i are left out, a simplified version of (2.35b):

$$\sigma_1 = \frac{S}{C}, \text{ in radians}$$

Then S_1 is computed using (2.35b), with $\sigma = \sigma_1$

Comparing the known S and S_1 :

$$\Delta S_1 = S - S_1$$

If ΔS_1 is not acceptable small, we have to do a new iteration.

2nd approximation or round:

$$\sigma_2 = \sigma_1 + \frac{\Delta S_1}{C}, \text{ in radians}$$

Then S_2 is computed using (2.35b), with $\sigma = \sigma_2$

Comparing the known S and S_1 :

$$\Delta S_2 = S - S_2$$

If ΔS_2 is not acceptable small we have to do a new iteration.

3rd approximation or round:

$$\sigma_3 = \sigma_2 + \frac{\Delta S_2}{C}, \text{ in radians}$$

Then S_3 is computed using (2.35b), with $\sigma = \sigma_3$

Comparing the known S and S_1 :

$$\Delta S_3 = S - S_3$$

If ΔS_3 is not acceptable small we have to do a new iteration.

The computations are aborted when the difference $S - S_i$ is acceptable small. This will normally happen already after the 3rd approximation: (2.35b) is converging fast.

2.4 Integration of the differential equations of dl

Equation (2.24) shows in differential form the relation between ellipsoidal longitude and the corresponding spherical longitude, dl and $d\lambda$:

$$dl = \frac{1}{V} \cdot d\lambda = \sqrt{1 - e^2 \cdot \cos^2 \beta} \cdot d\lambda$$

To be able to integrate equation (2.24) above, the reduced latitude β of point P_i must be substituted with the spherical angle (distance) σ . Reason: The reduced latitude β is dependent on σ , and we therefore must use σ as the variable, as we are integrating (2.24) between two values of σ .

We also have to find the mathematical relation between $d\lambda$ and $d\sigma$.

2.4.1 Manipulations on the dl equation (2.24)

Substituting the latitude β with the spherical angle (distance) σ and $d\lambda$ with $d\sigma$:

From (2.24) and adding and subtracting $d\lambda$:

$$dl = d\lambda - \left(1 - \sqrt{1 - e^2 \cdot \cos^2 \beta}\right) \cdot d\lambda \quad (2.36)$$

$d\lambda$ is expressed by $d\sigma$ in (2.19d):

$$\cos \beta \cdot d\lambda = \sin A \cdot d\sigma$$

In the right-angled triangle $P'_0 \cdot P' \cdot N'$ on figure 2.5, using the spherical sine proportion:

$$\sin A \cdot \cos \beta = \cos \beta_0$$

then:
$$d\lambda = \frac{\cos \beta_0}{\cos^2 \beta} d\sigma \quad (2.37)$$

(2.37) inserted in (2.36):

$$dl = d\lambda - \cos \beta_0 \cdot \left(\frac{1 - \sqrt{1 - e^2 \cdot \cos^2 \beta}}{\cos^2 \beta} \right) \cdot d\sigma \quad (2.38)$$

Deduced from (2.25) using $\cos^2 + \sin^2 = 1$:

$$\cos^2 \beta = 1 - \sin^2 \beta_0 \cdot \cos^2 \sigma$$

which inserted in (2.38) gives:

$$dl = d\lambda - \cos \beta_0 \cdot \left(\frac{1 - \sqrt{1 - e^2 \cdot (1 - \sin^2 \beta_0 \cos^2 \sigma)}}{1 - \sin^2 \beta_0 \cdot \cos^2 \sigma} \right) \cdot d\sigma \quad (2.39)$$

The axillary quantity or function k

The term $\sqrt{1 - e^2 \cdot \cos^2 \beta_0}$ is placed outside the fraction line and the following term is introduced, see (2.28), the same k as in the dS equation:

$$k^2 = \frac{e^2 \cdot \sin^2 \beta_0}{1 - e^2 \cdot \cos^2 \beta_0} \quad (2.40a)$$

(2.39) is by these operations converted to:

$$dl = d\lambda - \cos \beta_0 \cdot \sqrt{1 - e^2 \cos^2 \beta} \cdot \left(\frac{\sqrt{\frac{1 + k^2}{\tan^2 \beta_0}} - \sqrt{1 - k^2 \cdot \sin^2 \sigma}}{1 - \sin^2 \beta_0 \cdot \cos^2 \sigma} \right) d\sigma \quad (2.40b)$$

Binomial series instead of radical sign

It is not possible to integrate (2.40b) directly.

The root expressions are developed into series, and the denominator is converted to:

$$1 - \sin^2 \beta_0 \cdot \cos^2 \sigma = \sin^2 \beta_0 \cdot \left(\frac{1}{\tan^2 \beta_0} + \sin^2 \sigma \right)$$

Thus this parenthesis becomes equal to a parenthesis in the nominator.

Forming the terms of powers of $\cos \sigma$, the development of the series can be transformed to functions of a multiple of σ , similar to (2.33b) etc., for details see Grossmann, Walter (1972): Geodätische Rechnungen und Abbildungen or other textbooks.

The axillary quantities or functions k_1 and n

We introduce k_1 again, see (2.31) etc. According to (2.31):

$$k^2 = \frac{4k_1}{(1+k_1)^2}, \quad k = \frac{2\sqrt{k_1}}{1+k_1}$$

And we obtain from (2.28) and (2.29):

$$\frac{k^2}{\tan^2 \beta_0} = \frac{e^2 - k^2}{1 - e^2}$$

New variable, n , the *third flattening* derived from the semi axes of the ellipsoid:

$$n = \frac{a-b}{a+b}$$

and we obtain the first eccentricity of the ellipsoid, e , expressed by n :

$$e^2 = \frac{4n}{(1+n)^2}$$

The differential equation of dl is thereby converted to:

$$dl = d\lambda - \frac{e^2}{2} \cdot \cos \beta_0 \cdot \left(1 + n - \frac{1}{2}k_1 - \frac{1}{4}k_1^2 - \frac{1}{2}k_1 \cdot \cos 2\sigma + \frac{1}{4}k_1^2 \cdot \cos 4\sigma + \dots \right) d\sigma \quad (2.41)$$

2.4.2 Integration of the differential equation dl

Integration of (2.41) gives the final equations (2.42a) and (2.42b).

When the azimuth in P_1 , A_1 , also is known, we can by using the Neper's rules of a right-angled spherical triangle, compute λ_1 , σ_1 and β_1 by spherical trigonometry (see Geodesy part 1, page 3.6 or internet):

$$\tan \sigma_1 = \frac{-\cos A_1}{\tan \beta_1}$$

$$\tan \beta_0 = \frac{-1}{\sin \sigma_1 \cdot \tan A_1}$$

$$\cos \lambda_1 = \frac{\tan \beta_1}{\tan \beta_0}$$

$$\sigma_2 = \sigma_1 + \Delta \sigma$$

$$\tan \lambda_2 = \frac{\tan \sigma_2}{\cos \beta_0}$$

We have the integration limits of the integration of (2.41):

$$\Delta\lambda = \lambda_2 - \lambda_1 \quad \Delta\sigma = \sigma_2 - \sigma_1$$

(2.41) integrated:

(2.42a)

$$\Delta I = \Delta\lambda - \frac{e^2}{2} \cdot \cos \beta_0 \cdot \left\{ \left(1 + n - \frac{k_1}{2} - \frac{k_1^2}{4} \right) \Delta\sigma - \frac{1}{4} k_1 \cdot (\sin 2\sigma_2 - \sin 2\sigma_1) + \frac{1}{16} k_1^2 \cdot (\sin 4\sigma_2 - \sin 4\sigma_1) + \dots \right\}$$

The axillary quantities or functions R , R_1 , R_2 and R_3 :

To simplify the writing of (2.42a) R and R_i are introduced:

$R = \frac{e^2 \cos \beta_0}{2}$
$R_1 = 1 + n - \frac{k_1}{2} - \frac{k_1^2}{4}$
$R_2 = \frac{k_1}{4}$
$R_3 = \frac{k_1^2}{16}$

where $k_1 = \frac{1-w_0}{1+w_0} \quad n = \frac{a-b}{a+b}$

$$W_0 = \left(1 - e^2 \cdot \sin^2 B_0 \right)^{1/2} \quad \text{see (2.6)}$$

$$\tan B_0 = \frac{a}{b} \cdot \tan \beta_0$$

Finally we have the difference in ellipsoidal longitude, ΔI , expressed by the difference in spherical longitude $\Delta\lambda$ and the spherical distance, $\Delta\sigma$ (great circle curve = an angle):

$\Delta I = \Delta\lambda - R \cdot \left\{ R_1 \cdot \Delta\sigma - R_2 \cdot (\sin 2\sigma_2 - \sin 2\sigma_1) + R_3 \cdot (\sin 4\sigma_2 - \sin 4\sigma_1) + \dots \right\}$	(2.42b)
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NB: In (2.42b) and (2.43) the great circle curve, $\Delta\sigma$, is expressed in radians. If $\Delta\sigma$ is given in degrees, we must multiply $\Delta\sigma$ with the factor $(\pi/180)$ to obtain $\Delta\sigma$ in radians.

2.4.3 Computations from/to Δl and to/from $\Delta\sigma$

As described in chapter 2.2 we transfer the coordinates, azimuths and distances on the ellipsoid to the Bessel-Helmert sphere. On this sphere we calculate new spherical coordinates, azimuths and distances. Finally we transfer the spherical values to coordinates, azimuths and distances on the ellipsoid.

Regarding the longitude which is defined as l or L on the ellipsoid and $\Delta\sigma$ or σ on the sphere, we need to calculate Δl from $\Delta\sigma$ and $\Delta\sigma$ from Δl :

A: The longitude on the sphere is computed, $\Delta\lambda$, is computed. As a final result we want the unknown longitude, Δl , on the ellipsoid.

B: The longitude, Δl , on the ellipsoid is known. We need the longitude on the sphere, $\Delta\lambda$, for the calculations on the sphere.

A: The longitude Δl on the ellipsoid is unknown, $\Delta\lambda$ on the sphere is known:

Equations (2.42a) and (2.42b) will give Δl directly.

B: The longitude $\Delta\lambda$ on the sphere is unknown, Δl on the ellipsoid is known:

Check the equations (2.42a and b). The second term containing the parenthesis is independent of $\Delta\lambda$, and thus can be treated as a constant regarding $\Delta\lambda$.

Shifting the second term to the other side of the equal sign, and we get the equation going from Δl to $\Delta\lambda$:

$$\Delta\lambda = \Delta l + R \cdot \left\{ R_1 \cdot \Delta\sigma - R_2 \cdot (\sin 2\sigma_2 - \sin 2\sigma_1) + R_3 \cdot (\sin 4\sigma_2 - \sin 4\sigma_1) + \dots \right\} \quad (2.43)$$

NB: In (2.42b) and (2.43) the great circle curve, $\Delta\sigma$, is expressed in radians. If $\Delta\sigma$ is given in degrees, we must multiply $\Delta\sigma$ with the factor $(\pi/180)$ to obtain $\Delta\sigma$ in radians.

2.5 The 1st and 2nd geodetic main tasks. “Geodetiske hovedoppgaver”

The first geodetic main task (also named *the direct problem*), transferring of geographical coordinates from one point to another point:

Known: The coordinates of P_1 (B_1, L_1), the azimuth A_1 and the distance ΔS_{1-2} along the geodesic.

Unknown: The coordinates of P_2 (B_2, L_2) and the azimuth A_2

The second geodetic main task (also named *the inverse problem*), the inverse task of no. 1, to compute the geodesic between two known points and its azimuths at these points:

Known: The coordinates of P_1 (B_1, L_1) and P_2 (B_2, L_2)

Unknown: The azimuths A_1 and A_2 , the distance ΔS_{1-2}

We carry out the calculations on the Bessel-Helmert sphere, figure 2.7.

Spherical trigonometry is used in the calculations.

The equations (2.35) and (2.42) are used to go between the sphere and the ellipsoid concerning the distance along the geodesic and the difference in longitude.

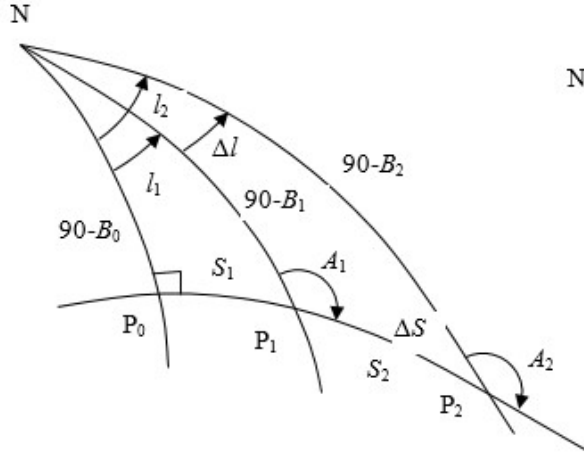


Figure 2.6 The ellipsoid

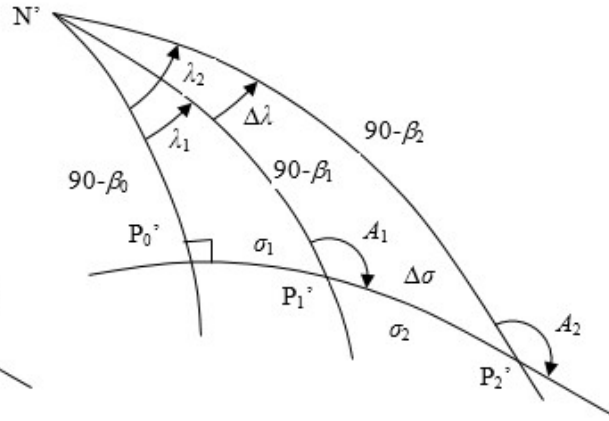


Figure 2.7 The sphere

2.5.1 Numerical solution of the First geodetic main task

With the first geodetic main task (also named the direct problem), we can transfer geographical coordinates from one point to another point:

Known: The coordinates of P_1 (B_1, L_1), the azimuth A_1 and the distance ΔS_{1-2} along the geodesic on the ellipsoid.

Unknown: The coordinates of P_2 (B_2, L_2) and the azimuth A_2 on the ellipsoid

The method:

- (1) Calculate the latitude on the sphere, the reduced latitude, of point P_1 from the geographical latitudes and the half axes of the ellipsoid:

$$\tan \beta_1 = \frac{b}{a} \cdot \tan B_1 = \sqrt{1-e^2} \cdot \tan B_1$$

- (2) From the spherical triangle $N'-P_0'-P_1'$ the spherical variables β_0 , σ_1 and λ_1 are computed. See also figure 2.7. Azimuth A_1 is known and β_1 were calculated in step (1):

$$2.1: \quad \tan \sigma_1 = \frac{-\cos A_1}{\tan \beta_1}$$

$$2.2: \quad \tan \beta_0 = \frac{-1}{\sin \sigma_1 \cdot \tan A_1}$$

$$2.3: \quad \cos \lambda_1 = \frac{\tan \beta_1}{\tan \beta_0}$$

$$2.4: \quad \text{A check: } \sin \lambda_1 = \frac{\sin \sigma_1}{\cos \beta_1}$$

$$2.5: \quad \tan B_0 = \frac{a}{b} \cdot \tan \beta_0$$

$$\text{Auxiliary quantities} \quad W_0 = (1 - e^2 \cdot \sin^2 B_0)^{1/2} \quad k_1 = \frac{1 - W_0}{1 + W_0}$$

(3) Calculating the distances

S is starting in point P_0 , the northernmost point on the geodesic, see figure 2.6.

3.1 The ellipsoidal distance S_1 is computed by using (2.35), from the calculated value of the spherical distance σ_1 in step (2.1).

NB: S is negative if point P is west of the northernmost point P_0 . On figure 2.6 S_1 and S_2 are positive.

3.2 The ellipsoidal distance S_2 is computed from S_1 and the known value ΔS_{1-2} along the geodesic:

$$S_2 = S_1 + \Delta S_{1-2}$$

The distance on the sphere (arc) σ_1 was calculated in step 2.1.

The distance σ_2 , is computed from S_2 which was computed in step 3.2, by using (2.35). Then:

$$\Delta\sigma = \sigma_2 - \sigma_1$$

(4) From the spherical triangle $N' - P'_0 - P'_2$ and the already calculated values and the known values we obtain the spherical variables λ_2 , β_2 and azimuth $A'_2 = A_2$

$$4.1 \quad \tan \lambda_2 = \frac{\tan \sigma_2}{\cos \beta_0}$$

$$4.2 \quad \tan A_2 = \frac{-1}{\sin \sigma_2 \cdot \tan \beta_0}$$

$$4.3 \quad \tan \beta_2 = \frac{\cos A_2}{\tan \sigma_2}$$

(5) The geographical latitude B_2 is calculated from the known equation, see step 1:

$$\tan B_2 = \frac{a}{b} \cdot \tan \beta_2$$

(6) The geographical longitude

$$\Delta\lambda = \lambda_2 - \lambda_1$$

$$\Delta\sigma = \sigma_2 - \sigma_1$$

6.1 Δl is computed by using (2.42b)

6.2 The geographical longitude L_2 can be computed:

$$L_2 = L_1 + \Delta l$$

2.5.2 Numerical solution of the Second geodetic main task

With the second geodetic main task (also named the *inverse problem*), the inverse task of no. 1, we can compute the geodesic between two known points and its azimuths at these points, see figure 2.6:

Known: The coordinates of $P_1 (B_1, L_1)$ and $P_2 (B_2, L_2)$

Unknown: The azimuths A_1 and A_2 , the distance ΔS_{1-2}

Method:

By using the formulas which is developed in chapter 2, the Second geodetic main task can only be solved by using a stepwise approach, an iteration.

Equation (2.43) is the problem, to calculate the spherical longitude, $\Delta\lambda$. The arc lengths σ_1 and σ_2 on the sphere are not known or calculated yet. We have to use a stepwise, iterative method to calculate first $\Delta\lambda$ and then λ_1 and λ_2 .

- (1) From the spherical triangle $N'-P'_0-P'_1$, see figure 2.7, we obtain the reduced latitude of the two points from the known geographical latitudes and the half axes of the ellipsoid:

$$\tan \beta_1 = \frac{b}{a} \cdot \tan B_1 \quad \tan \beta_2 = \frac{b}{a} \cdot \tan B_2$$

- (2) The first approximate value of the difference in spherical longitude, $\Delta\lambda$, is computed by (2.24):

$$2.1 \quad \Delta\lambda = V \cdot \Delta l = \frac{1}{\sqrt{1-e^2 \cdot \cos^2 \beta}} \cdot \Delta l$$

where the preliminary value of the latitude β is $\beta^1 = (\beta_1 + \beta_2) / 2$

- 2.2 From the spherical triangle $N'-P'_0-P'_1$, see figure 2.7: The first approximate values of the spherical variables $\lambda_1, \lambda_2, \beta_0, \sigma_1$ and σ_2 are obtained by using the approximate value of $\Delta\lambda$ and spherical trigonometry:

$$\tan \lambda_1 = \frac{\tan \beta_1 \cdot \cos \Delta\lambda - \tan \beta_2}{\tan \beta_1 \cdot \sin \Delta\lambda}$$

$$\tan \lambda_2 = \frac{-\tan \beta_2 \cdot \cos \Delta\lambda + \tan \beta_1}{\tan \beta_2 \cdot \sin \Delta\lambda}$$

$$\tan \beta_0 = \frac{\tan \beta_1}{\cos \lambda_1}$$

$$\tan \sigma_1 = \cos \beta_0 \cdot \tan \lambda_1$$

$$\tan \sigma_2 = \cos \beta_0 \cdot \tan \lambda_2$$

- (3) With the approximate values of β_0 , σ_1 and σ_2 from the first round described in step (2), a new value of $\Delta\lambda$ is computed with (2.43). This iterative process is repeated until there is no change in the $\Delta\lambda$ value.
- (4) When the final value of $\Delta\lambda$ is computed, the final values of the other variables are computed.
For β_0 , σ_1 , σ_2 , λ_1 and λ_2 , see the formulas in step (2).

The unknown azimuths:

$\tan A_1 = \frac{-1}{\sin \sigma_1 \cdot \tan \beta_0}$	$\tan A_2 = \frac{-1}{\sin \sigma_2 \cdot \tan \beta_0}$
--	--

- (5) The length of the geodesic.

- 5.1 With the computed values of σ_1 and σ_2 , the ellipsoidal lengths from point P_0 to P_1 (S_1) and to P_2 (S_2) are computed by using (2.35).
NB: S is negative if point P is west of the northernmost point P_0 . On figure 2.6 S_1 and S_2 are positive.

$S_i = C \left\{ \sigma_i + D_1 \cdot \sin 2\sigma_i + D_2 \cdot \sin 4\sigma_i + D_3 \cdot \sin 6\sigma_i + \dots \right\}$
--

- 5.2 The length of the geodesic is the difference:

$\Delta S_{1-2} = S_2 - S_1$

ΔS can also be computed directly by subtracting (2.35a) of S_1 from (2.35a) of S_2 :

$\Delta S = S_2 - S_1 = C \cdot \left\{ (\sigma_2 - \sigma_1) + D_1 \cdot (\sin 2\sigma_2 - \sin 2\sigma_1) - D_2 \cdot (\sin 4\sigma_2 - \sin 4\sigma_1) + D_3 \cdot (\sin 6\sigma_2 - \sin 6\sigma_1) \right\}$

2.6 Numerical examples of the geodetic main tasks

Numerical example no. 1:

See example exercise no. 3 (2018) in the course TBA4245. This example is complete, showing all the calculations.

Numerical example no. 2:

The Bessel-Helmert method.

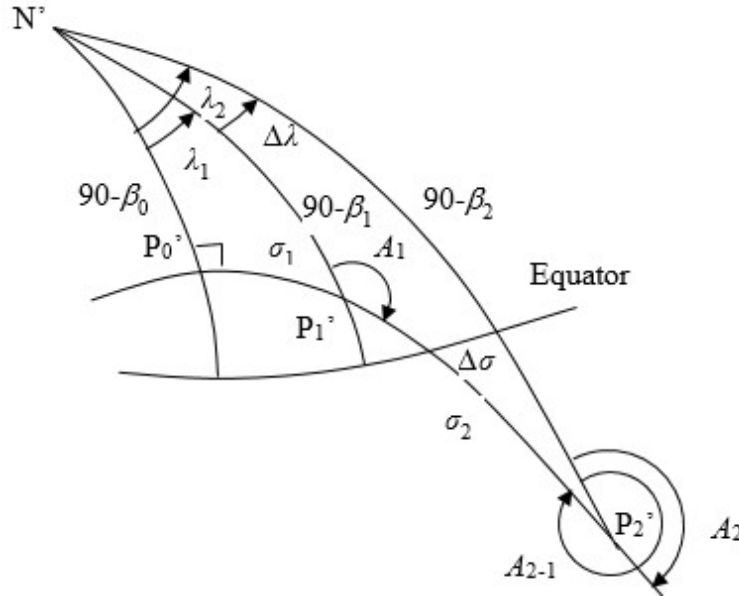


Figure 2.8: Numerical example, the sphere

(1) The first geodetic main task

The international ellipsoid:

Semi axis: $a = 6378388.000$ $b = 6356911.946$ First numerical eccentricity: $e = 0.0067226699$

Known:

Geographical latitude of P_1 : $B_1 = 50^\circ$ (ellipsoidal)Geographical longitude of P_1 : $L_1 = 10^\circ$ The length of the geodesic between P_1 and P_2 : $\Delta S = 15\,000\,000\text{m}$ Azimuth from P_1 to P_2 : $A_1 = 140^\circ$ Unknown: B_2 , L_2 and A_2

The method and the formulas are shown in chapter 2.5.1.

$$1 \quad \text{Reduced latitude of } P_1: \quad \beta_1 = 49.90481989^\circ (\text{spherical})$$

$$2 \quad \sigma_1 = 32.82036475^\circ = 0.5728234266 \text{ radians}$$

$$\lambda_1 = 57.30397070^\circ$$

$$\beta_0 = 65.54406265^\circ$$

$$B_0 = 65.61680460^\circ$$

$$W_0 = (1 - e^2 \sin^2 B_0)^{1/2} = 0.9972076544$$

$$k_1 = (1 - W_0) / (1 + W_0) = 0.001398124824$$

$$3 \quad \text{Equation (2.35):}$$

$$C = 6365815.259 \text{ m}$$

$$D_1 = 0.00069906190$$

$$D_2 = 1.222 \cdot 10^{-7}$$

$$D_1 = 5.69 \cdot 10^{-11}$$

$$S_1 = 3650\,541.563 \text{ m} \quad (2.35)$$

$$S_2 = S_1 + \Delta S = 18650\,541.463 \text{ m}$$

$$\sigma_2 = 2.93008310 \text{ radians} \quad (2.35) \text{ where } S_2 \text{ is known}$$

$$\sigma_2 = 167.8813952^\circ$$

$$\Delta\sigma = \sigma_2 - \sigma_1 = 135.061030^\circ$$

4

$$\lambda_2 = 152.58611190^\circ$$

$$\beta_2 =$$

$$\rightarrow A_2 = 294.7781901^\circ \quad \text{The azimuth in } P_2 \text{ of the geodesic}$$

5

$$\rightarrow B_2 = -62.95088994^\circ \quad P_2 \text{ is on the southern hemisphere}$$

6

Equation (2.43):

$$\Delta\lambda = \lambda_2 - \lambda_1 = 95.28214120^\circ$$

$$\Delta l = 95.09397199^\circ$$

$$\rightarrow L_2 = L_1 + \Delta l = 105.09397199^\circ$$

(2) The second geodetic main task

The same two points P_1 and P_2 as in the first geodetic main task above.

The international ellipsoid:

$$\text{Semi axis: } a = 6378388.000 \quad b = 6356911.946$$

$$\text{First numerical eccentricity: } e = 0.0067226699$$

Known:

$$\text{Geographical latitude of } P_1: \quad B_1 = 50^\circ \text{ (ellipsoidal)}$$

$$\text{Geographical longitude of } P_1: \quad L_1 = 10^\circ$$

$$\text{Geographical latitude of } P_2: \quad B_2 = -62.95088994^\circ \text{ (ell.)}$$

$$\text{Geographical longitude of } P_2: \quad L_1 = 105.09397199^\circ$$

$$\Delta l = 95.09397199^\circ$$

Unknown: The azimuths A_1 and A_2 , the length of the geodesic, ΔS

The method and the formulas are shown in chapter 2.5.2.

1 Reduced latitudes:

$$\beta_1 = +49.90481989^\circ \text{ (spherical)}$$

$$\beta_2 = -62.87254789^\circ (\text{spherical})$$

2

Equation (2.24):

First approximation:

$$\beta^1 = (\beta_1 + \beta_2) / 2 = -6.483864^\circ$$

$$\Delta\lambda = 95.41111828^\circ$$

3.

Second round, results not shown.

⋮

4

Last round, here the third round (third approximation):

Equation (2.43):

$$W_0 = (1 - e^2 \sin^2 B_0)^{1/2} = 0.9972076544$$

$$k_1 = (1 - W_0)/(1 + W_0) = 0.001398124824$$

$$R =$$

$$R_1 =$$

$$R_2 =$$

$$R_3 =$$

$$\Delta\lambda = 95.28214118^\circ \quad (2.43)$$

$$\lambda_1 =$$

$$\beta_0 = 65.54406266^\circ$$

$$B_0 = 65.61680461^\circ$$

$$\sigma_1 = 32.82036477^\circ$$

$$\sigma_2 = 167.88139530^\circ$$

$$\Delta\sigma = \sigma_2 - \sigma_1 = 135.06103050^\circ$$

$$\rightarrow A_1 = 140.0000000^\circ \quad \text{azimuth from } P_1 \text{ to } P_2$$

$$\rightarrow A_2 = 294.7781902^\circ$$

5

Equation (2.35):

$$\rightarrow \Delta S = 14\,999\,999.98 \text{ m}$$

The length of the geodesic between P_1 and P_2 differs from the numerical value in the first geodetic main task with 0.02 m. The calculations was carried out with a pocket calculator with limited accuracy, this is probably the cause of the difference of 2 cm, should have been (much closer to) 0.

Chapter 3: Not covered.

Old and approximate methods for calculations on the ellipsoid. Not covered.

CHAPTER 4**THE PROJECTION OF THE ELLIPSOID ON THE PLANE****Ellipsoid of revolution instead of using Spheroid**

Scientific literature (particularly geodesy) often uses “*ellipsoid*” in place of “*ellipsoid of revolution*” and only applies the adjective “tri-axial” when treating the general case.

Older literature uses “*spheroid*” in place of ‘*ellipsoid of revolution*’.

In this new version of the textbook “Geodesy part 2” we will use *ellipsoid* and mean ellipsoid of revolution. We will not use spheroid, spheroid was used in the old/original version of the textbook.

4.0 Introduction to chapter 4: The mapping, corrections, number of terms

The focus of “Geodesy part 2” is calculations on- and mapping of the ellipsoid of revolution.

Mapping

Chapter 4 explains how to convert from ellipsoidal latitude/longitude to/from north/east in the Gauss-Krüger map projection.

The equations are derived, and Holsen derives the equations based on the auxiliary quantities «hjelpstørrelsene», footpoint «fotpunkts-“ and isometric latitude «isometisk bredde».

Recent years new methods and most important, the number of terms from series expansion to find equations to calculate between a sphere, to/from a ellipsoid surface have been improved. (Karney 2013) ensures/claim precession less than one millimeter using his Matlab toolbox:

<http://www.mathworks.com/matlabcentral/fileexchange/39108-geodesics-on-an-ellipsoid-of-revolution>

Recent years, faster converging methods have been developed also here. (Karney 2011) claims to achieve a precession better than five nanometers (10^{-9} m) in a distance within 3900 km from the central meridian:

<http://www.mathworks.se/matlabcentral/fileexchange/15285-geodetic-toolbox>

Corrections

By using the equations of the map projection we can derive formulas for the meridian convergence, the corrections on the directions and distances and the scale factor. These formulas are for instance used when observed directions and distances on the ellipsoid are planned to be used on the plane for calculations of UTM coordinates.

The number of terms in the equations

The equations are power series with terms of various order. Higher order terms might not contribute to the final results, as they might be equal 0. We need more terms of higher order the further away the point we want to map is from the central meridian, as the size of the terms are very dependent on the distance from the central meridian (the north axis). The maximum difference or distance in longitude is ca. $3-3,5^\circ$ for the UTM projection.

Various textbooks have various ways of presenting the equations regarding the number of terms. In this new version of chapter 4 we have extended the number of terms compared to the old version.

Thus, the mapping equations should have the relevant terms for sub cm accuracy on rectangular coordinates and an accuracy of $0,0001''$ (arc second) = $0,000\ 00003^\circ$ on geographical coordinates.

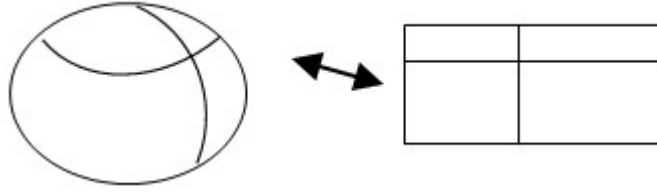
The same accuracies are obtained by the given equations for the meridian convergence, the corrections on the directions and distances and the scale factor.

Today, the series can be implemented on computer programs, and terms of higher order can easily be included in the equations.

Equations from the various textbooks and articles are compared and used, see the bibliography at the back of this chapter.

4.1 The General Mapping Equations

In this chapter we use an analytic approach to develop the equations of the map projection, and not using the geometrical approach as we did in the basic course (for instance using a cylinder to explain the UTM map projection).



A point on the ellipsoid having geographical (= ellipsoidal) coordinates, latitude and longitude (B, L), is projected on the plane where the projected point might be indicated in rectangular Cartesian coordinates (x, y). The relation between geographical coordinates and Cartesian coordinates on the plane is expressed by the mapping equations:

$$x = f_1(B, L) \quad , \quad x \text{ is normally northwards in Norway} \quad (4.1a)$$

$$y = f_2(B, L) \quad , \quad y \text{ is normally eastwards in Norway}$$

or vice versa:

$$B = F_1(x, y) \quad (4.1b)$$

$$L = F_2(x, y)$$

If we want a conformal representation of the surface of the ellipsoid on the plane, the equations must be formed to effect conformity. We start with looking at mapping of infinite small figures and then extend / integrate the found differential mathematical relations to figures or equations of finite size.

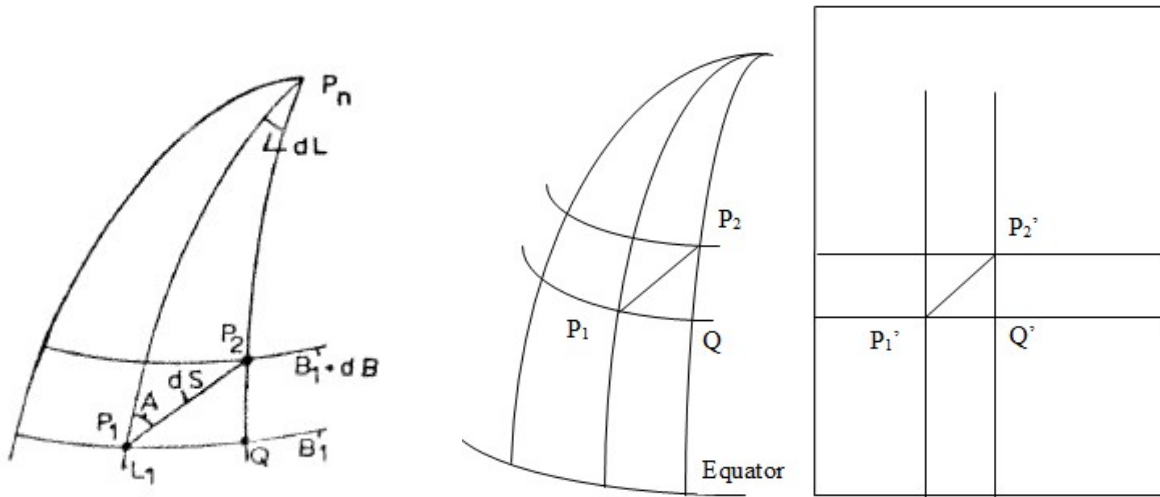


Figure 4.1 The infinite figure P_1 - P_2 - Q on the ellipsoid and on the plane

P_1 and P_2 are neighbouring points on the ellipsoid: $P_1(B_1, L_1)$ and $P_2(B_1 + dB, L_1 + dL)$

Their images on the plane are: $P'_1(x_1, y_1)$ and $P'_2(x_1 + dx, y_1 + dy)$

Distances on the ellipsoid:

Along the meridian: $P_2Q = R_M \cdot dB$

Perpendicular to the meridian: $P_1Q = (R_N \cdot \cos B) \cdot dL$

where: $R_M (= M)$ is the radius of the curvature along the Meridian through point P_1 .

and: $R_N (= N)$ is the radius of the curvature perpendicular to the Meridian through point P_1 , in the prime vertical.

The distance between P_1 and P_2 on the ellipsoid: $dS^2 = R_M^2 \cdot dB^2 + R_N^2 \cdot \cos^2 B \cdot dL^2$ (4.2)

The distance between P_1 and P_2 on the plane: $ds^2 = dx^2 + dy^2$ (4.3)

The square of the scale factor m :

$$m^2 = \frac{\text{distance on the plane}^2}{\text{distance on the ellipsoid}^2} = \frac{ds^2}{dS^2} = \frac{dx^2 + dy^2}{R_M^2 \cdot dB^2 + R_N^2 \cdot \cos^2 B \cdot dL^2} \quad (4.4)$$

In equation (4.4) dx and dy might be expressed by the latitude B and the longitude L , thus the scale factor m is expressed as a function of B and L alone.

4.2 Isometric Latitude, q

For development of the conformal mapping equations it is advantageous to use the isometric latitude q instead of the geographical latitude B .

The name "isometric" means: At any point on the ellipsoid equal increments in degrees of latitude B and longitude L , will give equal distance displacements in meters along the meridians and parallels respectively. See also the example in figure 4.4 and table 4.1 on page 67.

By introducing q we can simplify equation (4.4) and use the isometric latitude to more easily obtain the equations of the conformal mapping of a point on the ellipsoid to the projection plane. The isometric latitude is defined by a differential equation (see the denominator of equation (4.4) or (4.2)).

$$dS^2 = R_M^2 \cdot dB^2 + R_N^2 \cdot \cos^2 B \cdot dL^2 = R_N^2 \cdot \cos^2 B \cdot \left(\frac{R_M^2}{R_N^2 \cdot \cos^2 B} \cdot dB^2 + dL^2 \right)$$

Definition of isometric latitude, see the first term in the parenthesis above:

$dq = \frac{R_M}{R_N \cdot \cos B} \cdot dB$	(4.5)
--	-------

Another approach to define the isometric latitude q is to define q by using the azimuth, A , from P_1 to P_2 , see figure 4.1. We can define the equation of the azimuth in two ways:

1: $\tan A = \frac{R_N \cdot \cos B \cdot dl}{R_M \cdot dB}$	2: $\tan A = \frac{dl}{dq}$ per definition
--	--

The right hand sides of 1 and 2 above are equal, and we obtain equation (4.5).

From (4.2) when B is replaced with q :

$$dS^2 = R_N^2 \cdot \cos^2 B \cdot (dq^2 + dL^2) \quad (4.6a)$$

The line element on the ellipsoid, dS , expressed by (4.6a) is called the *isometric form* of the element.

The equation (4.4) by using q instead of B , see that the factor $(N^2 \cdot \cos^2 B)$ is common for the square of coordinate differentials dq and dL . This is important when we later use (4.6b) to develop the equations of the conformal map projection.

$$m^2 = \frac{dx^2 + dy^2}{R_N^2 \cdot \cos^2 B \cdot (dq^2 + dL^2)} \quad (4.6b)$$

4.2.1 Computing q when B is known and vice versa

In chapter 4.2.2 and 4.2.3 equations on closed form (not series) are shown for calculating the isometric latitude q from the geographic latitude B and vice versa.

In the first version of this text book (1973) with limited use of computers and data programs, the equations to go between q and B were in form of series developed from use of power series such as Taylor series, to obtain the wanted accuracy of the results. The problem was for mechanical computations that you needed tabulated values for every degree of B and the corresponding q as starting values for the calculations.

In this version of the text book we use closed equations, the old method described above is left out.

4.2.2 Computing q when B is known

By integration of (4.5) we obtain an equation where q is a function of the latitude B and the ellipsoid parameters. We recall some expressions derived in “Geodesy part 1” chapter 2.3 and also shown in chapter 2.1 in this text book:

$$\text{Radius of curvature in the Meridian: } R_M = \frac{a \cdot (1 - e^2)}{W^3} = \frac{c}{V^3} = \frac{a \cdot \sqrt{1 + e'^2}}{V^3} \quad (4.7a)$$

$$\text{Radius of curvature in the prime vertical: } R_N = \frac{a}{W} = \frac{c}{V} = \frac{a^2/b}{V} \quad (4.7b)$$

where the first and second numerical eccentricity of the ellipsoid, e and e' squared are:

$$e^2 = \frac{a^2 - b^2}{a^2} \quad e'^2 = \frac{a^2 - b^2}{b^2}$$

and the help functions W and V are:

$$W = (1 - e^2 \cdot \sin^2 B)^{1/2}$$

$$V = (1 + e'^2 \cdot \cos^2 B)^{1/2} = (1 - e^2 \cdot \cos^2 B)^{-1/2} = (1 + \eta^2)^{1/2} \quad (4.7c)$$

Substituting R_M and R_N in (4.5) by (4.7a) and (4.7b) respectively, we get the equation of dq in the following form, where e is the 1st numerical eccentricity of the ellipsoid:

$$dq = \frac{(1 - e^2) \cdot dB}{(1 - e^2 \cdot \sin^2 B) \cdot \cos B} = \frac{dB}{\cos B} - e^2 \cdot \frac{\cos B \cdot dB}{1 - e^2 \cdot \sin^2 B} \quad (4.8)$$

The last term of (4.8) can be split into two parts:

$$dq = \frac{dB}{\cos B} - \frac{e}{2} \cdot \frac{e \cdot \cos B \cdot dB}{1 + e \cdot \sin B} - \frac{e}{2} \cdot \frac{e \cdot \cos B \cdot dB}{1 - e \cdot \sin B} \quad (4.9)$$

The next and final step is to integrate (4.9) between latitudes $B = 0$ (Equator) and $B = B_P$ (point P).

For a sphere the 1st numerical eccentricity is $e = 0$, and thus the last two terms are equal to 0.

The integral of the *first term* of (4.9) is therefore equal to what we would get for a sphere, compare with the Mercator projection in “Geodesy part 1”, equation (3.51a and b), and with the Gauss-Krüger projection, equation (3.54b) in part 1. In (4.10) $\ln$ is the natural logarithm.

$$q = \ln \tan \left(\frac{\pi}{4} + \frac{B}{2} \right) \quad (4.10)$$

Integration of the *two last terms* of (4.9) between $B = 0$ (Equator) and $B = B_P$ (point P):

$$\begin{aligned} -\frac{e}{2} \cdot \int_0^B \frac{e \cdot \cos B}{1 + e \cdot \sin B} \cdot dB &= -\frac{e}{2} \cdot \ln(1 + e \cdot \sin B) \\ -\frac{e}{2} \cdot \int_0^B \frac{e \cdot \cos B}{1 - e \cdot \sin B} \cdot dB &= +\frac{e}{2} \cdot \ln(1 - e \cdot \sin B) \end{aligned}$$

and consequently, integration of the three terms in (4.9):

$$q = \ln \tan \left(\frac{\pi}{4} + \frac{B}{2} \right) + \frac{e}{2} \cdot \ln \frac{1 - e \cdot \sin B}{1 + e \cdot \sin B} \quad (4.11a)$$

The last term of (4.11a) we might interpret as an additive term for a sphere (the first term) to get the isometric latitude q of the ellipsoid.

In equations (4.11a) and (4.11b) $\ln$ is the natural logarithm and e is the 1st numerical eccentricity of the ellipsoid.

Alternative version of (4.11a):

$$q = \ln \left[\tan \left(\frac{\pi}{4} + \frac{B}{2} \right) \cdot \left(\frac{1 - e \cdot \sin B}{1 + e \cdot \sin B} \right)^{e/2} \right] \quad (4.11b)$$

For every value of the geographic (ellipsoidal) latitude B we can by using (4.11) compute the corresponding value of q .

From equation (4.11a) we can calculate the isometric latitude q at equator and at the north pole:

At equator: When $B = 0$, then $q = 0$

At the North pole: When $B = 90^\circ$, then $q = \text{infinite}$, which is not a practical value...

Numerical example:

The geographical latitude of point P: $B = 60^\circ$

The WGS84 ellipsoid:

The semi major and minor axes: $a = 6378137.0000 \text{ m}$ $b = 6356752.3142 \text{ m}$

The first numerical eccentricity: $e = 0,081819190929$

The isometric latitude, q , of point P by using equation (4.11a):

$$q = \ln \tan \left(\frac{\pi}{4} + \frac{60 \cdot \pi/180}{2} \right) + \frac{0,081819190929}{2} \cdot \ln \frac{1 - 0,081819190929 \cdot \sin 60}{1 + 0,081819190929 \cdot \sin 60}$$

$$q = \ln 3,73205080757 + 0,040909595464 \cdot \ln - 0,00580723515 \quad (\text{in radians})$$

$$q = 75,45612929 - 0,33273006 = 75,12339923 \text{ degrees} = 75^\circ 7' 24,2372''$$

Notice the use of the factor $\pi/180$ to go from denomination degrees to radians.

In the numerical example table 4.1 on page 67 the Norwegian Modified Bessel ellipsoid is used, its semi axes are: $a = 6377492,018 \text{ m}$ $b = 6356173,508 \text{ m}$

Using another ellipsoid than the WGS84 ellipsoid will give a small difference in the ellipsoidal term and thus also on the final result. Using the Modified Bessel ellipsoid we obtain the same value of q as in table 4.1:

$$q = 75,45612929 - 0,33173395 = 75,12439534^\circ = 75^\circ 7' 27,8232''$$

4.2.3 Computing B when q is known

Two methods: 1: closed formulas or 2: series expansions.

1. Closed formulas:

Obtaining B directly from equation (4.11) is a problem, as the equation on the right hand side is rather complex concerning B . A much used method in this case is to use the iteration method. We start with an initial approximate value of B , obtain a better value in the first iteration (round) and continue the iterations until we obtain the same value of B as in the previous iteration.

Equation for the starting value (first trial) of the geographic latitude, B . Notice that (4.12) is a simplified version of equation (4.13):

$$B_0 = 2 \cdot \tan^{-1}(e^q) - \frac{\pi}{2} \quad \text{where } \tan^{-1} = \text{inv tan} = \text{atan}, e \text{ is 1}^{\text{st}} \text{ num. eccentric.} \quad (4.12)$$

Equation of q for iterations, enter the new value of B in the next iteration. In (4.13) e is the 1st numerical eccentricity of the chosen ellipsoid and $\exp$ is the base ("grunntallet") of the natural logarithm, also named e , $e = 2,718281828459...$

$$B = 2 \cdot \tan^{-1} \left[\exp^q \cdot \left(\frac{1 + e \cdot \sin B}{1 - e \cdot \sin B} \right)^{e/2} \right] - \frac{\pi}{2} \quad (4.13)$$

See more about the method on <http://mathworld.wolfram.com/IsometricLatitude.html> (2015) or other internet places.

NB: Numbering of the equations:

In this new version of the text book our intention was to use the same numbering of the equations as in the old version. Therefore we jump from equation (4.13) in chapter 4.2.3 to equation (4.16) in this chapter. Equations numbered (4.14) and (4.15) are not used.

2. Series expansion:

Point P_0 has an isometric latitude q_0 calculated from the geographic latitude B_0 , as shown in (4.11a and b) above or in equation (4.49) of the Gauss-Krüger conformal map projection. Then, for point P with latitudes B and q , we can calculate $\Delta B = B - B_0$ from a known $\Delta q = q - q_0$ by using a power series of Δq .

ΔB expressed by Δq :

$$\Delta B = \left(\frac{dB}{dq}\right)_0 \cdot \Delta q + 1/2 \cdot \left(\frac{d^2B}{dq^2}\right)_0 \cdot \Delta q^2 + 1/6 \cdot \left(\frac{d^3B}{dq^3}\right)_0 \cdot \Delta q^3 + 1/24 \cdot \left(\frac{d^4B}{dq^4}\right)_0 \cdot \Delta q^4 + \dots \quad (4.16)$$

The differential coefficients in the terms of (4.16) can be derived from (4.5), and they are derived in point P_0 .

$$\Delta B = d_1 \cdot \Delta q + d_2 \cdot \Delta q^2 + d_3 \cdot \Delta q^3 + d_4 \cdot \Delta q^4 + \dots \quad (4.17)$$

where the coefficients d_i are as follows, NB the coefficients are dependent on the latitude and the parameters of the chosen ellipsoid/ellipsoid:

$d_1 = +\cos B_0 \cdot (1 + \eta_0^2)$	(4.18)
$d_2 = -1/2 \cdot \cos^2 B_0 \cdot \tan B_0 \cdot (1 + 4 \cdot \eta_0^2 + 3 \cdot \eta_0^4)$	
$d_3 = -1/6 \cdot \cos^3 B_0 \cdot (1 - \tan B_0^2 + 5 \cdot \eta_0^2 + 13 \cdot \eta_0^2 \cdot \tan B_0^2 + 7 \cdot \eta_0^4 - 27 \cdot \eta_0^4 \cdot \tan B_0^2 + \dots)$	
$d_4 = 1/24 \cdot \cos^4 B_0 \cdot \tan B_0 \cdot (5 - \tan B_0^2 + 56 \cdot \eta_0^2 - 40 \cdot \eta_0^2 \cdot \tan B_0^2 + \dots)$	
$d_5 = 1/120 \cdot \cos^5 B_0 \cdot (5 - 18 \cdot \tan B_0^2 + 18 \cdot \tan B_0^2 + \dots)$	

For the latitude and ellipsoid dependent function η in (4.18), see the formula collection in chapter 2.1.

Numerical example, closed formula:

We use the same example as in chapter 4.2.2 where we started with $B = 60^\circ$ and calculated the isometric latitude of point P: $q = 75,12339923^\circ = 75^\circ 7' 24,2372'' = 1,3111506618 \text{ rad}$

The WGS84 ellipsoid:

The semi major and minor axes: $a = 6378137.0000 \text{ m}$ $b = 6356752.3142 \text{ m}$

The first numerical eccentricity: $e = 0,081819190929$

Starting value of the spherical latitude, B , of point P by using equation (4.12):

$$B_0 = 2 \cdot \tan^{-1}(2,718281828459^{1,3111506618}) - \frac{\pi}{2} \quad , \text{ in radians}$$

$$B_0 = 1,044286624 \text{ rad} = 59,83321616^\circ$$

The first iteration gives, equation (4.13):

$$B_1 = 2 \cdot \tan^{-1} \left[2,718281828459^{1,3111506618} \cdot \left(\frac{1+0,081819190929 \cdot \sin 59,83321616}{1-0,081819190929 \cdot \sin 59,83321616} \right)^{0,081819190929/2} \right] - \frac{\pi}{2}$$

$$B_1 = 1,047535599 \text{ rad} = 60,01936869^\circ$$

Iterations no. 2, 3 and 4 using equation (4.13) give the following results, using B_1 in iteration no. 2, B_2 in iteration no. 3, etc.:

$$B_2 = 1,047198120 \text{ rad} = 60,00003257^\circ$$

$$B_3 = 1,047197552 \text{ rad} = 60,00000005^\circ$$

$$B_4 = 1,047197551 \text{ rad} = 60,00000000^\circ \quad , \text{ same value as in the example in chapter 4.2.2.}$$

4.3 The Conformal Projection of the Ellipsoid

To develop the mapping equations it is convenient to substitute the geographical latitude by the isometric latitude. See chapter 4.2.

The relation between geographical coordinates and Cartesian coordinates on the plane is expressed by the mapping equations:

$$x = f_1(q, L) \quad , x \text{ is normally northwards in Norway} \quad (4.19a)$$

$$y = f_2(q, L) \quad , y \text{ is normally eastwards in Norway} \quad (4.19b)$$

The square of the scale factor m is used to develop the equations (4.19), equation (4.4):

$$m^2 = \frac{\text{distance on the plane}^2}{\text{distance on the ellipsoid}^2} = \frac{ds^2}{dS^2} = \frac{dx^2 + dy^2}{R_M^2 \cdot dB^2 + R_N^2 \cdot \cos^2 B \cdot dL^2} \quad \text{is (4.4)}$$

In equation (4.4) we want to express dx and dy by B or q and L . Then we have the scale factor m expressed as a function of latitude and longitude alone.

In the equations the longitude l is used, l is the angle between the central meridian in the actual zone and the meridian through the actual point P.

The longitude L is the angle between the Greenwich Meridian and the meridian through point P.

Example:

- Point P, longitude referred to Greenwich: $L = 10,12345678^\circ$ east Greenwich
- UTM, zone 32 (covering southern part of Norway), central meridian (north axis): $L = 9^\circ$
- Point P, longitude l : $l = 10,12345678 - 9 = +1,12345678^\circ$

The differential values of L and l are the same in size, as $l = L_P - 9$ (zone 32): $dL = dl$

The line element ds on the plane can be expressed by the differentials dq and dl . We use figure 4.2 below to see the relations between the differentials, dx , dy , dq and dl .

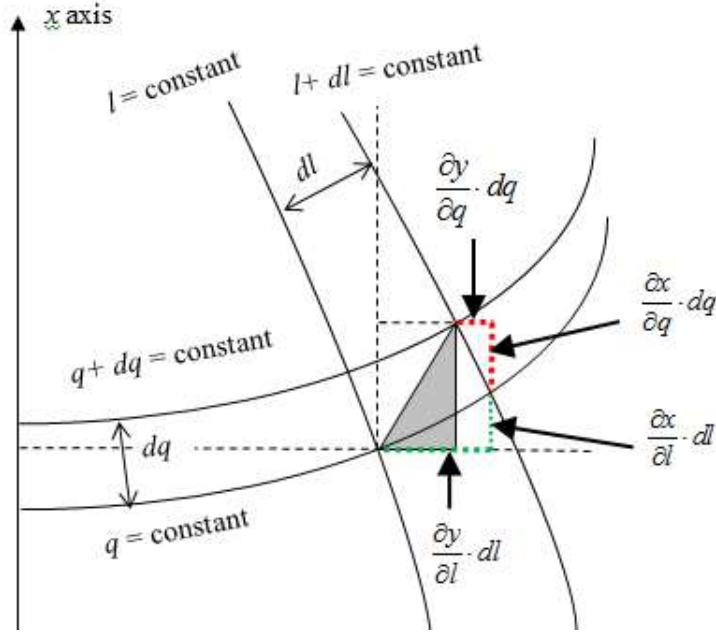


Figure 4.2

We decompose dl and dq in the x and y direction and thus finding the components of dl and dq in the x and y direction. See the figure. The coefficients $\frac{\partial x}{\partial q}$ etc. are the gradients of the lines.

$$dx = \frac{\partial x}{\partial q} \cdot dq + \frac{\partial x}{\partial l} \cdot dl \quad (4.20a)$$

$$dy = \frac{\partial y}{\partial q} \cdot dq + \frac{\partial y}{\partial l} \cdot dl \quad (4.20b)$$

In equation (4.4), also shown on page 62, we need $ds^2 = dx^2 + dy^2$. Squaring (4.20a and b):

$$dx^2 = \left(\frac{\partial x}{\partial q}\right)^2 \cdot dq^2 + 2 \cdot \frac{\partial x}{\partial q} \cdot \frac{\partial x}{\partial l} \cdot dq \cdot dl + \left(\frac{\partial x}{\partial l}\right)^2 \cdot dl^2$$

$$dy^2 = \left(\frac{\partial y}{\partial q}\right)^2 \cdot dq^2 + 2 \cdot \frac{\partial y}{\partial q} \cdot \frac{\partial y}{\partial l} \cdot dq \cdot dl + \left(\frac{\partial y}{\partial l}\right)^2 \cdot dl^2$$

and we get $ds^2 = dx^2 + dy^2$:

$$ds^2 = \left[\left(\frac{\partial x}{\partial q}\right)^2 + \left(\frac{\partial y}{\partial q}\right)^2 \right] \cdot dq^2 + 2 \cdot \left[\frac{\partial x}{\partial q} \cdot \frac{\partial x}{\partial l} + \frac{\partial y}{\partial q} \cdot \frac{\partial y}{\partial l} \right] \cdot dq \cdot dl + \left[\left(\frac{\partial x}{\partial l}\right)^2 + \left(\frac{\partial y}{\partial l}\right)^2 \right] \cdot dl^2 \quad (4.21)$$

We introduce the following designations E , F and G ("fundamentalstørrelser E , F og G ") to simplify writing the equation (4.21), and they are useful for developing the equations for the general Gaussian conformal mapping.

E , F and G are the *Gaussian coefficients*:

$$\boxed{E = \left[\left(\frac{\partial x}{\partial q} \right)^2 + \left(\frac{\partial y}{\partial q} \right)^2 \right] \quad F = \left[\frac{\partial x}{\partial q} \cdot \frac{\partial x}{\partial l} + \frac{\partial y}{\partial q} \cdot \frac{\partial y}{\partial l} \right] \quad G = \left[\left(\frac{\partial x}{\partial l} \right)^2 + \left(\frac{\partial y}{\partial l} \right)^2 \right]} \quad (4.22)$$

Then (4.21) becomes:

$$ds^2 = E \cdot dq^2 + 2 \cdot F \cdot dq \cdot dl + G \cdot dl^2 \quad (4.23)$$

Equation (4.23) is called the first fundamental quadratic form.

The right-hand side of (4.23) and of (4.6a) substituted in (4.4):

$$m^2 = \frac{ds^2}{dS^2} = \frac{dx^2 + dy^2}{R_M^2 \cdot dB^2 + R_N^2 \cdot \cos^2 B \cdot dL^2} = \frac{E \cdot dq^2 + 2 \cdot F \cdot dq \cdot dl + G \cdot dl^2}{R_N^2 \cdot \cos^2 B \cdot (dq^2 + dl^2)} \quad (4.24)$$

The azimuth of the line dS on the ellipsoid is A , see figure 4.1, and $\tan A$ is:

$$\tan A = \frac{P_1 Q}{P_2 Q} = \frac{dl}{dq} \quad (\text{per definition}) \quad (4.25)$$

The nominator and the denominator of the right-hand side of (4.24) is divided by dq^2 , and using $\tan A$ from (4.25):

$$m^2 = \frac{(E \cdot dq^2 + 2 \cdot F \cdot dq \cdot dl + G \cdot dl^2)/dq^2}{R_N^2 \cdot \cos^2 B \cdot (dq^2 + dl^2)/dq^2} = \frac{E + 2 \cdot F \cdot \tan A + G \cdot \tan^2 A}{R_N^2 \cdot \cos^2 B \cdot (1 + \tan^2 A)} = \frac{E + 2 \cdot F \cdot \tan A + G \cdot \tan^2 A}{R_N^2 \cdot \cos^2 B / \cos^2 A}$$

and then multiplying by $\cos^2 A$:

$$m^2 = \frac{(E + 2 \cdot F \cdot \tan A + G \cdot \tan^2 A) \cdot \cos^2 A}{(R_N^2 \cdot \cos^2 B / \cos^2 A) \cdot \cos^2 A}$$

$$\boxed{m^2 = \frac{E \cdot \cos^2 A + 2 \cdot F \cdot \sin A \cdot \cos A + G \cdot \sin^2 A}{R_N^2 \cdot \cos^2 B}} \quad (4.26)$$

By using equation (4.26) we can calculate how the scale factor m is varying with the azimuth from a point.

Conformal map projection: Conformity means that the scale at a point on the plane relative to the corresponding point on the ellipsoid must be equal in all directions. That is, the scale must be independent of the azimuth A to obtain a conformal map projection.

As described in «Geodesy part 1», an infinite (differential) circle on the ellipsoid must be mapped as a circle on the plane (and not as an ellipse, where the scale is varying with the azimuth). This characteristic is described by using the *Tissot's indicatrix* (ellipse), see “Geodesy part 1”, chapter 3.8.

To eliminate the azimuth from equation (4.26) the following conditions to E , F and G must be fulfilled:

1: Choosing $F = 0$ makes the second term in the numerator to be 0, and (4.26) becomes:

$$m^2 = \frac{E \cdot \cos^2 A + 0 + G \cdot \sin^2 A}{R_N^2 \cdot \cos^2 B}$$

2: Setting $E = G$ eliminates the azimuth from the remaining two terms, as $\cos^2 A + \sin^2 A = 1$:

$$m^2 = \frac{E}{R_N^2 \cdot \cos^2 B} = \frac{G}{R_N^2 \cdot \cos^2 B} \quad (4.27)$$

Choosing $F = 0$ and $E = G$ is synonymous with, check equations (4.22):

$F = 0 = \frac{\partial x}{\partial q} \cdot \frac{\partial x}{\partial l} + \frac{\partial y}{\partial q} \cdot \frac{\partial y}{\partial l} = 0$	(4.28a)
$E = G = \left(\frac{\partial x}{\partial q}\right)^2 + \left(\frac{\partial y}{\partial q}\right)^2 = \left(\frac{\partial x}{\partial l}\right)^2 + \left(\frac{\partial y}{\partial l}\right)^2$	(4.28b)

If we had used the geographic latitude B instead of the isometric latitude q , E would not have been equal G : $E \neq G$ for a conformal map projection when using B .

By using q instead of B the development of the map projection equations is simplified, see below.

The equations (4.28) set the conditions for a conformal mapping in the differential mode. To develop the practical Gaussian map projection equations we must develop the equations (4.28) to a form which is possible to integrate.

These differential equations are named the *Cauchy-Riemann differential equations* (4.31) and they can be derived from (4.28).

Multiplying (4.28a) by $2i$, where i is the imaginary unit, $i = \sqrt{-1}$, and moving the last term in (4.28a) to the right-hand side:

$$2 \cdot \frac{\partial x}{\partial q} \cdot \left(i \cdot \frac{\partial x}{\partial l}\right) = -2 \cdot \left(i \cdot \frac{\partial y}{\partial q}\right) \cdot \frac{\partial y}{\partial l} \quad (4.29a)$$

Multiplying (4.28b) by i^2 where i is the imaginary unit, $i^2 = -1$, moving two terms and changing the signs of all terms:

$$\begin{aligned} -i^2 \cdot \left(\frac{\partial x}{\partial q}\right)^2 + i^2 \cdot \left(\frac{\partial x}{\partial l}\right)^2 &= -i^2 \cdot \left(\frac{\partial y}{\partial l}\right)^2 + i^2 \cdot \left(\frac{\partial y}{\partial q}\right)^2 \\ \left(\frac{\partial x}{\partial q}\right)^2 + \left[i \cdot \frac{\partial x}{\partial l}\right]^2 &= \left(\frac{\partial y}{\partial l}\right)^2 + \left[i \cdot \frac{\partial y}{\partial q}\right]^2 \end{aligned} \quad (4.29b)$$

Adding (4.29a) and (4.29b):

$$\left(\frac{\partial x}{\partial q}\right)^2 + 2 \cdot \frac{\partial x}{\partial q} \cdot \left(i \cdot \frac{\partial x}{\partial l}\right) + \left[i \cdot \frac{\partial x}{\partial l}\right]^2 = \left(\frac{\partial y}{\partial l}\right)^2 - 2 \cdot \left(i \cdot \frac{\partial y}{\partial q}\right) \cdot \frac{\partial y}{\partial l} + \left[i \cdot \frac{\partial y}{\partial q}\right]^2 \quad (4.29c)$$

Compare (4.29c) with the general equation of the square of the sum of two variables, a and b :

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Applying the general equation above on both sides of (4.29c):

$$\left(\frac{\partial x}{\partial q} + i \cdot \frac{\partial x}{\partial l}\right)^2 = \left(\frac{\partial y}{\partial l} - i \cdot \frac{\partial y}{\partial q}\right)^2 \quad (4.30a)$$

Taking the square root of (4.30a):

$$\frac{\partial x}{\partial q} + i \cdot \frac{\partial x}{\partial l} = \pm \left(\frac{\partial y}{\partial l} - i \cdot \frac{\partial y}{\partial q}\right) \quad (4.30b)$$

Separating the real and the imaginary terms in (4.30b) gives two solutions.

The mapping is conformal when the following differential equations are used:

$$1: \quad \frac{\partial x}{\partial q} = -\frac{\partial y}{\partial l} \quad \text{and} \quad \frac{\partial x}{\partial l} = +\frac{\partial y}{\partial q} \quad (4.31a)$$

$$2: \quad \frac{\partial x}{\partial q} = +\frac{\partial y}{\partial l} \quad \text{and} \quad \frac{\partial x}{\partial l} = -\frac{\partial y}{\partial q} \quad (4.31b)$$

The equations (4.31) are called the *Cauchy-Riemann differential equations*. They give a conformal mapping.

(4.31a) effects an “inverse” conformal mapping as the scale factor becomes negative, $m < 0$.

For geodetic purposes it is (4.31b) which is of interest, as (4.31b) gives a conformal mapping where the scale factor is positive, $m > 0$. It is a condition that the scale factor must be positive.

The existence of the *Cauchy-Riemann differential equations* implies that a conformal mapping is effected by an *analytical function* of this form:

$$x + i \cdot y = F(q + i \cdot l) \quad (4.32)$$

$$\text{With } x + i \cdot y = z \quad \text{and } q + i \cdot l = w \quad \text{then } z = F(w) \quad (4.33)$$

The derivative of an analytical function is independent of the direction in which the derivation is made.

Numerical example of q , l and B

On the ellipsoid equal angles Δq and Δl correspond to equal curve elements (equal in meters). Hence, by using constant values of Δq and Δl we might construct a network of squares on the ellipsoid. This will not be the case when using equal angles ΔB and Δl . An example is given in table 4.1 where $\Delta q = \Delta l = 0,010$ gon ($0,009^\circ$).

From (4.5), the definition of q , we have: $\Delta B \cdot R_M = \Delta q \cdot R_N \cdot \cos B$

R_M and R_N are the two principal radii of curvature.

Notice that $\Delta B < \Delta q$, except for areas close to Equator, see table 4.1 below.

The values in table 4.1 are valid for the ellipsoid which is used in the old Norwegian datum NGO1948. This system for coordinate referencing is replaced with the new datum EUREF89. The semi axes of the ellipsoid used in NGO1948, the Modified Bessel Ellipsoid, are:

$$a = 6377492,018 \text{ m} \quad b = 6356173,508 \text{ m}$$

The values of the isometric latitude q are calculated from the geographic latitude B by using equation (4.11).

Table 4.1 NB: Values in the old Norwegian datum NGO1948

B	q Degrees, minutes, seconds	R_N approximate	R_M approximate	$\Delta q \cdot R_N \cdot \cos B$ northward	$\Delta l \cdot R_N \cdot \cos B$ eastward	$\Delta q = \Delta l$ in gon	ΔB in gon
0°	0°	6377 492 m	6334 926 m	1001,8 m	1001,8 m	0,010	0,01006
60°	75° 07' 27,8232"	6393 514 m	6382 793 m	502,1 m	502,1 m	0,010	0,0050
70°	99° 04' 18,8582"	6396 369 m	6391 345 m	343,6 m	343,6 m	0,010	0,0034
80°	139° 12' 33,1108"	6398 233 m	6396 938 m	174,5 m	174,5 m	0,010	0,0017
85°	179° 01' 42,7734"	6398 719 m	6398 392 m	87,6 m	87,6 m	0,010	0,0009

By equal steps of q and l the corresponding steps of B will decrease with increasing latitude, and the surface of the ellipsoid will be divided into squares whose side lengths decrease with increasing latitude, see also figure 4.4.

If this network is mapped on the plane by the simplest of all analytical functions:

$$x + i \cdot y = q + i \cdot l$$

Its projection will consist of squares of equal size. This projection is nothing else than a Mercator projection.

Notice that the Mercator projections are normally used for geographical maps in small scale originally used for sailing. The Belgian/German cartographer Gerhard Kramer (Latinised to Gerardus Mercator) gave name to this projection, and he used this projection in 1569 for a world map (nautical chart).

The Mercator projection is a normal cylindrical map projection. By normal we mean that the cylinder is “standing” and thus the equator is the common line for the cylinder and the sphere. See the example below.

For the *Gauss-Krüger projection* which is used for detail maps, the Central meridian in the actual zone is the line with no distortions (not Equator), see the following chapters where the equations of the Gauss-Krüger map projection is shown and discussed.

4.3.1 Example, Mercator conformal map projection

See chapters 3.9.1 and 3.9.2 in “Geodesy part 1” for more about this projection.

For the Mercator conformal projection of a sphere with radius R the equations for the mapping are:

$$x = q \cdot R = R \cdot \ln \tan \left(\frac{\pi}{4} + \frac{B}{2} \right), \text{ the first term of equation (4.11a), the sphere term.}$$

$$y = l \cdot R$$

In the equations above we must multiply q and l with the actual radius of the sphere, R , as the analytical function $x + i \cdot y = q + i \cdot l$ is presented with a unit radius, $R = 1$.

In this projection E , F and G from equation (4.22) are:

$$\frac{\partial x}{\partial q} = R \quad \left(\frac{\partial x}{\partial q} \right)^2 = R^2 \quad \frac{\partial x}{\partial l} = 0 \quad \frac{\partial y}{\partial q} = 0 \quad \frac{\partial y}{\partial l} = R \quad \left(\frac{\partial y}{\partial l} \right)^2 = R^2$$

$$E = G = R^2 \quad \text{and} \quad F = 0 \quad \Rightarrow \text{We have a conformal projection! See the conditions for (4.28).}$$

The projection of the geographical network on the sphere is shown in figure 4.3. A normal (standing) cylinder is touching along the equator. The equator is mapped as a straight line when the cylinder is cut along a north-south going line (a generatrix) on the cylinder. The parallels (with constant latitude) are mapped as straight lines parallel to equator. Their distance from the equator are generally dependent on the characteristic of the map, here we have conformal mapping as we use an analytical function for the mapping. The meridians are imaged as parallel lines at equal distances and they are normal to the equator.

The orthogonal (x,y) network on the plane represent the (B,L) network on the sphere.

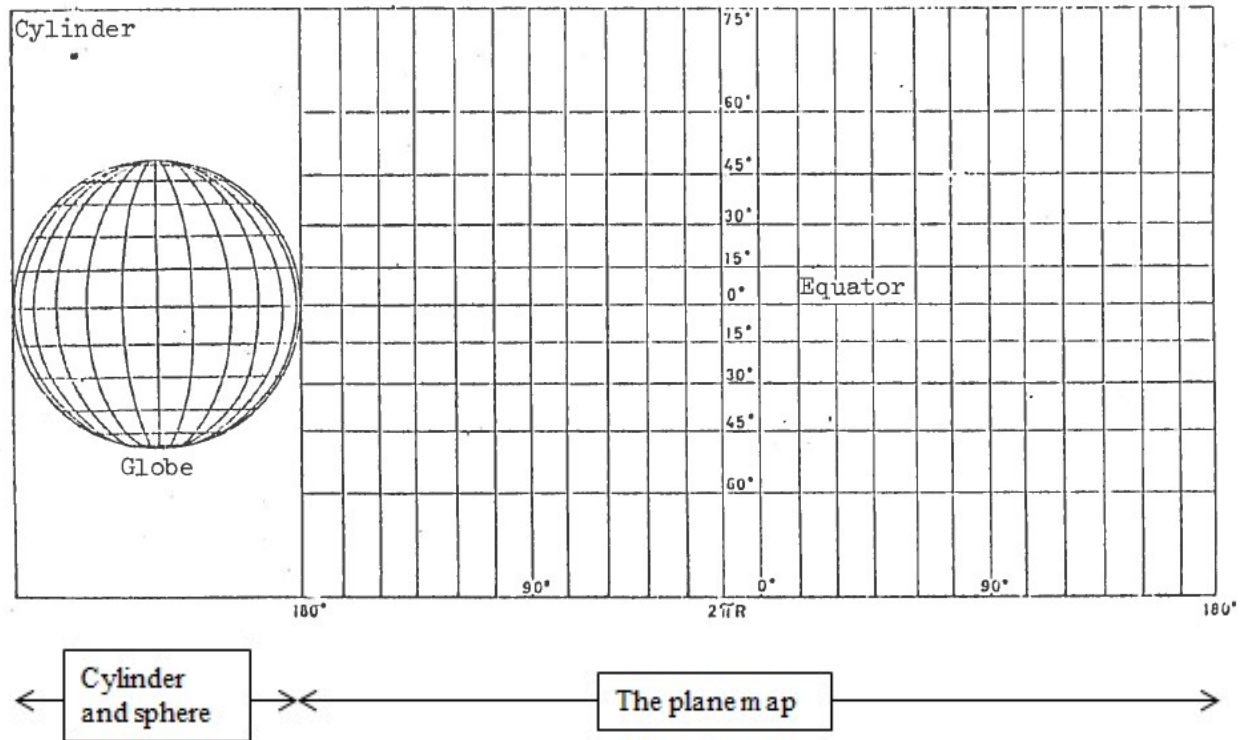


Figure 4.3

The poles are getting no real mapping, as the x value for $B = 90^\circ$ is infinite. This can be visualised by using the isometric latitude q . We will have infinite squares with sides $\Delta q = \Delta l$ in a zone when going from equator to the north pole. See figure 4.4, which is illustrating the conformal mapping in this example. In the map projection plane the infinite numbers of squares will be of same size.

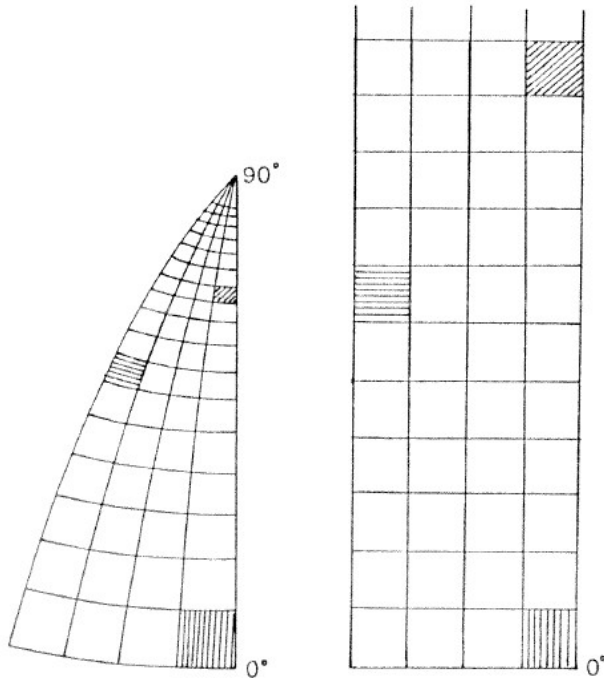


Figure 4.4 Squares on the sphere and on the plane map

4.4 The conformal projection of Gauss-Krüger

For Geodetic maps which are detail maps, not world maps, we use the conformal *Gauss-Krüger* map projection.

This projection is also named the *Transverse Mercator* projection, as it can be interpreted geometrically as a cylinder projection, where the cylinder is lying: A transversal, “lying”, (not normal) cylinder projection with a meridian as a common line between the ellipsoid and the cylinder.

But, as described in the previous chapters, the Gauss-Krüger projection is defined analytically and the projection equations are derived from an analytical function.

For the general Gauss-Krüger projection, the Central meridian in the actual zone is the line with no distortions, see the following chapters where the equations of the Gauss-Krüger map projection is shown and discussed. The ellipsoid is partitioned into zones (for instance 60 zones of 6° longitude in the UTM projection). The meridians and the parallel circles are crossing each other with an angle of 90° both on the ellipsoid and on the plane, due to the method of conformal mapping.

The objective is to map a point with geographic coordinates (B, l) or (B, L) on the ellipsoid into a point with Cartesian, rectangular coordinates (x, y) on the plane, and vice versa.

For the general Gauss-Krüger projection the following four conditions must be fulfilled:

1. Conformal projection: An angle between two geodesics on the ellipsoid is mapped without corrections into the plane, the angle is preserved.
2. The x axis (north axis) on the plane shall be the image of a chosen meridian, the central meridian of the zone.
3. The central meridian shall be mapped in scale 1 : 1 (the scale factor is 1). No scale distortion.
4. Origin in equator, the mapping of equator is the east axis (y axis)

Facts about the *Gauss-Krüger* (the Transverse Mercator) projection and the grids covering it (Osborne):

- Only one grid line runs true north–south. The grid line coincident with the central meridian is true: all other meridians deviate from the grid lines. The UTM projection is a set of 60 distinct Transverse Mercator projections each covering a width of 6° in latitude: the grid lines run true north–south only on the central meridians at 3° E, 9° E, 15° E, . . .
- The scale on the maps derived from Transverse Mercator projections is not uniform: it is a function of the position. For example maps of the Norwegian series N50 have a nominal scale of 1:50 000: this value is only exact on two slightly curved lines almost parallel to the central meridian and a distant approximately 180 km east and west of it. The scale on the central meridian is constant, but it is slightly less than the nominal value of 1, the UTM scale factor is 0,9996.

The above facts are unknown to the majority of map users. They are the subject of this chapter together with the presentation of equations relating latitude and longitude to grid coordinates.

4.4.1 Map projections in Norway

The following variants or modifications of the general Gauss-Krüger projection are used in Norway:

- The UTM (Universal Transversal Mercator) projection is used in the national reference system or datum EUREF89. The four conditions above:
 - 1: Ok
 - 2: Ok
 - 3: Not met: The central meridian is mapped with a factor 0,9996, and thus UTM coordinates are shrank with the factor 0,9996 compared to the conformal Gauss-Krüger coordinates (x,y) on the plane. In addition has UTM an offset of 500 000 m for the east coordinate ($E = 500\,000$ m on the central meridian).
 - 4: Ok
- The NTM (Norwegian Transversal Mercator) projection is a secondary projection and is used on larger construction sites instead of UTM to avoid the problems with the UTM factor of 0,9996. NTM is a projection linked to EUREF89. The four conditions above:
 - 1: Ok
 - 2: Ok
 - 3: Ok, but NTM has an offset of 100 000 m for the east coordinate ($E = 100\,000$ m on the central meridian).
 - 4: Not met: The origin is at 58° North, and there is an offset (starting value) in the origin of $N = 1\,000\,000$ m on the north coordinate.

- The Gauss-Krüger projection in NGO1948 in the old national datum NGO1948 (now substituted with EUREF89). The four conditions above:
 - 1: Ok
 - 2: Ok
 - 3: Ok
 - 4: Not met: The origin is at 58° North.

4.4.2 About the equations of the Gauss-Krüger projection

We use the mapping equations (4.32), $x + i \cdot y = F(q + i \cdot l)$, where l is the difference in longitude between the central meridian and the meridian going through a point P which is to be imaged, see figure 4.5.

If $L = 10,12345678$ degrees east of Greenwich for point P
and the point P is in UTM zone 32 with central meridian $L_0 = 9$ degrees E Gr.w.
then $l = L - L_0 = 1,12345678$ degrees east of the central meridian.

Condition no. 2: The x axis (north axis) on the plane shall be the image of a chosen meridian, the central meridian of the zone. With the origin at the intersection between the central meridian and equator we have for $l = 0$:

$$y = 0 \quad \text{and} \quad x_{y=0} = F(q) \quad (4.34)$$

Let G be the length of the arc of the central meridian from the origin at equator to a point with latitude B . Then:

$$x = G = \int_0^B R_M \cdot dB = F(q) \quad (4.35)$$

R_M is the radius of curvature along the meridian. For the computation of G , see “Geodesy Part 1”, chapter 2.6.

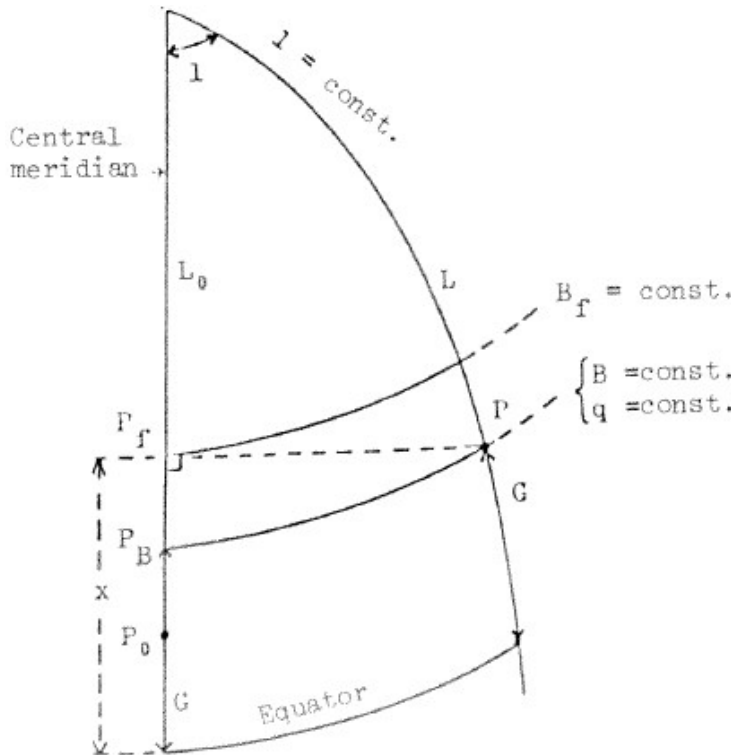


Figure 4.5

Condition no. 3 (a scale of 1 : 1 along the central meridian) gives important differential relations between various variables along the central meridian.

The variables:

- | | |
|---|-----------------------------|
| 1. In the projection: | dx |
| Arc of the meridian on the ellipsoid: | |
| 2. New symbol: | dG |
| 3. By using the geographic latitude B : | $R_M \cdot dB$ |
| 4. By using the isometric latitude q : | $R_N \cdot \cos B \cdot dq$ |

These four variables above are equal in size (in meters):

$$dx = dG = R_M \cdot dB = R_N \cdot \cos B \cdot dq$$

which leads to the following useful relations for use of developing the coefficients of the mapping equations, “dividing” by dq :

$$\frac{dx}{dq} = \frac{dG}{dq} = R_M \cdot \frac{dB}{dq} = \frac{dF(q)}{dq} = F'(q) = R_N \cdot \cos B \quad (4.36)$$

Symbols used in (4.36): The derivative: $\frac{\partial F}{\partial q} = F'$. On the central meridian: $F(q)_0 = G_0$

We can get the derivative of an analytical function by deriving it with respect to the real part (or the imaginary part) of the independent variable.

With (4.36) we already have the first derivative of $F(q)$ at a point on the central meridian. At that point on the central meridian (the north axis) we can develop the conformal mapping equation (4.32), $x + i \cdot y = F(q + i \cdot l)$, in a Taylor series.

Point P_0 on the central meridian:

The longitude l is small (normally less than 3°), while the latitudes B and q are large (B is between 58° and 71° in Norway). Therefore we must use a point P_0 on the central meridian to develop the Taylor series in, and P_0 must have north coordinates (B, q, x) that are close to or equal to the north coordinates of point P , which is the point to be mapped. See figure 4.5.

From the Taylor series we get the mapping equations of the projection by separating the real and the imaginary parts of the Taylor series.

For surveying purposes the width of a zone is generally limited to 6° of longitude, which corresponds to the width of an UTM zone. Taking into account that neighboring zones to a certain degree overlap, the absolute value of the latitude l on figure 4.5 will be less than $3,5^\circ$.

We therefore will obtain sufficient accuracy with moderate numbers of terms in the Taylor series. The series will converge fast, we can abort the series after few terms.

4.4.3 x and y computed from known B and l

Starting point, see also figure 4.5:

- Using the analytical function of the conformal Gauss-Krüger projection: $x + i \cdot y = F(q + i \cdot l)$
- Developing this analytical function at an arbitrary or random point P_0 on the central meridian:
Using the coordinate values of P_0 in the coefficients in the terms of the series: $B_0, q_0, l_0 = 0$.

The relation between the points P and P₀:

$$\begin{aligned} \Delta B &= B - B_0 & B &= B_0 + \Delta B \\ \Delta q &= q - q_0 & q &= q_0 + \Delta q \\ l &= L - L_0 & L &= L_0 + l \end{aligned}$$

Inserting the above formulas into the analytical function $x + i \cdot y = F(q + i \cdot l)$ we obtain:

$$x + i \cdot y = F(q_0 + \Delta q + i \cdot l)$$

Developing the right hand side of the analytical function $x + i \cdot y = F(q_0 + \Delta q + i \cdot l)$ into a Taylor series at P₀ on the Central meridian where $l = 0$:

Generally making a Taylor series of $F(r + s)$ will converge fast (few terms in the series), when the term s is small in size compared to the term r . In this actual case $r = q_0$ and $s = \Delta q + i \cdot l$.

Taylor series on the right hand side, where index 0 indicates values of point P₀:

$$\begin{aligned} x + i \cdot y &= \\ &= F(q)_0 + F'(q)_0 \cdot (\Delta q + i \cdot l) + 1/2 \cdot F''(q)_0 \cdot (\Delta q + i \cdot l)^2 + 1/6 \cdot F'''(q)_0 \cdot (\Delta q + i \cdot l)^3 + \dots \end{aligned} \quad (4.37)$$

According to (4.35) the distance from equator along the central meridian to point P₀ equal to:

$$G_0 = F(q)_0$$

Then the Taylor series becomes:

$$\begin{aligned} x + i \cdot y &= \\ &= G_0 + \left(\frac{dG}{dq}\right)_0 \cdot (\Delta q + i \cdot l) + 1/2 \cdot \left(\frac{d^2G}{dq^2}\right)_0 \cdot (\Delta q + i \cdot l)^2 + 1/6 \cdot \left(\frac{d^3G}{dq^3}\right)_0 \cdot (\Delta q + i \cdot l)^3 + \dots \end{aligned} \quad (4.38)$$

Simplifications:

1. We choose where to place point P₀ on the central meridian. When we choose P₀ to have the *same latitude* as point P which is to be mapped, then $\Delta q = q - q_0 = 0$. We develop the series at point P_B (see figure 4.5), then ΔB and Δq are equal to zero, and the series for x and y will be powers of the longitude l only.
2. We indicate the coefficients in (4.38) by $a_1, a_2, a_3, \dots$, and $(x - G_0)$ by Δx .

Then equation (4.38) becomes:

$$x + i \cdot y = G_0 + a_1 \cdot i \cdot l + a_2 \cdot (i \cdot l)^2 + a_3 \cdot (i \cdot l)^3 + a_4 \cdot (i \cdot l)^4 + \dots \quad (4.39a)$$

$$\Delta x + i \cdot y = a_1 \cdot i \cdot l + a_2 \cdot (i \cdot l)^2 + a_3 \cdot (i \cdot l)^3 + a_4 \cdot (i \cdot l)^4 + \dots \quad (4.39b)$$

The coefficients a_i can be derived from the relations between various variables along the central meridian (4.36) and the definition of isometric latitude in (4.5). The values of point P₀ = P_B on the central meridian are entered into the coefficients.

$$dq = \frac{R_M}{R_N \cdot \cos B} \cdot dB \quad \Rightarrow \quad \frac{dB}{dq} = \frac{R_N \cdot \cos B}{R_M}$$

$$\frac{dG}{dB} = R_M$$

The coefficients a_i :

Symbols used in the equations (4.40):

The principal radius of curvature, R_N , is dependent on the latitude B , see equations in chapter 2.1. For η (eta), see equations in chapter 2.1, η is dependent on the latitude.

From comparing (4.38) and (4.39):

$$a_1 = \left(\frac{dG}{dq} \right)_0 = \left(\frac{dG}{dB} \cdot \frac{dB}{dq} \right)_0 = F'(q)_0$$

$$\boxed{a_1 = R_N \cdot \cos B} \quad (4.40a)$$

The coefficient a_2 is developed from a_1 , generally the coefficient a_{i+1} is developed from a_i :

$$a_2 = 1/2 \cdot F''(q)_0 = 1/2 \cdot \left(\frac{d^2 G}{dq^2} \right)_0 = 1/2 \cdot \left(\frac{d}{dB} \frac{dG}{dq} \cdot \frac{dB}{dq} \right)_0 = 1/2 \cdot \left(\frac{d}{dB} (R_N \cdot \cos B) \cdot \frac{R_N \cdot \cos B}{R_M} \right)_0$$

$$\boxed{a_2 = -\frac{1}{2} \cdot R_N \cdot \cos^2 B \cdot \tan B} \quad (4.40b)$$

$$a_3 = 1/6 \cdot \left(\frac{d^3 G}{dq^3} \right)_0 = 1/6 \cdot \left(\frac{d}{dB} \frac{d^2 G}{dq^2} \cdot \frac{dB}{dq} \right)_0 = 1/6 \cdot \left(\frac{d}{dB} \left(-\frac{1}{2} \cdot R_N \cdot \cos^2 B \cdot \tan B \right) \cdot \frac{R_N \cdot \cos B}{R_M} \right)_0$$

$$\boxed{a_3 = -\frac{1}{6} \cdot R_N \cdot \cos^3 B \cdot (1 - \tan^2 B + \eta^2)} \quad (4.40c)$$

$$a_4 = 1/24 \cdot \left(\frac{d^4 G}{dq^4} \right)_0 = 1/24 \cdot \left(\frac{d}{dB} \frac{d^3 G}{dq^3} \cdot \frac{dB}{dq} \right)_0$$

$$\boxed{a_4 = +\frac{1}{24} \cdot R_N \cdot \cos^4 B \cdot \tan B \cdot (5 - \tan^2 B + 9 \cdot \eta^2 + 4 \cdot \eta^4)} \quad (4.40d)$$

$$a_5 = 1/120 \cdot \left(\frac{d^5 G}{dq^5} \right)_0 = 1/120 \cdot \left(\frac{d}{dB} \frac{d^4 G}{dq^4} \cdot \frac{dB}{dq} \right)_0$$

$$\boxed{a_5 = +\frac{1}{120} \cdot R_N \cdot \cos^5 B \cdot (5 - 18 \cdot \tan^2 B + \tan^4 B + \dots)} \quad (4.40e)$$

In (4.39a) we separate the real and the imaginary terms, and we get series with the real terms for x and a series with the imaginary terms for y , and inserting $i^2 = -1$:

$$x + i \cdot y = G_0 + a_1 \cdot i \cdot l + a_2 \cdot (-1) \cdot l^2 + a_3 \cdot (-1) \cdot i \cdot l^3 + a_4 \cdot (-1)^2 \cdot l^4 + a_5 \cdot (-1)^4 \cdot i \cdot l^5 + \dots$$

Separating real and imaginary terms:

$x = G_0 - a_2 \cdot l^2 + a_4 \cdot l^4 + \dots$	(4.41)
$y = a_1 \cdot l - a_3 \cdot l^3 + a_5 \cdot l^5 + \dots$	

In (4.41) the coefficients a_i are referred to the latitude B , the latitude of the point P which we are going to image.

NB: In (4.41) the latitude l is expressed in radians. If we have l in degrees we must multiply l with the factor $(\pi/180)$ to obtain l in radians.

Inserting for a_i from the (4.40) equations into (4.41), l in radians:

$x = G_0 + \frac{R_N}{2} \cdot \cos^2 B \cdot \tan B \cdot l^2 + \frac{R_N}{24} \cdot \cos^4 B \cdot \tan B \cdot (5 - \tan^2 B + 9 \cdot \eta^2 + 4 \cdot \eta^4) \cdot l^4 + \dots$	(4.42)
$y = R_N \cdot \cos B \cdot l + \frac{R_N}{6} \cdot \cos^3 B \cdot (1 - \tan^2 B + \eta^2) \cdot l^3 +$ $+ \frac{R_N}{120} \cdot \cos^5 B \cdot (5 - 18 \cdot \tan^2 B + \tan^4 B_0 + 14 \cdot \eta^2 - 58 \cdot \tan^2 B \cdot \eta^2 \dots) \cdot l^5 + \dots$	

Additional terms to (4.42):

In the NTNU course TBA4245 (2018) we are also using the text book Hofmann-Wellenhof et al “GNSS” (2008). In this book you will find additional terms to (4.42) for the x and y equations (series expansions). The terms with a_6 , a_7 and a_8 are shown. See equation (8.21) in “GNSS”. We must consider using these additional terms especially if the east value (the l value) is large as it is close to the border of the zone. An example using all the terms is given in the example exercise no. 4 (2015) in the course TBA4245.

NB: In “GNSS” the axes, compared to this book, are switched in the map projection plane..., the y axis is towards north, the x axis is towards east.

Numerical example:

See the example exercise no. 4 (2017) in the course TBA4245.

4.4.4 B and l computed from known x and y

The opposite way compared to chapter 4.4.2., where the analytical function is $x + iy = F(q + i \cdot l)$. Here the analytical function is:

$$q + i \cdot l = F(x + i \cdot y) \quad (4.43)$$

Analog to (4.34) and (4.36):

With the origin at the intersection between the central meridian and equator we have for $q_{l=0}$:

$$q_{l=0} = F(x) = F(G) \quad (4.44)$$

$$F'(x) = F'(G) = \frac{dq}{dG} = \frac{1}{R_N \cdot \cos B}$$

Starting point, see also figure 4.5:

- Using the analytical function of the conformal Gauss-Krüger projection: $q + i \cdot l = F(x + i \cdot y)$
- Developing this analytical function at an arbitrary or random point P_0 on the central meridian: Using values of P_0 in the coefficients in the terms of the series ($B_0, q_0, l_0 = 0$).

The relation between the points P and P_0 :

$$\begin{aligned} \Delta x &= x - x_0 = x - G_0 & x &= G_0 + \Delta x \\ \Delta B &= B - B_0 & B &= B_0 + \Delta B \\ \Delta q &= q - q_0 & q &= q_0 + \Delta q \\ l &= L - L_0 & L &= L_0 + l \end{aligned}$$

Inserting the above formulas into the analytical function $q + i \cdot l = F(x + i \cdot y)$ we obtain:

$$q + i \cdot l = F(x_0 + \Delta x + i \cdot y)$$

Developing the right hand side of the analytical function $q + i \cdot l = F(x_0 + \Delta x + i \cdot y)$ into a Taylor series at P_0 on the central meridian where $l = 0$:

Generally making a Taylor series of $F(r + s)$ will converge fast (few terms in the series), when the term s is small in size compared to the term r . In this actual case $r = x_0$ and $s = \Delta x + i \cdot y$.

Taylor series on the right hand side, where index 0 indicates values of point P_0 :

$$\begin{aligned} q + i \cdot l &= \\ &= F(x)_0 + F'(q)_0 \cdot (\Delta x + i \cdot y) + 1/2 \cdot F''(q)_0 \cdot (\Delta x + i \cdot y)^2 + 1/6 \cdot F'''(q)_0 \cdot (\Delta x + i \cdot y)^3 + \dots \end{aligned}$$

Then the Taylor series becomes:

$$\begin{aligned} q + i \cdot l &= \\ &= q_0 + \left(\frac{dq}{dG}\right)_0 \cdot (\Delta x + i \cdot y) + 1/2 \cdot \left(\frac{d^2q}{dG^2}\right)_0 \cdot (\Delta x + i \cdot y)^2 + 1/6 \cdot \left(\frac{d^3q}{dG^3}\right)_0 \cdot (\Delta x + i \cdot y)^3 + \dots \end{aligned} \quad (4.45)$$

Simplifications:

1. We choose where to place point P_0 on the central meridian. When we choose P_0 to have the same x coordinate (on the plane) as point P which is to be mapped, then $\Delta x = x - x_0 = 0$. We develop the series at point P_f (see figure 4.5) then Δx is equal to zero, and the series for q and l will be powers of the plane east coordinate y only. P_f is named the *footpoint* with geographical coordinates ($B_f, l = 0$) or ($q_f, 0$). Also used name: footprint.
2. We indicate the coefficients in (4.45) by $b_1, b_2, b_3, \dots$, and $(q - q_0)$ by Δq .

Then equation (4.45) becomes:

$$q + i \cdot l = q_0 + b_1 \cdot i \cdot y + b_2 \cdot (i \cdot y)^2 + b_3 \cdot (i \cdot y)^3 + b_4 \cdot (i \cdot y)^4 + \dots \quad (4.46a)$$

$$\Delta q + i \cdot l = b \cdot i \cdot y + b_2 \cdot (i \cdot y)^2 + b_3 \cdot (i \cdot y)^3 + b_4 \cdot (i \cdot y)^4 + \dots \quad (4.46b)$$

The coefficients b_i :

The coefficients b_i is derived by a similar procedure as for the coefficients a_i in equations (4.40), by using the relations between various variables along the central meridian (4.36) and the definition of isometric latitude in (4.5). The values of point $P_0 = P_f$ on the central meridian are entered into the coefficients.

From comparing (4.45) and (4.46):

$$b_1 = \left(\frac{dq}{dG} \right)_0 = F'(q)_0$$

The coefficient b_2 is developed from b_1 , generally the coefficient b_{i+1} is developed from b_i :

$b_1 = \frac{1}{a_1} = \frac{1}{R_{Nf} \cdot \cos B_f}$	(4.47a)
---	---------

$b_2 = + \frac{\tan B_f}{2 \cdot R_{Nf}^2 \cdot \cos B_f}$	(4.47b)
--	---------

$b_3 = + \frac{1 + 2 \cdot \tan^2 B_f + \eta_f^2}{6 \cdot R_{Nf}^3 \cdot \cos B_f}$	(4.47c)
---	---------

$b_4 = + \frac{\tan B_f \cdot (5 + 6 \cdot \tan^2 B_f + \eta_f^2 - 24 \cdot \eta_f^4)}{24 \cdot R_{Nf}^4 \cdot \cos B_f}$	(4.47d)
---	---------

$b_5 = + \frac{5 + 28 \cdot \tan^2 B_f + 24 \cdot \tan^4 B_f + 6 \cdot \eta_f^2 + 8 \cdot \tan^2 B_f \cdot \eta_f^2}{120 \cdot R_{Nf}^5 \cdot \cos B_f}$	(4.47e)
--	---------

Symbols used in equations (4.47):

The principal radius of curvature, R_N , is dependent on the latitude B , see equations in chapter 2.1. For η (eta), see equations in chapter 2.1, η is dependent on the latitude and the ellipsoid.

In (4.46b) we separate the real and the imaginary terms, and we get a series with the real terms for Δq and a series with the imaginary terms for l , and inserting $i^2 = -1$:

$$\Delta q + i \cdot l = b_1 \cdot i \cdot y + b_2 \cdot (-1) \cdot y^2 + b_3 \cdot (-1) \cdot i \cdot y^3 + b_4 \cdot (-1)^2 \cdot y^4 + b_5 \cdot (-1)^4 \cdot i \cdot y^5 + \dots$$

Separating real and imaginary terms, see also (4.50) for B computed from y :

$$\Delta q = -b_2 \cdot y^2 + b_4 \cdot y^4 + \dots \quad (4.48a)$$

$l = b_1 \cdot y - b_3 \cdot y^3 + b_5 \cdot y^5 + \dots$	(4.48b)
---	---------

See a note on page 79 about additional terms to (4.48b) and (4.50).

NB: In (4.48) the longitude l and the isometric latitude q are expressed in radians. If we want degrees instead of radians, we must multiply l and q with the factor $(180/\pi)$ to obtain l and q in degrees.

In (4.48) the coefficients b_i are referred to the latitude B_f , the latitude of the *footpoint* P_f on the central meridian, see figure 4.5. P_f has the same north coordinate in the plane, x , as the point P which we are going to image, $\Delta x = x_P - x_{P_f} = 0$.

The geographic latitude B_f of the latitude footpoint P_f

As we start with the rectangular coordinates x and y on the map projection plane, we have to calculate the geographic latitude B_f of the footpoint P_f from the x and y coordinates for use in calculating the coefficients in (4.47).

For calculating the value of B_f in this version of the book, you will find the equations in the text book “GNSS” by Hofmann-Wellenhof et al (2008): Equations (8.27-8.30).

From isometric latitude q to geographic latitude B :

From (4.48) we can calculate the difference in isometric latitude Δq between the footpoint P_f and point P . We want, however, to calculate $\Delta B = B - B_f$ to obtain geographic latitude of point P from the latitude of the footpoint P_f . The isometric latitude is of no interest in practical life.

For the coefficients d_i , see (4.18), and according to (4.17) we obtain:

$$\begin{aligned}\Delta B &= B - B_f = d_1 \cdot \Delta q + d_2 \cdot \Delta q^2 + d_3 \cdot \Delta q^3 + d_4 \cdot \Delta q^4 + \dots \\ B &= B_f + d_1 \cdot \Delta q + d_2 \cdot \Delta q^2 + d_3 \cdot \Delta q^3 + d_4 \cdot \Delta q^4 + \dots\end{aligned}\tag{4.49}$$

Substituting Δq from (4.48) into (4.49) we get the geographic latitude B from x and y :

$$B = B_f + f_2 \cdot y^2 + f_4 \cdot y^4 + f_6 \cdot y^6 + \dots\tag{4.50}$$

where the coefficients f_i are:

$$\begin{aligned}f_2 &= -\frac{\tan B_f}{2 \cdot R_{Nf}^2} \cdot (1 + \eta_f^2) \\ f_4 &= +\frac{\tan B_f}{24 \cdot R_{Nf}^4} \cdot (5 + 3 \cdot \tan^2 B_f + 6 \cdot \eta_f^2 - 6 \cdot \eta_f^2 \cdot \tan^2 B_f - 3 \cdot \eta_f^4 - 9 \cdot \tan^2 B_f \cdot \eta_f^4)\end{aligned}$$

Symbols used in the equations of the f coefficients above:

The principal radius of curvature, R_N , is dependent on the *footpoint* latitude B_f , see equations in chapter 2.1. For η (eta), see equations in chapter 2.1, η is dependent on the footpoint latitude and the ellipsoid.

NB: In (4.50) the latitude B and B_f are expressed in radians. If we want degrees instead of radians we must multiply B and B_f with the factor $(180/\pi)$ to obtain B and B_f in degrees.

In (4.50) the coefficients b_i are referred to the latitude B_f , the latitude of the footpoint P_f on the central meridian, see figure 4.5. P_f has the same north coordinate in the plane, x , as the point P which we are going to image, $\Delta x = x_P - x_{Pf} = 0$.

See equation (4.48b) for the calculation of the longitude l from x and y , it is copied here (4.48b):

$$l = b_1 \cdot y - b_3 \cdot y^3 + b_5 \cdot y^5 + \dots$$

Additional terms to (4.48b) and (4.50):

In the NTNU course TBA4245 (2017) we are also using the text book Hofmann-Wellenhof et al “GNSS” (2008). In this book you will find additional terms for the B and l equations (series expansions) (4.48) and (4.50). The terms with f_6 , b_7 and f_8 are shown. See equation (8.26) in “GNSS”. We must consider using these additional terms especially if the east value (the y value) is large as it is close to the border of the zone. An example using all the terms is given in the example exercise no. 4 (2017) in the course TBA4245.

NB: In “GNSS” the axes, compared to this book, are switched in the map projection plane..., the y axis is towards north, the x axis is towards east.

Numerical example:

See the example exercise no. 4 (2017) in the course TBA4245.

4.5 Meridian convergence. Scale factor. Corrections on observations.

Previously in chapter 4 the equations of the Gauss-Krüger conformal map projection are shown. Now we will use these equations to derive necessary functions and correction equations for use in calculations on the map projection plane:

- *Meridian convergence, c* : The angle between true north and grid north.
- *The correction of directions, δ* : The angle between the mapping of the geodesic and the straight line on the plane.
- *The scale factor, m* : At a point, or the ratio between infinitesimal distance on the plane and on the ellipsoid.
- *The correction of distances, d* : The difference between a distance on the plane and on the ellipsoid.

We need values of the meridian convergence and the reduction of the directions to go from the *azimuth* on the ellipsoid to the *bearing* on the plane. We normally do the calculations and the estimation of coordinates on the plane. Therefore we must also correct the distances measured by use of GNSS (GPS) and total stations from a distance on the ellipsoid (or more precise the real measured distance) to a distance on the plane (in the map projection).

The relation between the azimuth, A , meridian convergence, c , and the correction to the horizontal direction, δ , can clearly be seen from figure 4.6.

We have a conformal projection. The angle A on the plane is the angle from the image of the meridian through P and to the image of a geodesic originating at P , A is the azimuth of the geodesic.

From figure 4.6, where t is the bearing (“retningsvinkel”) of the straight line originating/starting in point P, we can see:

$$A = c + t - \delta$$

The equations of azimuth and correction of the horizontal angle give an accuracy of 0,001" (arc second) = 0,000 0003°.

NB: To avoid false signs always use a sketch/figure, and remember that the images of the meridian and the geodesics have their concave sides turned towards the x axis.

4.5.1 The Gauss-Krüger meridian convergence on the plane

A meridian is a curve on the ellipsoid with constant longitude, and a meridian goes through the north and south pole of the ellipsoid. The meridian through point P is mapped as a curve on the plane.

The distance between the central meridian (the north axis) and the mapped meridian is decreasing when we go towards the poles. See the east values in table 4.2 for various latitudes.

The meridian convergence, c , is the angle from true north (the mapped meridian) to the grid north (a parallel to the north axis on the plane).

The meridian convergence is positive when starting at the northern part of the meridian through P ($l = \text{constant}$) and going clockwise to the northern part of the $y = \text{constant}$ line (grid north). See figure 4.6.

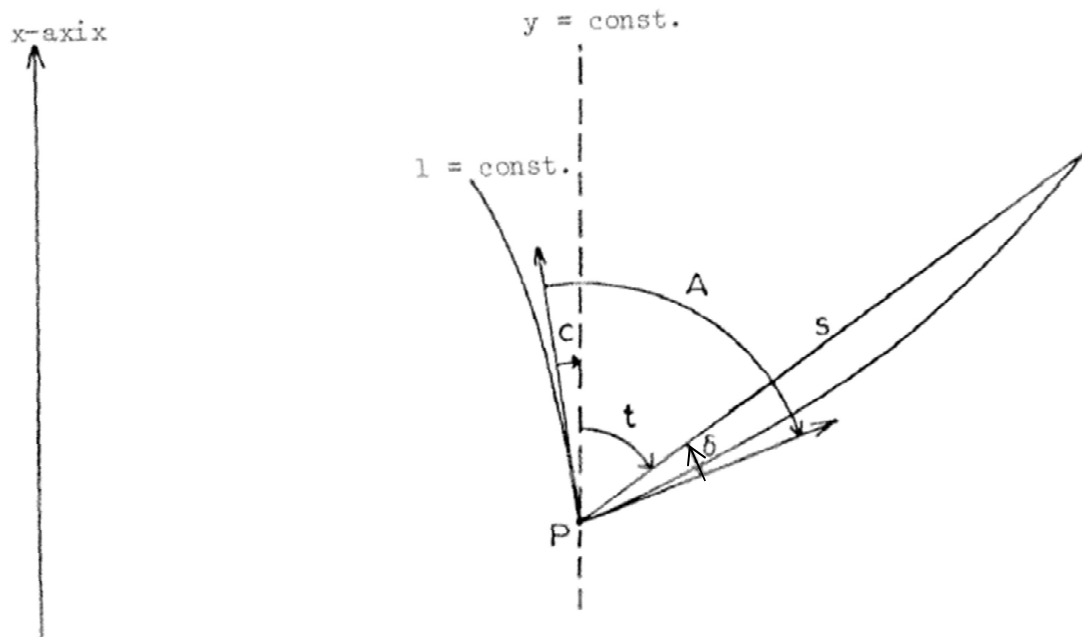


Figure 4.6

Numerical example:

The Meridian convergence and coordinates on the map projection plane (UTM) along the meridian $L = 12^\circ$ East Greenwich, on the WGS84 ellipsoid:

This meridian is at the border between the UTM zones 32 and 33, as zone 32 is between 6° and 12° east of Greenwich. Thus the maximum values of the meridian convergence are shown in table 4.2. For the mainland of Norway this maximum value of c is ca. 3 gon.

Table 4.2

Latitude B	North in km and UTM	East in km and UTM	Meridian convergence, in gon/grads
90°	10001.966	500.0	
84°	9329.0	535.0	3.315...
80°	8883.1	558.1	3.282...
70°	7768.7	614.5	3.132...
60°	6655.2	667.3	2.887...
50°	5542.9	715.0	2.554...
40°	4432.1	756.1	2.143...
30°	3322.6	789.4	1.667...
20°	2214.3	813.9	1.141...
10°	1106.9	828.9	0.579...
0°	0	834.0	0.00000

The UTM east values in table 4.2 show that the width of the UTM zones is (offset of 500 km eastward, $E = 500$ km at the north axis):

$$\begin{aligned}
 0^\circ: & \quad (834.0 - 500) \cdot 2 = 667 \text{ km} && \text{(at equator)} \\
 60^\circ: & \quad (667.3 - 500) \cdot 2 = 334 \text{ km} && \text{(southern part of Norway, Hønefoss)} \\
 84^\circ: & \quad (535.0 - 500) \cdot 2 = 70 \text{ km} && \text{(northernmost latitude of UTM)}
 \end{aligned}$$

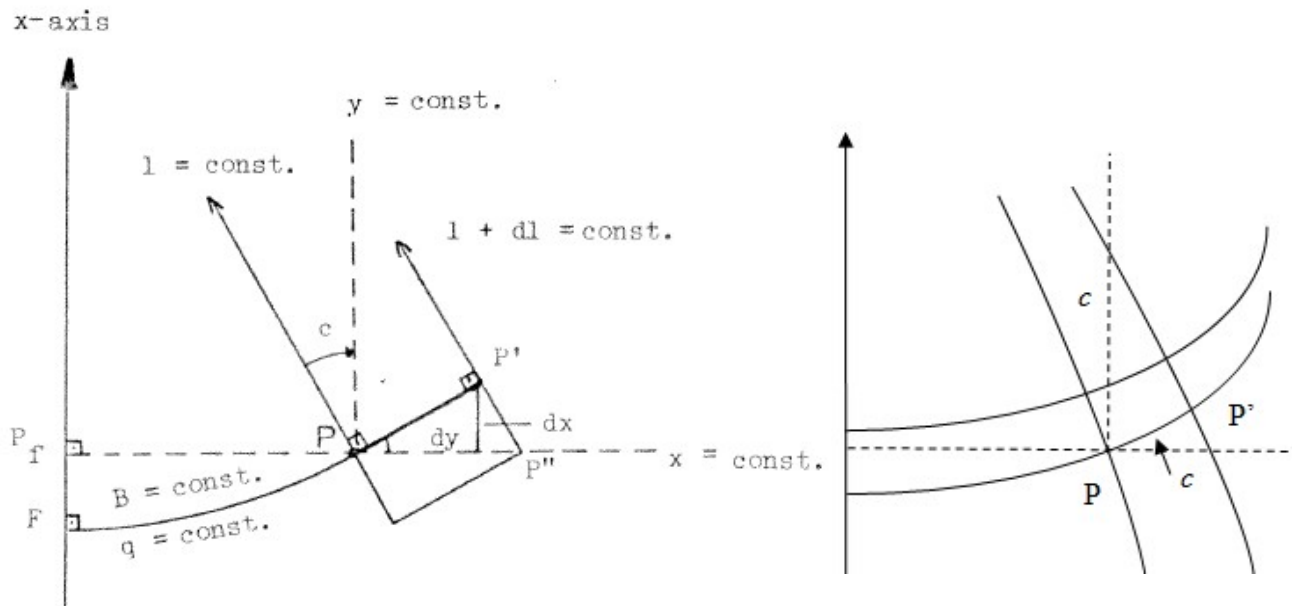


Figure 4.7

Development of the Meridian convergence equation(s)

The Gauss-Krüger meridian convergence is defined as the angle at point P from true north to the grid north. It is indicated by c on figure 4.7. The figure P-P'-P'' is an infinitesimal triangle (infinite small triangle). The lines $l = \text{constant}$ and $l + dl = \text{constant}$ are images of the meridians at longitudes l (going through P) and $l + dl$ (going through P') respectively from the central meridian. The line F-P is the image of the parallel circle through P on the ellipsoid. Point P_f is the footpoint.

There are two separate equations of c , depending on which coordinates that are known:

A: The geodetic coordinates B and l are known

B: The rectangular coordinates x and y in the plane are known.

A: Computation of the meridian convergence c when B and l are known

From figure 4.7 we see:

$$\tan c = \frac{dx}{dy} \quad (4.51a)$$

We want to express $\tan c$ as functions of q and l instead of x and y . This is similar to obtaining the scale factor in chapter 4.3, see figure 4.2 and equation (4.20) on page 63.

If we develop (4.51a) along the line P-P' where q is constant, i.e. $dq = 0$, then (4.20a and b) becomes:

$$dx = \frac{\partial x}{\partial l} \cdot dl \quad \text{and} \quad dy = \frac{\partial y}{\partial l} \cdot dl$$

Inserted into (4.51a):

$$\tan c = \frac{dx}{dy} = \frac{\frac{\partial x}{\partial l} \cdot dl + 0}{\frac{\partial y}{\partial l} \cdot dl + 0} = \frac{\frac{\partial x}{\partial l}}{\frac{\partial y}{\partial l}} \quad (4.51b)$$

and using the connections in the *Cauchy-Riemann differential equations* (4.31b):

$$\tan c = \frac{dx}{dy} = \frac{\frac{\partial x}{\partial l}}{\frac{\partial y}{\partial l}} = -\frac{\frac{\partial y}{\partial q}}{\frac{\partial x}{\partial q}} \quad (4.51c)$$

The next step is to use the Gauss-Krüger conformal map projection equation (4.41) to find the partial derivatives of x and y with respect to l in (4.51b). The projection equations (4.41):

$$x = G_0 - a_2 \cdot l^2 + a_4 \cdot l^4 + \dots$$

$$y = a_1 \cdot l - a_3 \cdot l^3 + a_5 \cdot l^5 + \dots$$

Then the two partial derivatives of (4.51b) are:

$$\frac{\partial x}{\partial l} = \frac{\partial G}{\partial l} - \frac{\partial(a_2 \cdot l^2)}{\partial l} + \frac{\partial(a_4 \cdot l^4)}{\partial l} + \dots = 0 - 2 \cdot a_2 \cdot l + 4 \cdot a_4 \cdot l^3 + \dots$$

$$\frac{\partial y}{\partial l} = \frac{\partial(a_1 \cdot l)}{\partial l} - \frac{\partial(a_3 \cdot l^3)}{\partial l} + \dots = a_1 - 3 \cdot a_3 \cdot l^2 + \dots$$

Inserted into (4.51b):

$$\tan c = \frac{\frac{\partial x}{\partial l}}{\frac{\partial y}{\partial l}} = \frac{-2 \cdot a_2 \cdot l + 4 \cdot a_4 \cdot l^3 + \dots}{a_1 - 3 \cdot a_3 \cdot l^2 + \dots}$$

Substituting the coefficients a_i by the expressions in (4.40) we get a tangent equation for the Gauss-Krüger meridian convergence c .

NB: It is more practical to use the c = equation (4.52b) instead of (4.52a).

$$\tan c = \sin B \cdot l + 1/3 \cdot \sin B \cdot \cos^2 B \cdot (1 + \tan^2 B + 3 \cdot \eta^2 + 2 \cdot \eta^4) \cdot l^3 + \frac{1}{15} \cdot \sin B \cdot \cos^4 B \cdot (2 + 4 \cdot \tan^2 B + 2 \cdot \tan^4 B) \cdot l^5 + \dots \quad (4.52a)$$

An alternative equation of c is obtained by using the series expansion: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} + \dots$ where $x = \tan c$, to go from (4.52a) to (4.52b):

$$c = \tan c - \frac{\tan^3 c}{3} + \frac{\tan^5 c}{5} + \dots$$

In this final equation only terms up to l^5 are shown, terms of higher order can be neglected:

$$c = \sin B \cdot l + \frac{1}{3} \cdot \sin B \cdot \cos^2 B \cdot (1 + 3 \cdot \eta^2 + 2 \cdot \eta^4) \cdot l^3 + \frac{1}{15} \cdot \sin B \cdot \cos^4 B \cdot (2 - \tan^2 B) \cdot l^5 + \dots \quad (4.52b)$$

In (4.52a) and (4.52b) c and l are in radians. If we want degrees instead of radians we must multiply c with the factor $(180/\pi)$ to obtain c in degrees. NB: $l = 0$ on the central meridian, see figure 4.5 on page 71. B is the latitude of point P which we are going to image. For the latitude and ellipsoid dependent function η , see the formula collection in chapter 2.1.

B: Computation of the meridian convergence c when x and y are known

If the rectangular coordinates (x and y) in the map projection plane are given, the geodetic coordinates (B and l) often are determined first by use of equations (4.48b) and (4.50). Then the meridian convergence can be calculated by (4.52a) or (4.52b).

Deriving the equation (4.54) with known x and y coordinates is done quite similar to the derivation of (4.52a).

On figure 4.7 the line P-P' is the image of a line element (or an arc element) of the parallel circle on the ellipsoid through point P.

$$P-P' = m \cdot R_N \cdot \cos B \cdot dl \quad \text{where } m \text{ is the scale factor in P.}$$

The line P'-P'' is the image of an arc element of the meridian through P'.

$P'-P'' = -m \cdot R_M \cdot dB$ Negative because of going southwards from P' to P'' , positive northw.

Meridian convergence, derived along the line $P-P'$ where $dx = 0$, and dq and dl are dependent on dy only, see (4.5) for the definition of isometric latitude q (is used here):

$$\tan c = \frac{-m \cdot R_M \cdot dB}{m \cdot R_N \cdot \cos B \cdot dl} = -\frac{dq}{dl} = -\frac{\frac{\partial q}{\partial y} \cdot dy}{\frac{\partial l}{\partial y} \cdot dy} \quad (4.53)$$

The next step is to use the Gauss-Krüger conformal map projection equation (4.48) to find the partial derivatives of q and l with respect to y in (4.53). The projection equations (4.48):

$$q + i \cdot l = q_0 + b \cdot i \cdot y + b_2 \cdot (i \cdot y)^2 + b_3 \cdot (i \cdot y)^3 + b_4 \cdot (i \cdot y)^4 + \dots$$

$$q = q_0 - b_2 \cdot y^2 + b_4 \cdot y^4 + \dots$$

$$l = b_1 \cdot y - b_3 \cdot y^3 + b_5 \cdot y^5 + \dots$$

Then the two partial derivatives of (4.53) are:

$$\frac{\partial q}{\partial y} = -\left(\frac{\partial q_0}{\partial y} - \frac{\partial(b_2 \cdot y^2)}{\partial y} + \frac{\partial(b_4 \cdot y^4)}{\partial y} + \dots\right) = 0 + 2 \cdot b_2 \cdot y - 4 \cdot b_4 \cdot y^3 + \dots$$

$$\frac{\partial l}{\partial y} = \frac{\partial(b_1 \cdot y)}{\partial y} - \frac{\partial(b_3 \cdot y^3)}{\partial y} + \dots = b_1 - 3 \cdot b_3 \cdot y^2 + \dots$$

Inserted into (4.53):

$$\tan c = \frac{\frac{\partial q}{\partial y}}{\frac{\partial l}{\partial y}} = \frac{2 \cdot b_2 \cdot y - 4 \cdot b_4 \cdot y^3 + \dots}{b_1 - 3 \cdot b_3 \cdot y^2 + \dots}$$

Substituting the coefficients b_i by the expressions in (4.47) we obtain a tangent equation for the Gauss-Krüger meridian convergence c .

We can calculate the meridian convergence from x and y with the following equation (4.54a). NB the term with y^5 is not shown, use (4.54b) instead, is also more practical in use as we obtain c directly:

$$\tan c = \frac{\tan B_f}{R_{Nf}} \cdot y - \frac{\tan B_f}{3 \cdot R_{Nf}^3} \cdot (1 - \eta_f^2) \cdot y^3 + \dots$$

(4.54a)

The subscript f in (4.54) means that the latitude dependent quantities refer to the latitude of the footpoint P_f , the point on the central meridian with the same plane x coordinate as point P . See on page 78 for how the latitude B_f is calculated. For the latitude and ellipsoid dependent functions η and R_N , see the formula collection in chapter 2.1.

An alternative equation of c is obtained by using the series expansion: $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} + \dots$ where $x = \tan c$, to go from (4.54a) to (4.54b):

$$c = \tan c - \frac{\tan^3 c}{3} + \frac{\tan^5 c}{5} + \dots$$

In this final equation only terms up to l^5 are shown, terms of higher order can be neglected:

(4.54b)

$$c = \frac{\tan B_f}{R_{Nf}} \cdot y - \frac{\tan B_f}{3 \cdot R_{Nf}^3} \cdot (1 + \tan^2 B_f - \eta_f^2) \cdot y^3 + \frac{\tan B_f}{15 \cdot R_{Nf}^5} \cdot (2 + 5 \cdot \tan^2 B_f + 3 \cdot \eta_f^2) \cdot l^5 + \dots$$

Numerical example meridian convergence:

See the example exercise no. 5 (2017) in the course TBA4245.

4.5.2 Corrections of the direction of the geodesic

The correction of the horizontal direction, δ , can clearly be seen from figure 4.8.

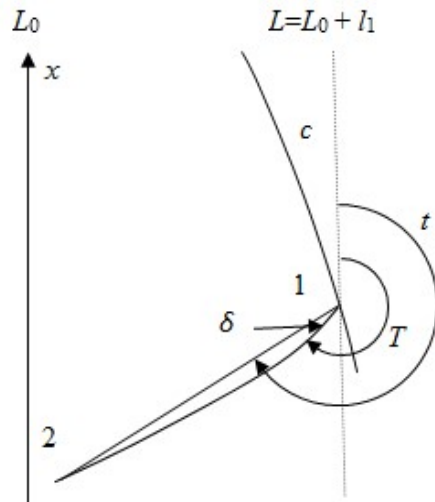


Figure 4.8 On the plane

Definitions of bearings:

- The angle t is the plane bearing (“retningsvinkel”) of the straight line originating/starting in point 1 and ending in point 2.
- The angle T is the plane bearing (“retningsvinkel”) of the curved line, the image of a geodesic originating at point 1 and ending in point 2.

Definition of the sign of δ :

Usually we correct the ellipsoidal horizontal directions (computed or measured) to directions of straight lines on the projection.

Therefore, the correction of the direction of the geodesic, δ , is defined as the bearing of the straight line 1-2, t , minus the bearing of the image of the geodesic in point 1, T , see also figure 4.8 above:

$$t = T + \delta \quad \delta = t - T$$

The image of the geodesic on the plane:

This section is copied from “Geodesy part 1”, page 112.

The projection of the geodesic will in almost all cases turn the concave side against the x axis, as shown in figure 4.9.

Because the concave side of the image curve of the geodesic is turned towards the north axis, we can easily check the sign of δ by using a figure like figure 4.8 or 4.9.

A special case is when the line is crossing the x axis:

Be aware of that the projection of the geodesic normally does not cross the north axis at the same point as the straight line between the same end points. Figure 4.9 only illustrates the principle of the curvature of the line.

The corrections of the horizontal direction may in some cases get a sign in contrary to the sign using (4.55).

For instance: What is the numerical value of δ_1 when $y_1 = -\frac{1}{2}y_2$?

Answer, using the first term of (4.55): $\delta_1 = 0$

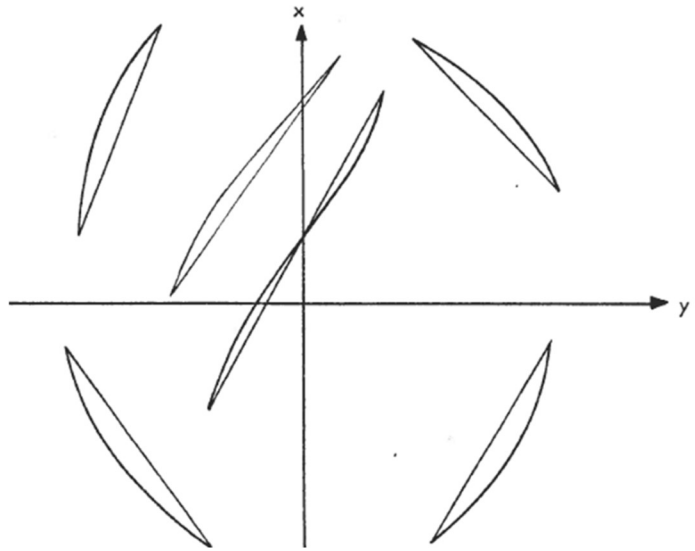


Figure 4.9 Is figure 3.24 in “Geodesy part 1”

Correction of directions, with extra terms:

In “Geodesy part 1” we developed an equation for this correction, (3.64c), and this spherical correction is the first term of equation (4.55). This equation gives sufficient accuracy, also when working within the UTM system for $S < 100$ km. This equation is a result of abbreviated series (sphere formula), and it can be extended by taking into account further terms of the series.

See figure 4.8 where point 1 is the station or starting point and point 2 is the end point. The correction δ from the ellipsoid to the plane in point 1 for the line 1-2 is, see next page for comments:

$$\delta_{1-2} = t_{1-2} - T_{1-2} = -\frac{\rho}{6 \cdot R_1^2} \cdot (2 \cdot y_1 + y_2) \cdot (x_2 - x_1) + \frac{\rho}{12 \cdot R_1^4} \cdot (3 \cdot y_1^2 \cdot y_2 - y_1^3) \cdot (x_2 - x_1) \quad (4.55)$$

In the textbook “Map projections” by Bugayevskiy and Snyder (1995) you will find a third term:

$$\delta_{3rd \text{ term}} = -\frac{e^2 \cdot \sin 2B_1}{2 \cdot R_1^3} \cdot y_1^2 \cdot (y_2 - y_1) \quad (4.56)$$

The magnitude of this term is normally not significant different from 0, and (4.56) can be left out for almost all of the calculations.

In equation (4.55):

- The coordinate y is the east value and x is the north value, both the Gaussian coordinates on the plane.
- The factor ρ is used to obtain grads or gon and not radians as denomination of the correction, $\rho = 200 \text{ gon} / \pi$.
- The radius of the curvature, R_1 , is the *radius of the Gaussian sphere in point 1* (the from point), see chapter 2.5.1 in “Geodesy part 1”:

$$R_1 = \sqrt{R_{M1} \cdot R_{N1}}$$
 where R_M and R_N are the two principal radii of curvature in point 1. See in chapter 2.1 for formulas of R_M and R_N .

The first term is identical to equation (3.64c) in “Geodesy part 1”, the sphere term. The second term can be considered as the “ellipsoidal correction” term.

Numerical example:

The order of magnitude of the first and second term of (4.55) is demonstrated by a numerical example:

$$y_1 = y_2 = 180 \text{ km}$$

$$R_1 = 6390 \text{ km}$$

$$x_2 - x_1 = 60 \text{ km}$$

$$\Rightarrow \text{The correction: } \delta = -0,008\,4192 + 0,000\,0022 = -0,008\,4170 \text{ gon}$$

In this case the second term amounts to 0,0022 mgon.

Numerical example, correction of the direction:

See example exercise no. 5 (2018) in the course TBA4245. It is also covering UTM coordinates.

4.5.3 Scale factor, m

One of the conditions at start of chapter 4.4 in developing the equations of the conformal Gauss-Krüger map projections, is that the central meridian shall be mapped in scale 1 : 1 (the scale factor is 1), no scale distortion along the north axis (the central meridian).

Outside the common line on the plane we have seen that the scale is increasing with the north value in the Mercator conformal projection (where equator is the common line) in figure 4.3, and increasing with the east value in the Gauss-Krüger conformal map projection (where a meridian is the common line).

The local scale factor of a point is of interest in some cases. We can develop equations that will give us the scale factor in a point, from the equations of the map projection.

There are two separate equations to use for calculating the scale factor, depending on which coordinates that are known:

A: The geodetic coordinates B and l on the ellipsoid are known

B: The rectangular coordinates x and y on the plane are known.

Scale factor, m: At a point, or the ratio between an (infinitesimal) distance on the plane and on the ellipsoid.

We used the scale factor to develop the equations of the conformal Gauss-Krüger map projections, see equation (4.27) on page 65, where E and G are the Gaussian coefficients, R_N is the principle radius of curvature and B is the geographic latitude: Equation of m^2 , compare to (4.22) and (4.27):

$$m^2 = \frac{E}{R_N^2 \cdot \cos^2 B} = \frac{G}{R_N^2 \cdot \cos^2 B} = \frac{1}{R_N^2 \cdot \cos^2 B} \cdot \left[\left(\frac{\partial x}{\partial q} \right)^2 + \left(\frac{\partial y}{\partial q} \right)^2 \right]$$

Introducing another G , this time the length of an arc of the central meridian, see (4.36):

$$m^2 = \frac{1}{R_N^2 \cdot \cos^2 B} \cdot \left[\left(\frac{\partial x}{\partial G} \right)^2 \cdot \left(\frac{\partial G}{\partial q} \right)^2 + \left(\frac{\partial y}{\partial G} \right)^2 \cdot \left(\frac{\partial G}{\partial q} \right)^2 \right] = \frac{1}{R_N^2 \cdot \cos^2 B} \cdot \left[\left(\frac{\partial x}{\partial G} \right)^2 + \left(\frac{\partial y}{\partial G} \right)^2 \right] \cdot \left(\frac{\partial G}{\partial q} \right)^2$$

According to (4.36): $\frac{dG}{dq} = R_N \cdot \cos B$, and then:

$$m^2 = \left(\frac{\partial x}{\partial G} \right)^2 + \left(\frac{\partial y}{\partial G} \right)^2 \quad (4.57)$$

A: Scale factor at a point from the known geodetic coordinates B and l

We have (4.57).

The next step is to use the Gauss-Krüger conformal map projection equation (4.41) to find the partial derivatives of x and y with respect to G in (4.57). The projection equations (4.41):

$$x = G_0 - a_2 \cdot l^2 + a_4 \cdot l^4 + \dots$$

$$y = a_1 \cdot l - a_3 \cdot l^3 + a_5 \cdot l^5 + \dots$$

The partial derivatives of x and y with respect to G :

$$\frac{\partial x}{\partial G} = \frac{\partial G}{\partial G} - \frac{\partial(a_2)}{\partial q} \cdot \frac{dq}{dG} \cdot l^2 + \frac{\partial(a_4)}{\partial q} \cdot \frac{dq}{dG} \cdot l^4 + \dots$$

$$\frac{\partial y}{\partial G} = \frac{\partial(a_1)}{\partial q} \cdot \frac{dq}{dG} \cdot l - \frac{\partial(a_3)}{\partial q} \cdot \frac{dq}{dG} \cdot l^3 + \dots$$

From (4.40) we derive:

$$\frac{da_1}{dq} = 2 \cdot a_2 \quad \frac{da_2}{dq} = 3 \cdot a_3$$

$$\frac{da_3}{dq} = 4 \cdot a_4 \quad \frac{da_4}{dq} = 5 \cdot a_5$$

With a_i from (4.40) and $\frac{dG}{dq} = R_N \cdot \cos B$ according to (4.36) we get, only terms up to l^3 is shown here:

$$\frac{\partial x}{\partial G} = \frac{\partial G}{\partial G} - \frac{\partial(a_2)}{\partial q} \cdot \frac{dq}{dG} \cdot l^2 + \dots = 1 + \frac{1}{2} \cdot \cos^2 B \cdot (1 - \tan^2 B + \eta^2) \cdot l^2 + \dots$$

$$\frac{\partial y}{\partial G} = \frac{\partial(a_1)}{\partial q} \cdot \frac{dq}{dG} \cdot l + \dots = -\cos B \cdot \tan B \cdot l - \frac{1}{6} \cdot \cos^3 B \cdot \tan B \cdot (5 - \tan^2 B + \dots) \cdot l^3 + \dots$$

Inserted into (4.57):

$$m^2 = 1 + \cos^2 B \cdot (1 + \eta^2) \cdot l^2 + \frac{1}{3} \cdot \cos^4 B \cdot (2 - \tan^2 B + \dots) \cdot l^4 + \dots$$

To find the square root of the right-hand side of m^2 we develop it in binomial series, e.g.:

$$\sqrt{1+v} = 1 + \frac{1}{2} \cdot v - \frac{1}{8} \cdot v^2 + \dots$$

This binomial series applied on the m^2 equation above, where v is equal to the two last terms, $m = 1 + v$:

$$m = 1 + \frac{1}{2} \cos^2 B \cdot (1 + \eta^2) \cdot l^2 + \frac{1}{24} \cdot \cos^4 B \cdot (5 - 4 \cdot \tan^2 B - 6 \cdot \eta^2 + \dots) \cdot l^4 + \dots \quad (4.58)$$

In (4.58) the longitude l is expressed in radians. NB: $l = 0$ on the central meridian, see figure 4.5 on page 71. The factor $(1/\rho)$ must be used to obtain radians and not degrees as denomination of the longitude l . $\rho = 180 \text{ degrees} / \pi$ and $l_{\text{rad}} = l^\circ / \rho$

B is the latitude of the point P which we are going to image. For the latitude and ellipsoid dependent function η , see the formula collection in chapter 2.1.

B: Scale factor at a point using the known rectangular coordinates x and y on the plane

If the rectangular coordinates (x and y) in the map projection plane are given, the geodetic coordinates (B and l) often are determined first by use of equations (4.48b) and (4.50). Then the scale factor can be calculated by (4.58).

Deriving equation (4.59) with known x and y coordinates is done quite similar to the derivation of (4.58).

We start with (4.57), and we make use of a similar procedure as used under method A above.

The next step is to use the Gauss-Krüger conformal map projection equation (4.48) to find the partial derivatives of q and l with respect to y in (4.56). The projection equations (4.48) and (4.50):

$$q = q_0 - b_2 \cdot y^2 + b_4 \cdot y^4 + \dots$$

$$B = B_F + f_2 \cdot y^2 + f_4 \cdot y^4 + f_6 \cdot y^6 + \dots$$

$$l = b_1 \cdot y - b_3 \cdot y^3 + b_5 \cdot y^5 + \dots$$

An after some “manipulations” similar to the method described in method A, we can use the following equations, (4.59a) and (4.59b), to calculate the scale factor from x and y :

$$m = 1 + \frac{1 + \eta_f^2}{2 \cdot R_{Nf}^2} \cdot y^2 + \frac{1 + 2 \cdot \eta_f^2}{24 \cdot R_{Nf}^4} \cdot y^4 + \frac{1 + 3 \cdot \eta_f^2}{720 \cdot R_{Nf}^6} \cdot y^6 + \dots \quad (4.59a)$$

The subscript f in (4.59a) and (4.59b) means that the latitude dependent quantities refer to the latitude of the footpoint P_f , the point on the central meridian with the same plane x coordinate as point P . See page 78 for how the latitude B_f is calculated. For the latitude and ellipsoid dependent functions η and R_N , see the formula collection in chapter 2.1.

The Gaussian mean radius of curvature, see the formulas in chapter 2.1:

$$R_f^2 = R_{Mf} \cdot R_{Nf} = \frac{c}{V^3} \cdot \frac{c}{V} = \frac{1}{V^2} \cdot \frac{c^2}{V^2} = \frac{R_{Nf}^2}{1 + \eta_f^2}$$

Substituting the Gaussian mean radius of curvature into (4.59a) gives an alternative equation to (4.59a):

$$m = 1 + \frac{1}{2 \cdot R_f^2} \cdot y^2 + \frac{1}{24 \cdot R_f^4} \cdot y^4 + \frac{1}{720 \cdot R_f^6} \cdot y^6 + \dots \quad (4.59b)$$

The last term in (4.59b) is copied from the textbook “Map projections” by Bugayevskiy and Snyder (1995). The magnitude of this 4th term in (4.59a) and (4.59b) is normally not significant different from 0, and can be left out for almost all of the calculations.

Numerical example, scale factor:

Control point: Moholt column. Given, known coordinates:

	UTM	Gauss-Krüger	Latitude, longitude
North	$N = 7031952,892$	$x = 7034766,799$	$B = 63,40895119^\circ$
East	$E = 571469,041$	$y = 71497,640$	$L = 10,43111038^\circ$

We must use the Gauss-Krüger coordinates in the equations, not the UTM coordinates. They are obtained by adjusting the UTM coordinates for the scale factor of 0,9996 and for the offset of 500 000 m on the UTM east value. See coordinate values in the table above.

The longitude l in the equations, is from the central meridian of UTM zone 32:

$$l = L - 9 = 1,43111038^\circ$$

The parameters of the ellipsoid used in the Norwegian datum EUREF89:

$$\begin{aligned} a &= 6378137,0000 \\ b &= 6356752,3141 \\ e^2 &= 0,0066943800355 \end{aligned}$$

For the latitude of point P_f , the footpoint, equation (8.27) in “GNSS” by Hofmann-Wellenhof (2008) is used:

$$B_f = 63,41611492^\circ$$

Equation (4.58), the scale factor m from B and l , the values of each term are shown:

$$m = 1 + 0,000\,062\,5857 - 0,000\,000\,0356 = 1,000\,062\,550$$

Equation (4.59b), the scale factor m from x and y , the values of each term are shown. Notice that the 4th and last term is not contributing to the value of m :

$$m = 1 + 0,000\,062\,5778 + 6,527^{-10} + 2,72^{-15} = 1,000\,062\,578$$

There is a small difference between the values of m in (4.58) and (4.59b). With 7 digits behind the decimal point the value is the same.

Numerical example, scale factor

See example exercise no. 5 (2018) in the course TBA4245. It is also covering UTM coordinates.

4.5.4 Correction of distances, d

Definition of the sign of d :

Usually we correct the ellipsoidal distances (computed or measured) to distances on the plane. Therefore, the correction to the distance of the geodesic, d , is here defined as the distance of the straight line 1-2 in the map projection plane, s , minus the distance 1-2 on the ellipsoid, S :

$$s = S + d \quad d = s - S \quad S = s - d$$

In “Geodesy part 1” we developed an equation for this correction, (3.65), and this spherical correction is the first term of equation (4.60a). The equation (3.65) gives sufficient accuracy (for $S < 100$ km). The equation (3.65) in “Geodesy part 1” is a result of abbreviated series (sphere formula), and it can be extended by taking into account further terms of these series.

The correction on the ellipsoidal distance when we go from the ellipsoid to the Gauss-Krüger conformal map projection is:

$$d = \frac{s}{6 \cdot R_m^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_2^2) - \frac{s \cdot \eta_m^2 \cdot \tan B_m}{6 \cdot R_m^3} \cdot (y_2^2 - y_1^2) \cdot (x_2 - x_1) - \frac{s \cdot 5 \cdot y_m^4}{24 \cdot R_m^4} \quad (4.60a)$$

In equation (4.60a):

- The coordinate y is the east value and x is the north value, both the Gaussian coordinates on the plane.
- Subscript m means values in the middle point of the line.
- The radius of the curvature, R_m , is the *radius of the Gaussian sphere in the middle point* of the line 1-2, see chapter 2.5.1 in “Geodesy part 1”:

$$R_m = \sqrt{R_{Mm} \cdot R_{Nm}}, \text{ where } R_M \text{ and } R_N \text{ are the two principal radii of curvature in the middle point of the line 1-2.}$$

See in chapter 2.1 for formulas of R_M and R_N .

Comments on equation (4.60a):

On long distances and/or with large y values (east values) we must be more precise on which values we use in the correction equation and how many terms we must use.

The first term we recognise as the "sphere formula".

In addition we have 2 correction terms which normally are very small in absolute values:

- The term in the middle is the "ellipsoidal correction" term.
- In (4.60a) the 3rd and last term is copied from "Geodätische Rechnungen und Abbildungen" by Grossmann (1964).

4.5.5 Correction of distances, UTM coordinates

We have to take into account that UTM is decreased by a constant factor 0.9996 compared with the Gauss-Krüger map projection.

The Gauss-Krüger distance s is the theoretically correct distance to use in the formula (and not D_{UTM} which is computed from the UTM coordinates), where $s = D_{UTM} / 0,9996$

The UTM coordinates must also be corrected to avoid the "UTM traps":

- Reduce the east values with 500 000 m
- Rectify the coordinates for the shrinkage ($k = 0.9996$).

Starting point is formula (4.60a):

$$s - S = \frac{s}{6R^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_2^2) + \text{two extra terms in (4.60a)}$$

$$\frac{D_{UTM}}{0,9996} - S = \frac{D_{UTM} / 0,9996}{6R^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_2^2) + \text{two extra terms in (4.60a)}$$

$$-S = \frac{D_{UTM} / 0,9996}{6R^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_2^2) - \frac{0,9996 \cdot D_{UTM} + 0,0004 \cdot D_{UTM}}{0,9996} + \text{two extra terms in (4.60a)}$$

Finally, the equation for correction of the distance, when UTM coordinates is used; notice the extra UTM term (the 2nd term):

(4.60b)

$$d_{UTM} = D_{UTM} - S =$$

$$d_{UTM} = \frac{D_{UTM}/0,9996}{6 \cdot R_m^2} \cdot (y_1^2 + y_1 \cdot y_2 + y_2^2) - \frac{0,0004 \cdot D_{UTM}}{0,9996} - \frac{s \cdot \eta_m^2 \cdot \tan B_m}{6 \cdot R_m^3} \cdot (y_2^2 - y_1^2) \cdot (x_2 - x_1) - \frac{s \cdot 5 \cdot y_m^4}{24 \cdot R_m^4}$$

In equation (4.60b):

- The coordinate y is the east value and x is the north value, both the Gaussian coordinates on the plane.
- D_{UTM} is the distance calculated from the UTM coordinates of point 1 and 2.
- For the two extra terms in (4.60b), see (4.60a)
- The radius of the curvature, R_m , is the radius of the Gaussian sphere in the middle point of the line 1-2, see chapter 2.5.1 in "Geodesy part 1":

$$R_m = \sqrt{R_{M_m} \cdot R_{N_m}}, \text{ where } R_M \text{ and } R_N \text{ are the two principal radii of curvature in the middle point of the line 1-2.}$$

See in chapter 2.1 for formulas of R_M and R_N .

Numerical example no. 1, correction of the distance:

$$\begin{array}{llll}
 y_1 = 160000 \text{ m} & y_2 = 204000 \text{ m} & y_m = 182000 \text{ m} & x_2 - x_1 = 60000 \text{ m} \\
 s = 74404.30 \text{ m} & B = 64 \text{ degrees} & R = 6390000 \text{ m} &
 \end{array}$$

$$\begin{array}{ll}
 \text{Sphere term:} & +30.3245 \text{ m} \\
 \text{The ellipsoid correction term:} & -0.0001 \text{ m (not contributing)} \\
 \text{The third and last term (Grossman):} & -0.0102 \text{ m} \\
 \text{The Gauss-Krüger correction: } d = s - S = & +30.3142 \text{ m}
 \end{array}$$

The numerical value of the UTM term in the example above is

$$0,0004 \cdot (D_{UTM} / 0.9996) = -29.762 \text{ m}$$

The total UTM correction on the distance S : $d_{UTM} = 30,314 - 29,762 = + 0,552 \text{ m}$

Numerical example no. 2, correction of the distance

See example exercise no. 5 (2018) in the course TBA4245. It is also covering UTM coordinates.

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